

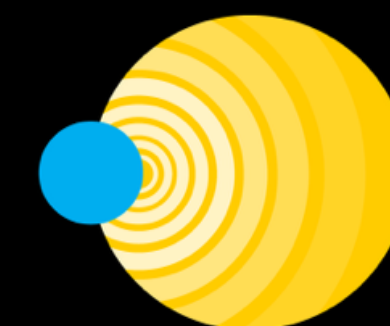
BREAKTHROUGH LISTEN

On Detecting Interstellar Scintillation in Narrowband Radio SETI

BRYAN BRZYCKI

UNIVERSITY OF CALIFORNIA BERKELEY

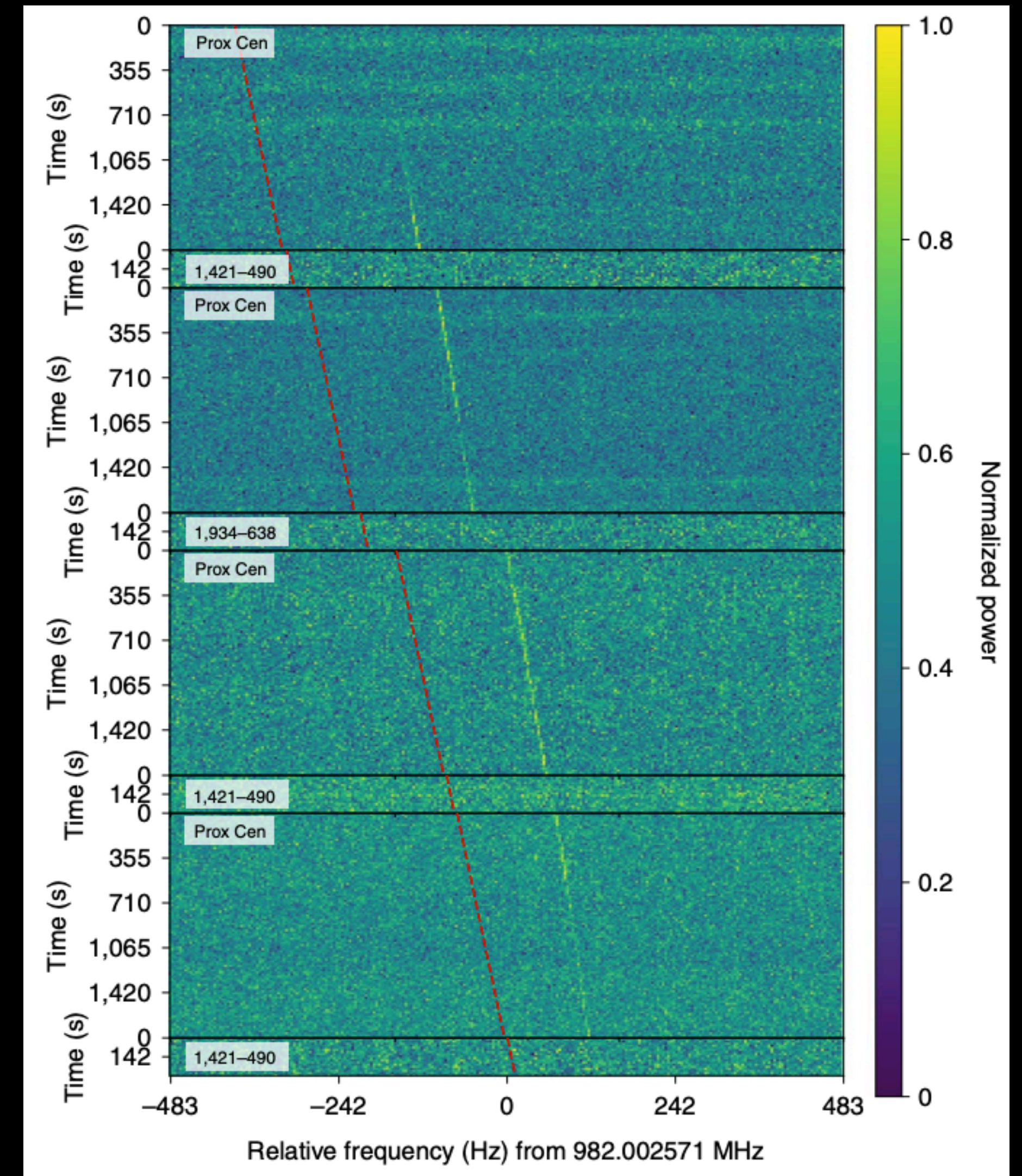
AAS, JANUARY 12, 2023



BERKELEY SETI
RESEARCH CENTER

Standard filters used for radio technosignature candidates

- Narrowband (vs. astrophysical sources)
- Non-zero drift rate (vs. RFI)
- Sky localization (vs. RFI)



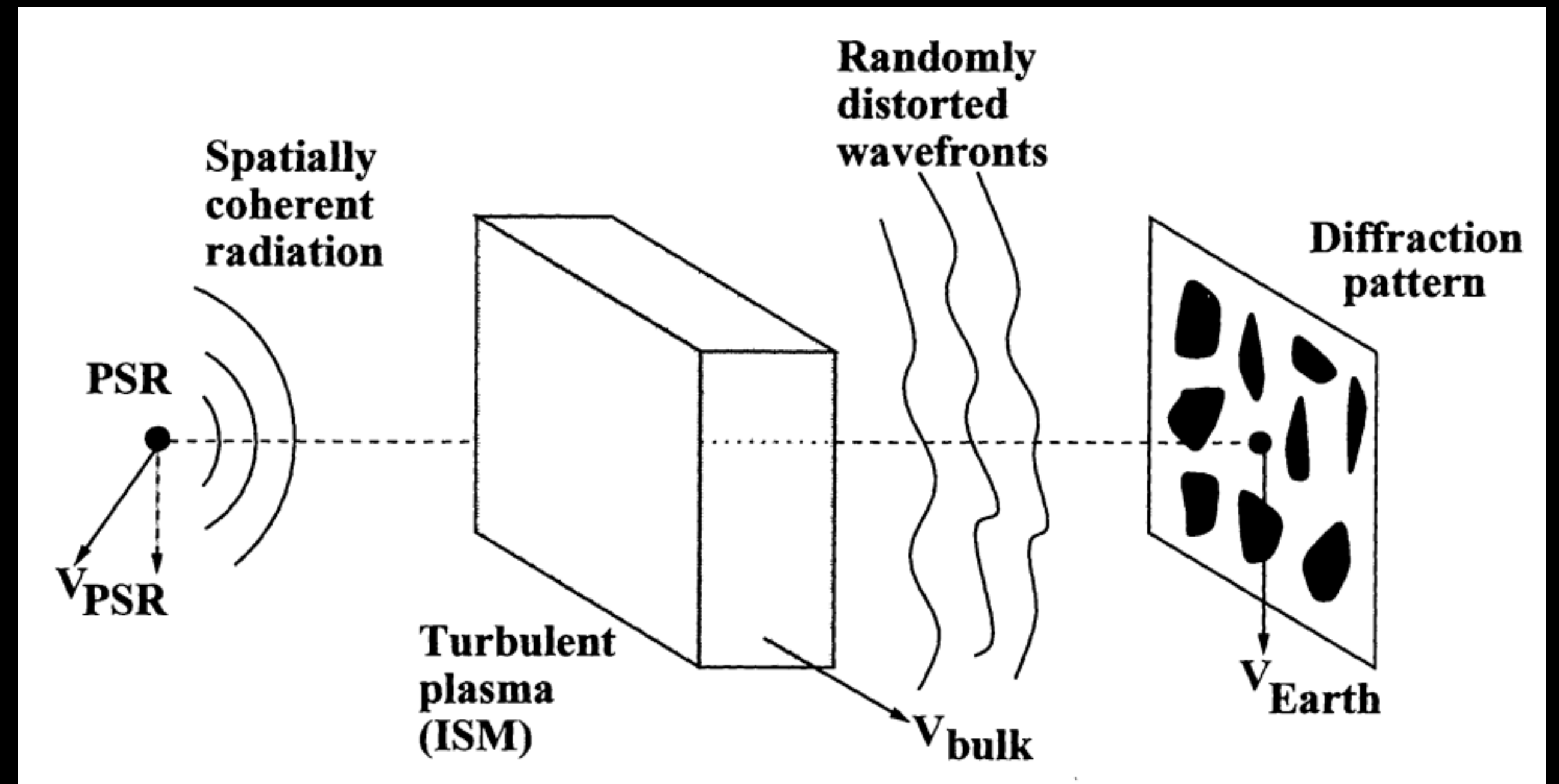
Can we use astrophysical phenomena as a filter to distinguish technosignatures from RFI?



ESA

Diffraction scintillation in the ISM

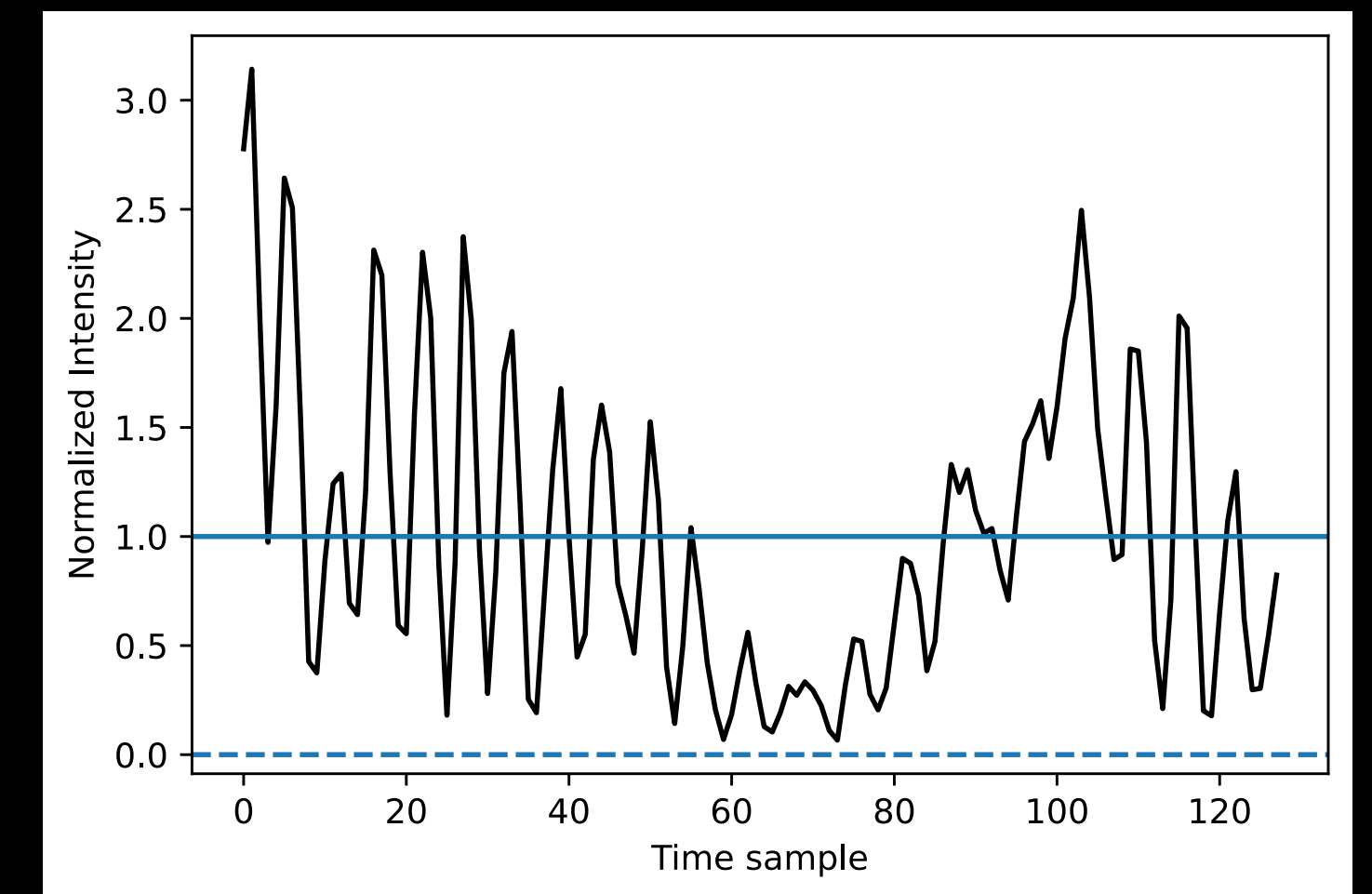
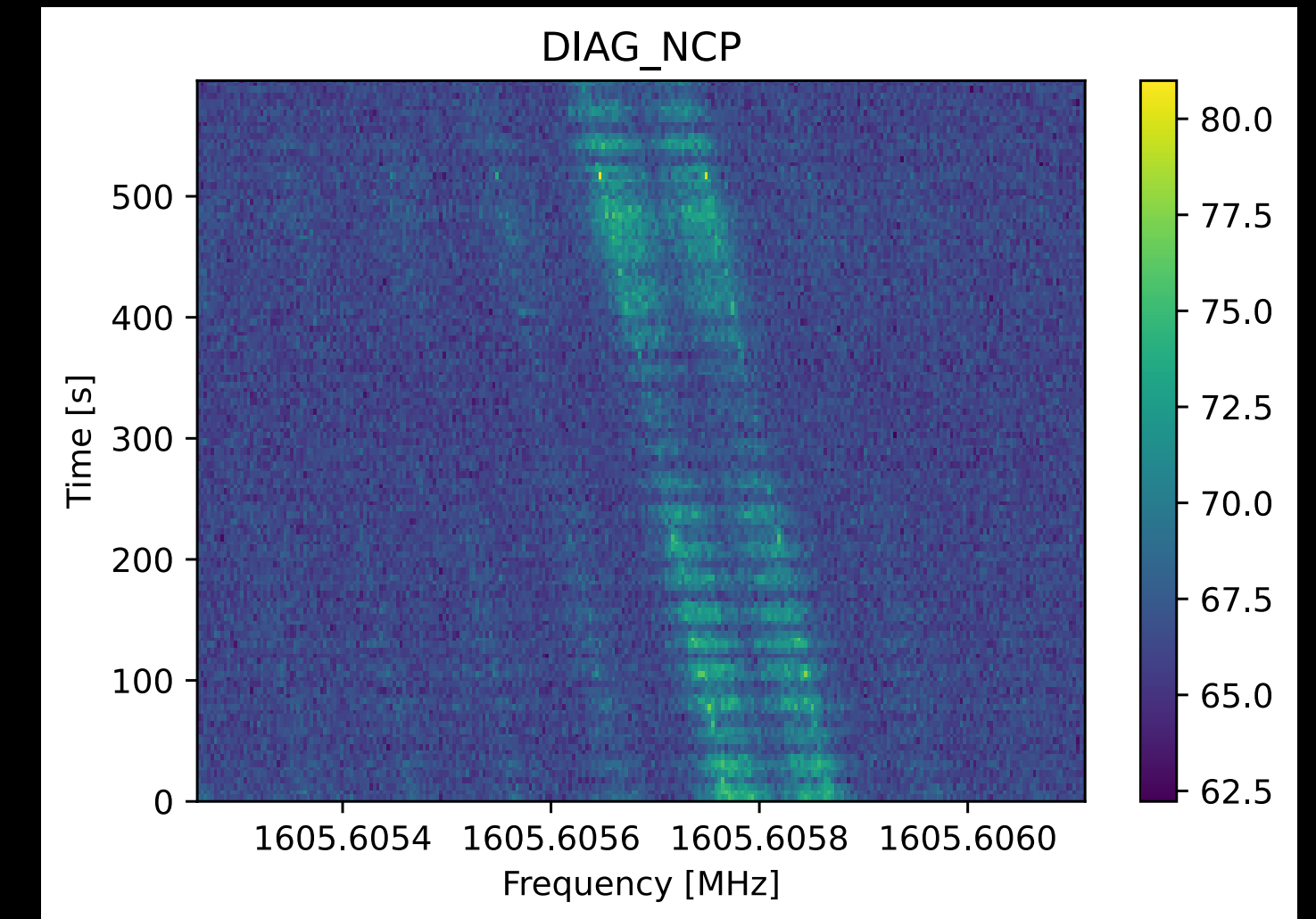
- Electron density fluctuations in ionized plasma give rise to phase fluctuations
- Interference pattern at observer plane with characteristic spatial and spectral scales
- Can lead to 100% intensity modulation on characteristic temporal scales Δt_d , especially towards the galactic center

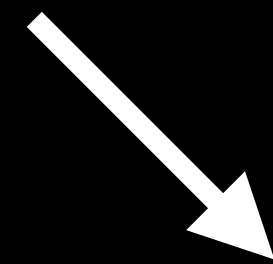
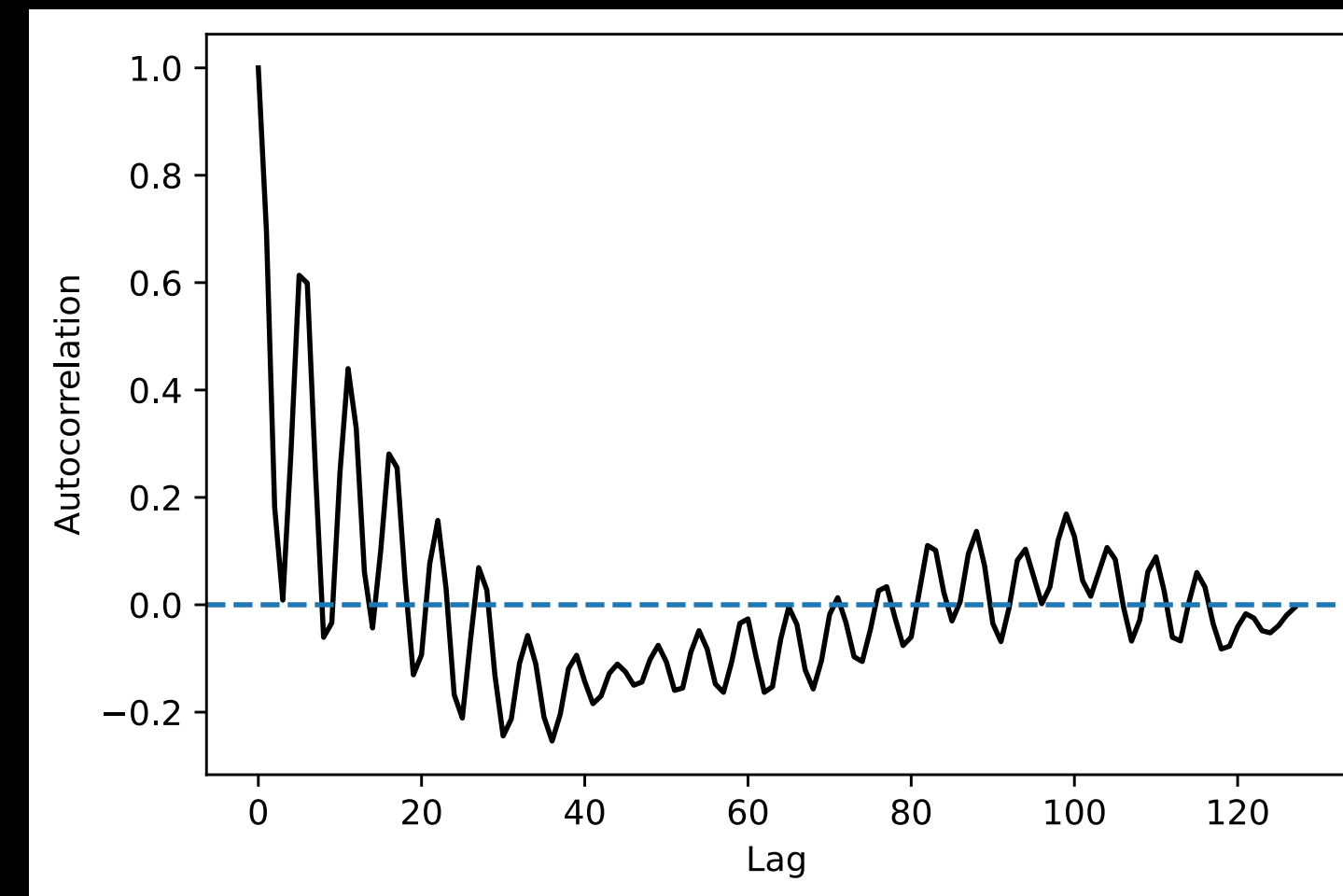
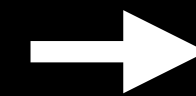
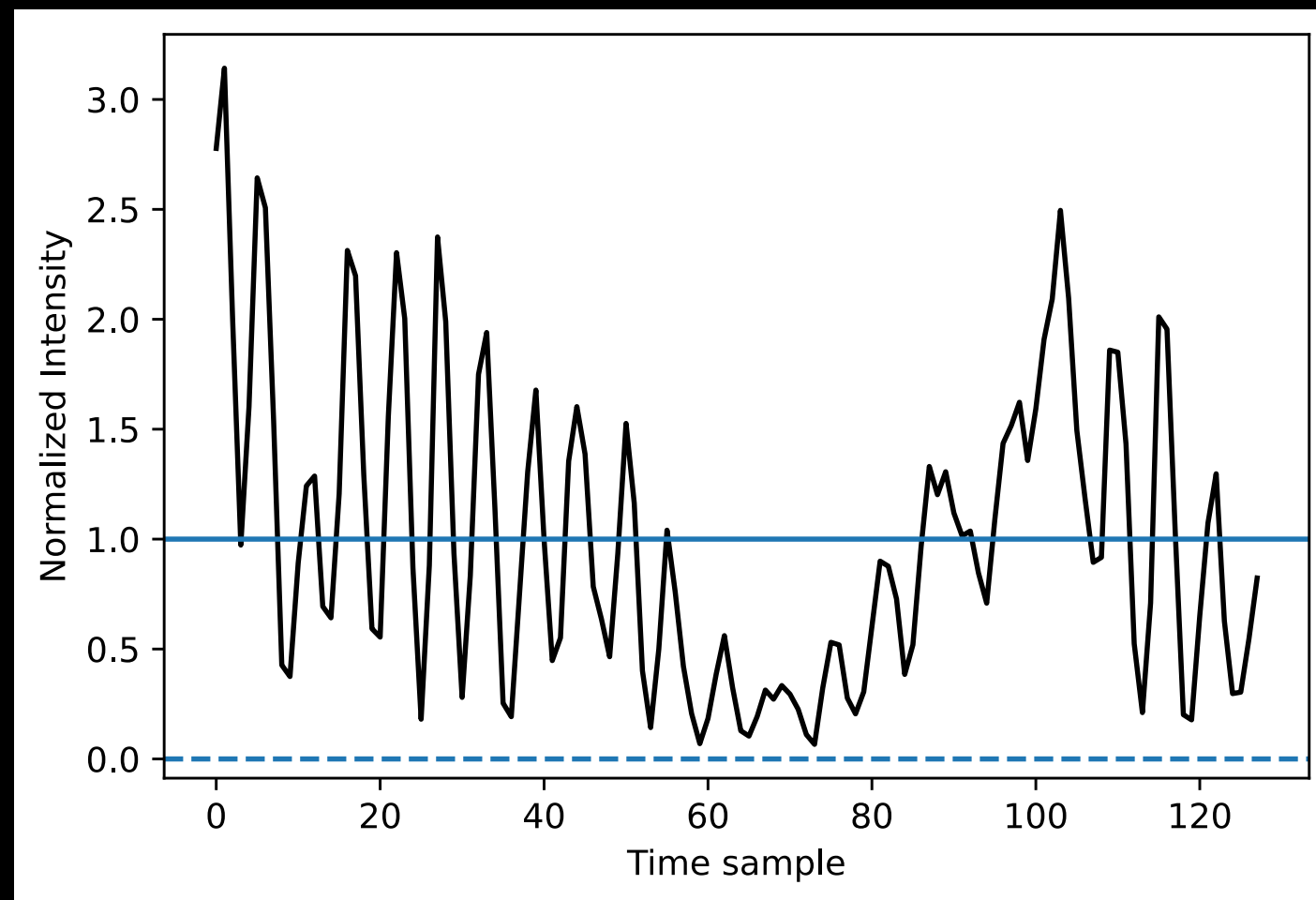
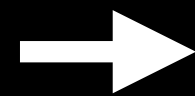
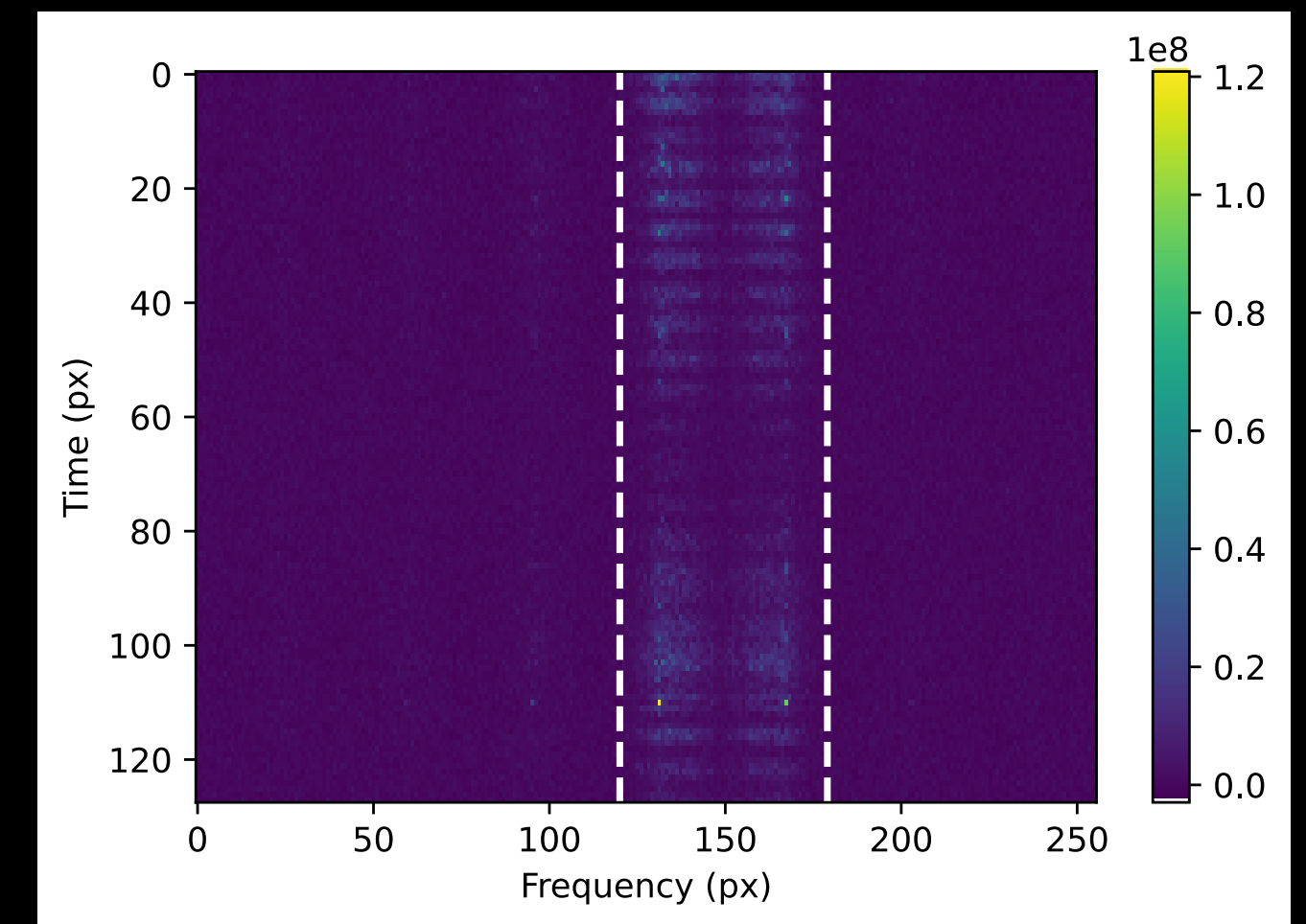
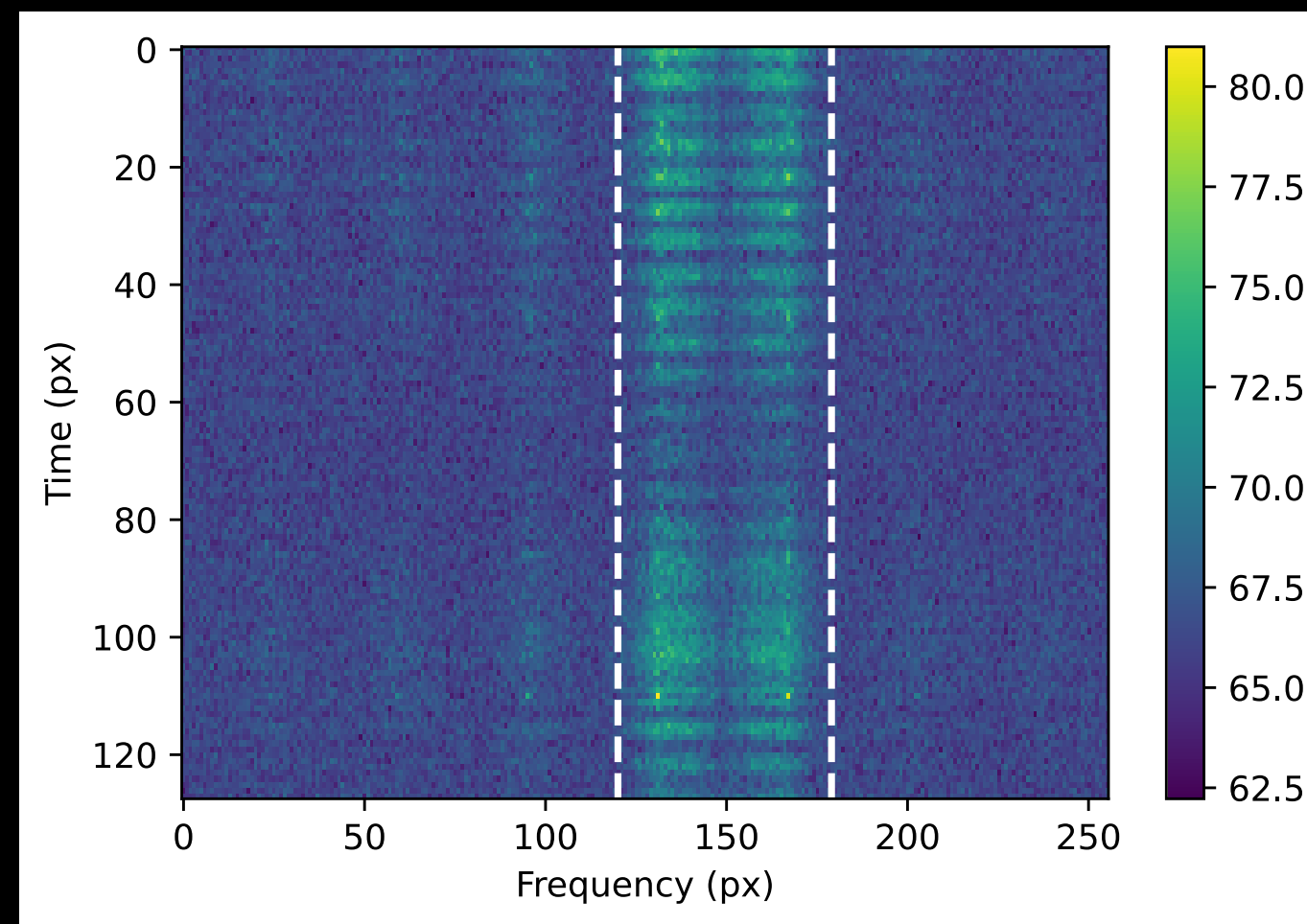
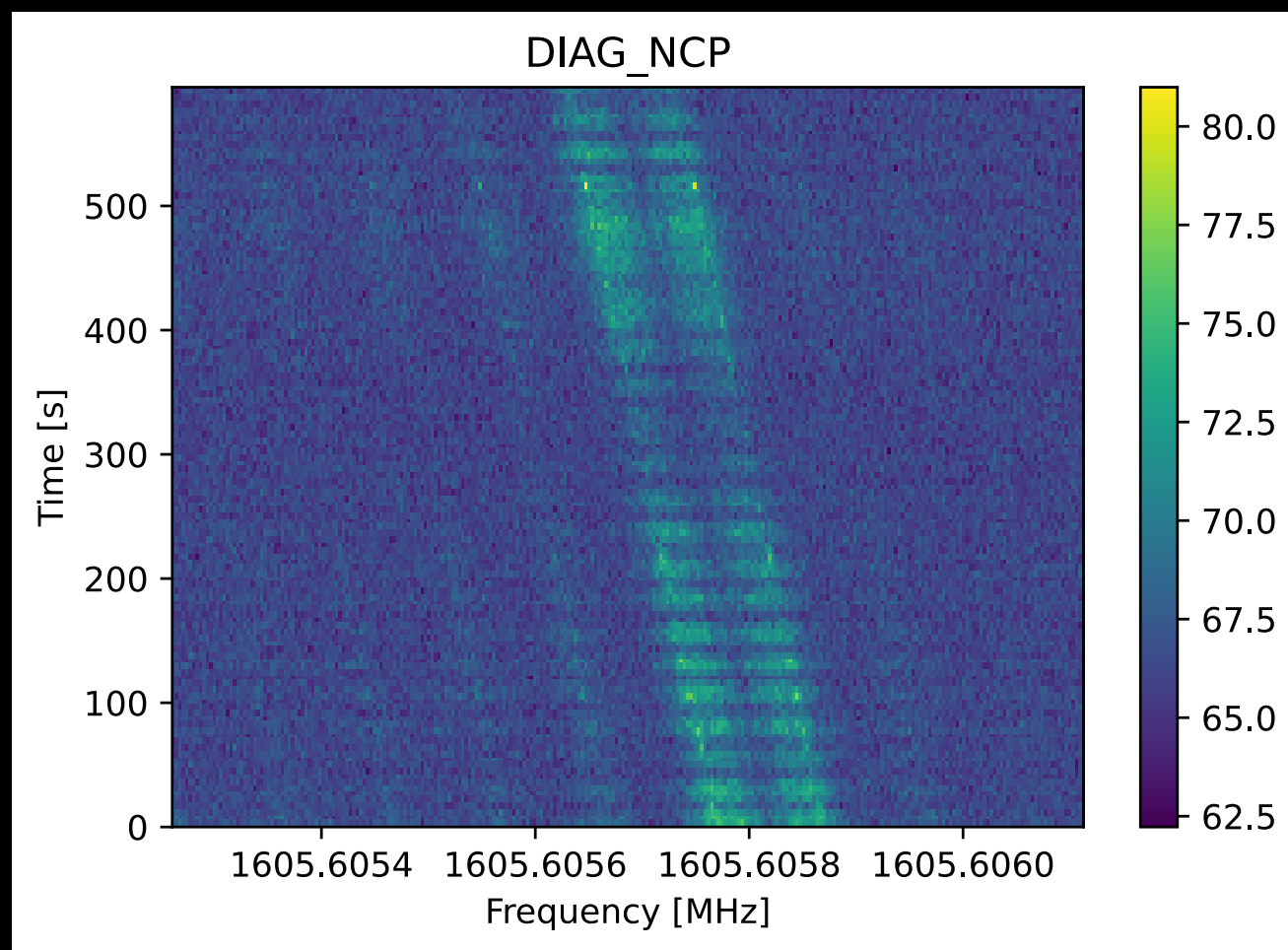


Cordes 2002

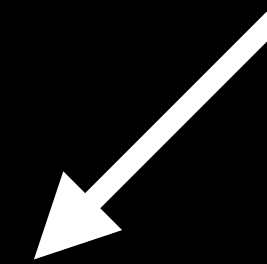
So how might we detect scintillation?

- Estimate intensity time series from detected signals for analysis
- Since scintillation is stochastic, identify measurable statistics for asymptotic behavior
 - E.g. standard deviation, Kolmogorov-Smirnoff statistic, fit to autocorrelation function
- Would existing RFI modulation confound real scintillation?
 - Develop process for creating synthetic scintillated data
 - Compare statistical distributions of detected signals with those of synthetic scintillated signals





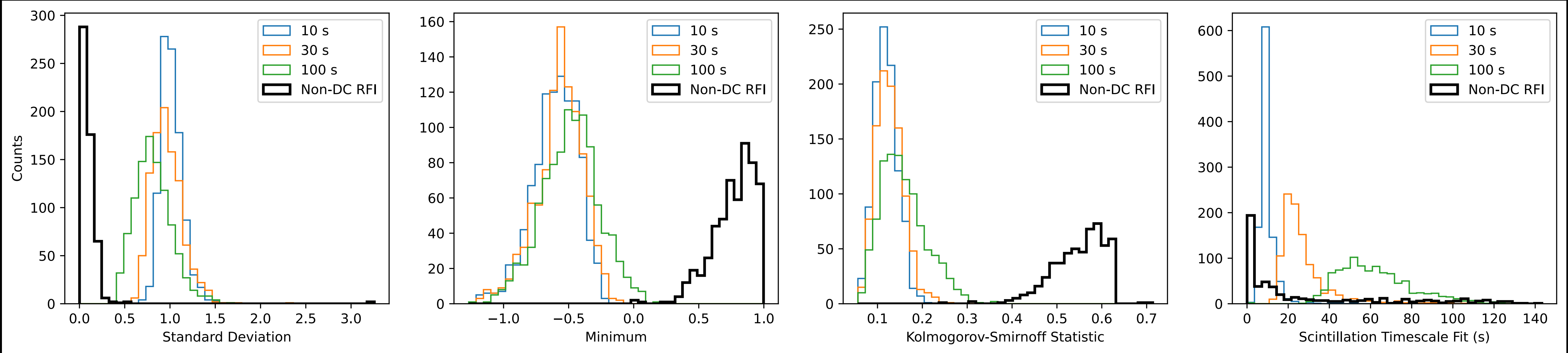
Diagnostic statistics



GBT RFI vs. synthetic scintillated signals

C band

S/N > 25



Standard Deviation

Minimum

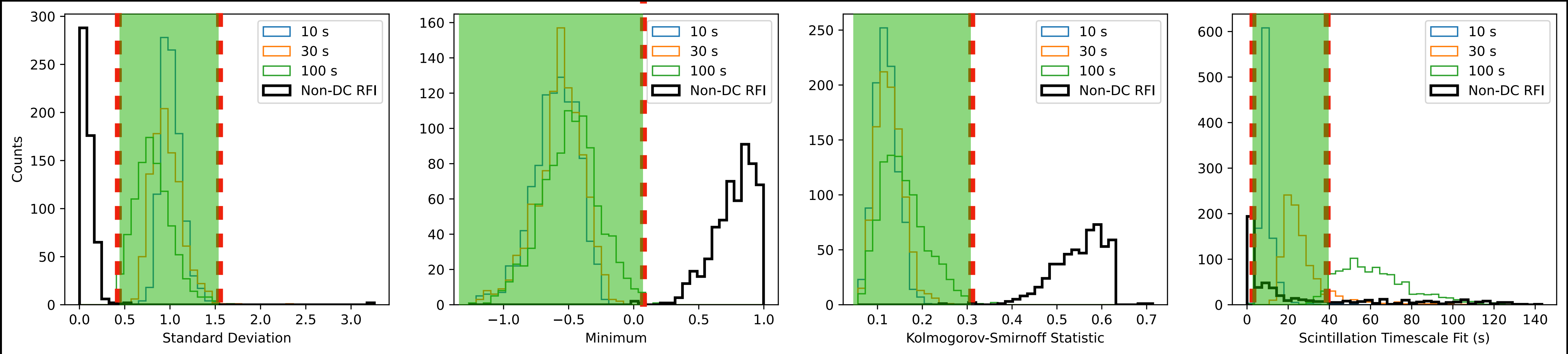
Kolmogorov-Smirnoff Statistic

Scintillation Timescale Fit

GBT RFI vs. synthetic scintillated signals

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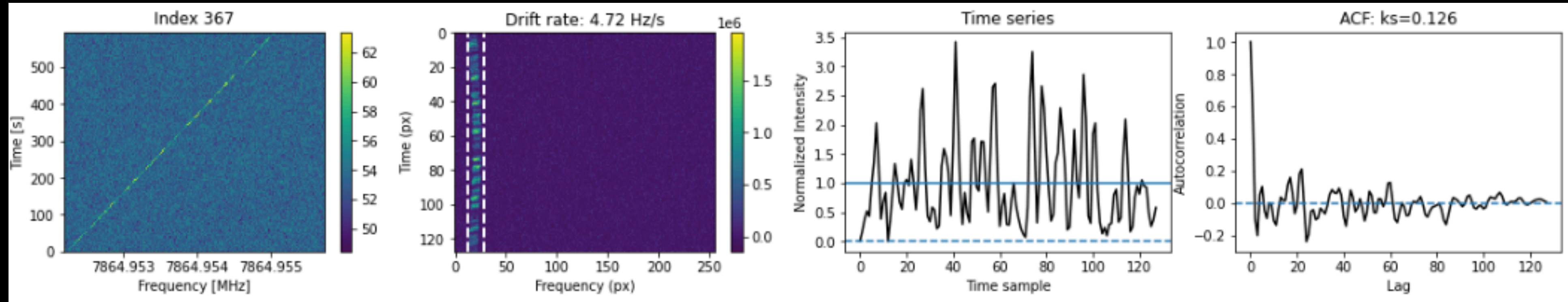
Scintillation Timescale Fit

Summary

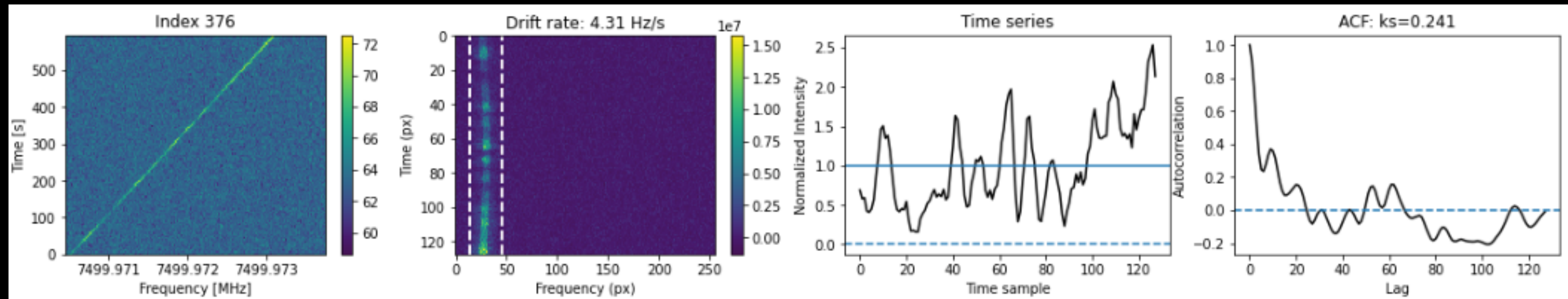
- Developed a methods framework for scintillation analysis, with accompanying codebase
- Based on the RFI environment, we get better thresholding at higher frequencies
- Better extraction & classification methods may lead to improvements
- Looking forward: dedicated survey to search for scintillated signals towards the Galactic Center

Extra Slides

What signals pass these thresholds?



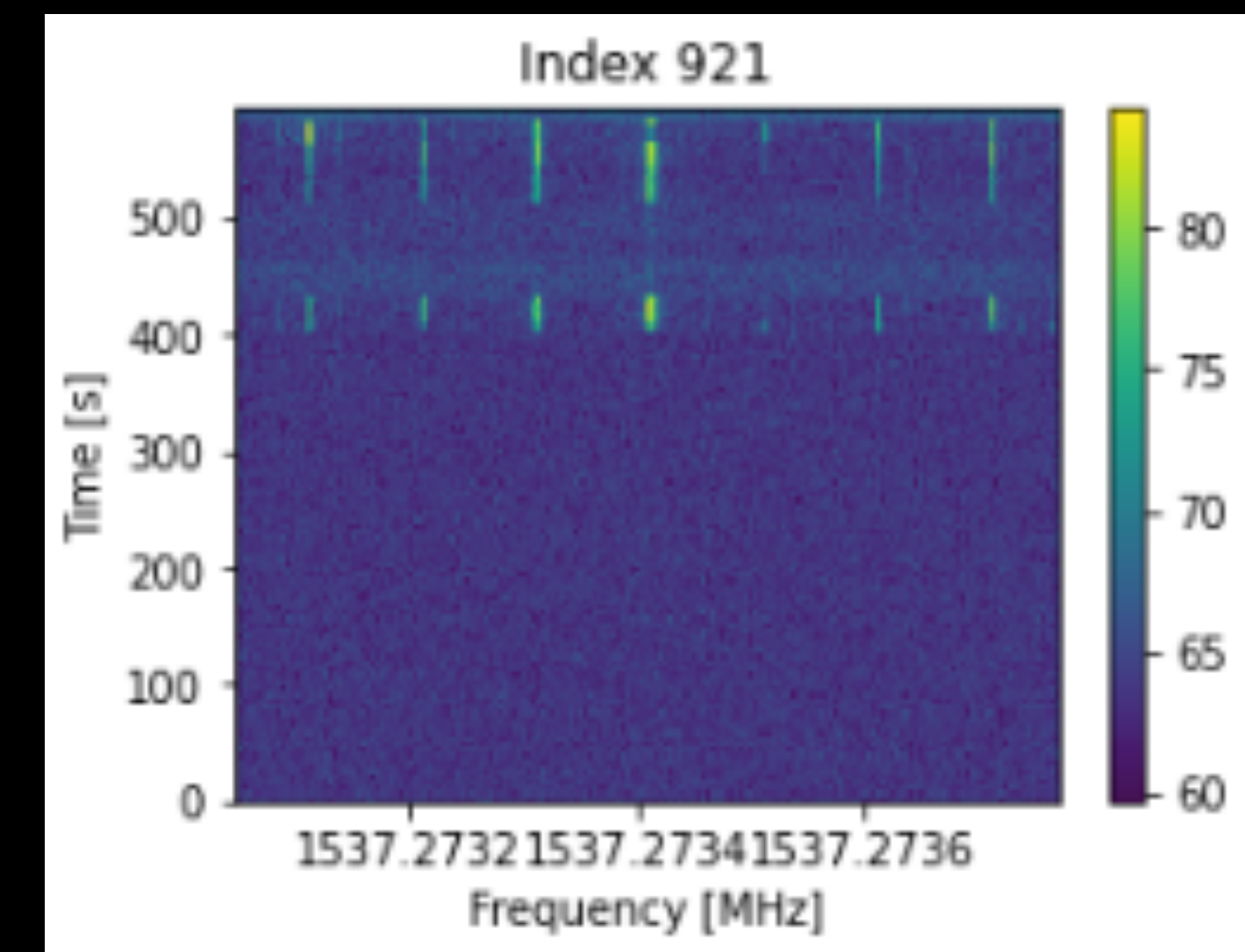
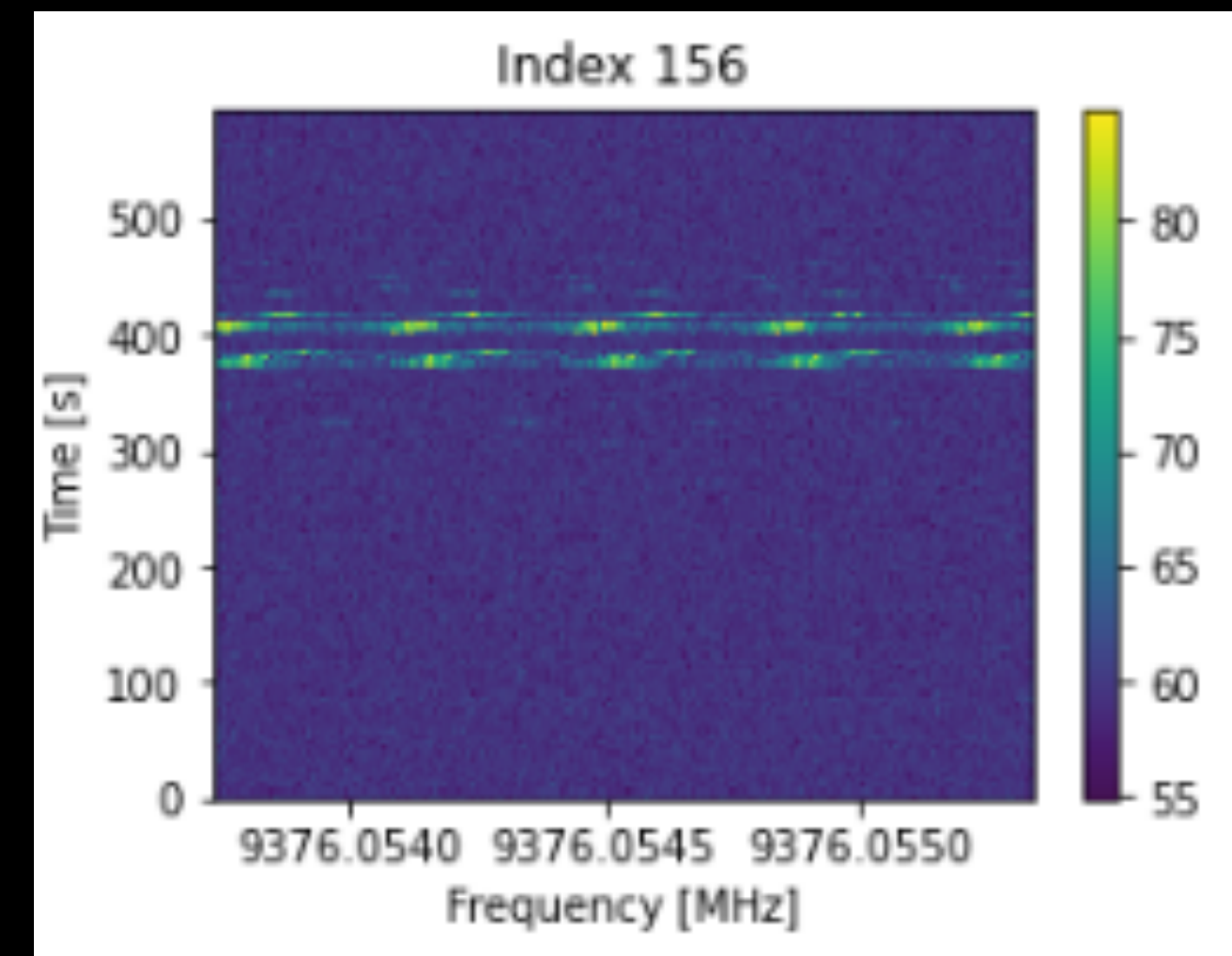
Timescale fit ~ 2 s



Timescale fit ~ 60 s

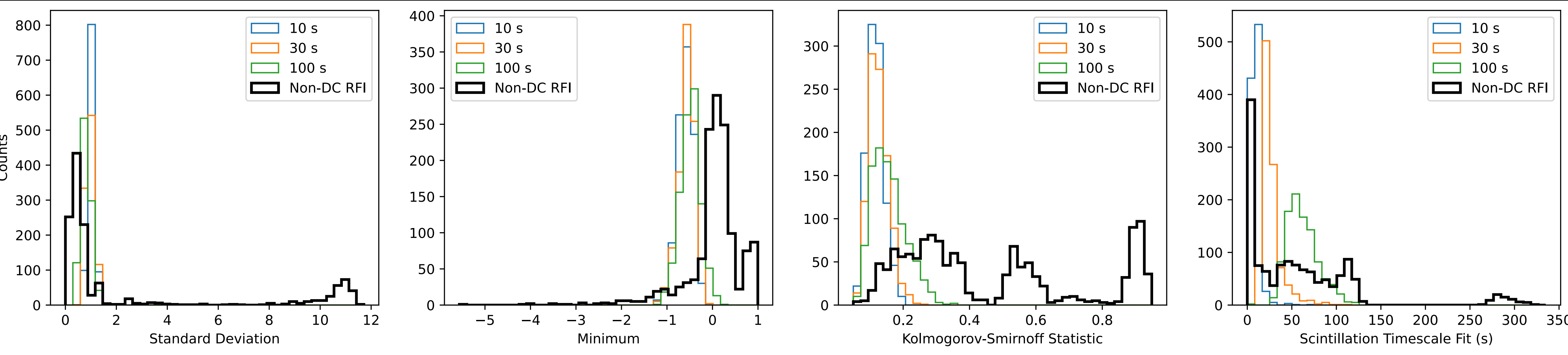
Limitations from RFI analysis

- L and S bands in particular are very noisy
- Non-narrowband signals detected just because they are above the SNR threshold
- Difficult to apply a one-size-fits-all bounding box method
- Perhaps ML can help!



L band

S/N = 25



Std. Dev.

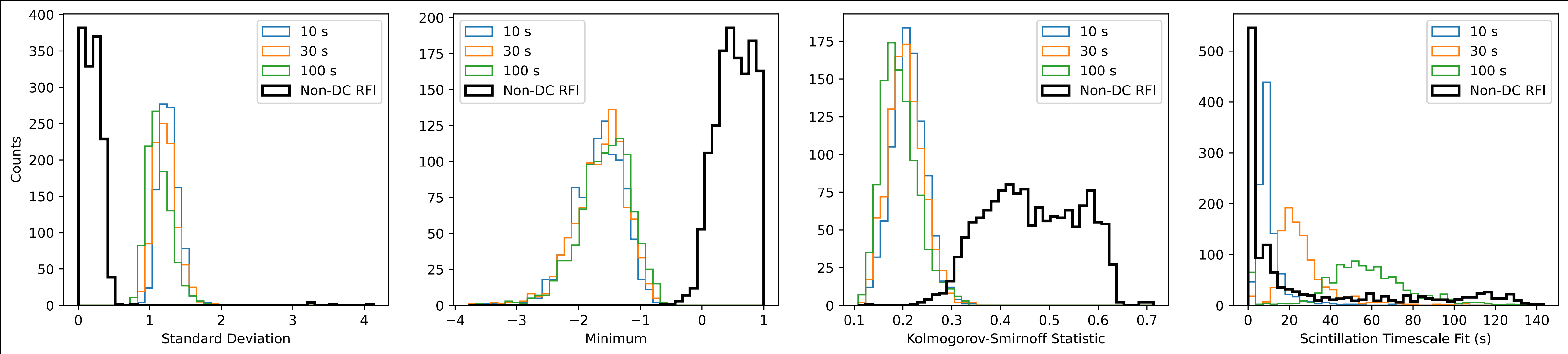
Minimum

KS Statistic

Timescale Fit

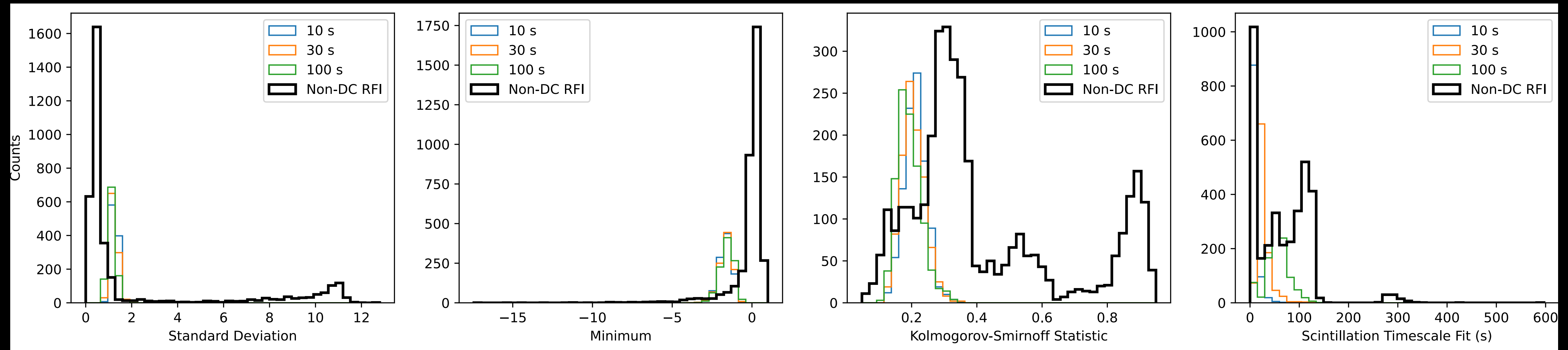
C band

S/N = 10

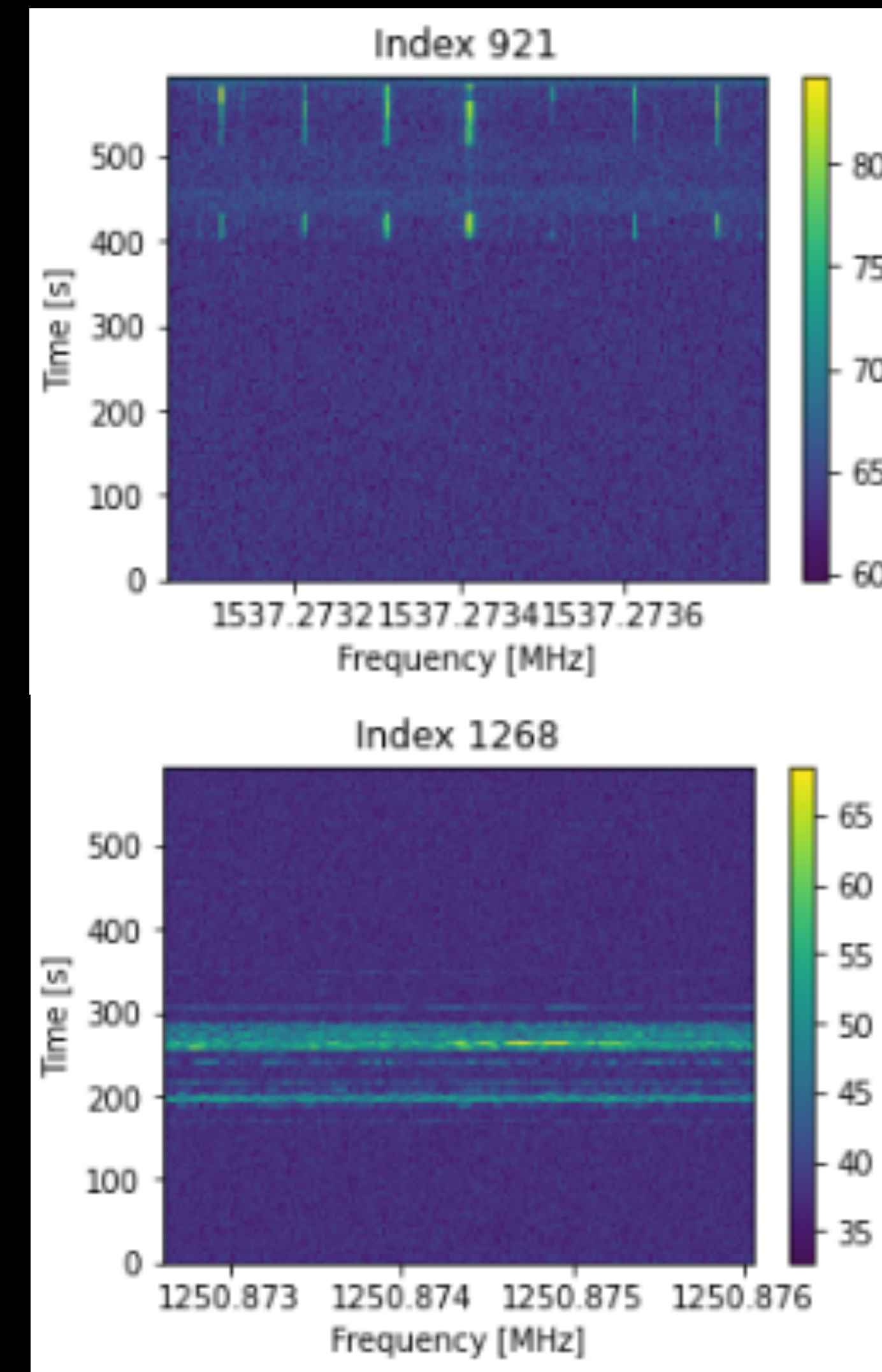
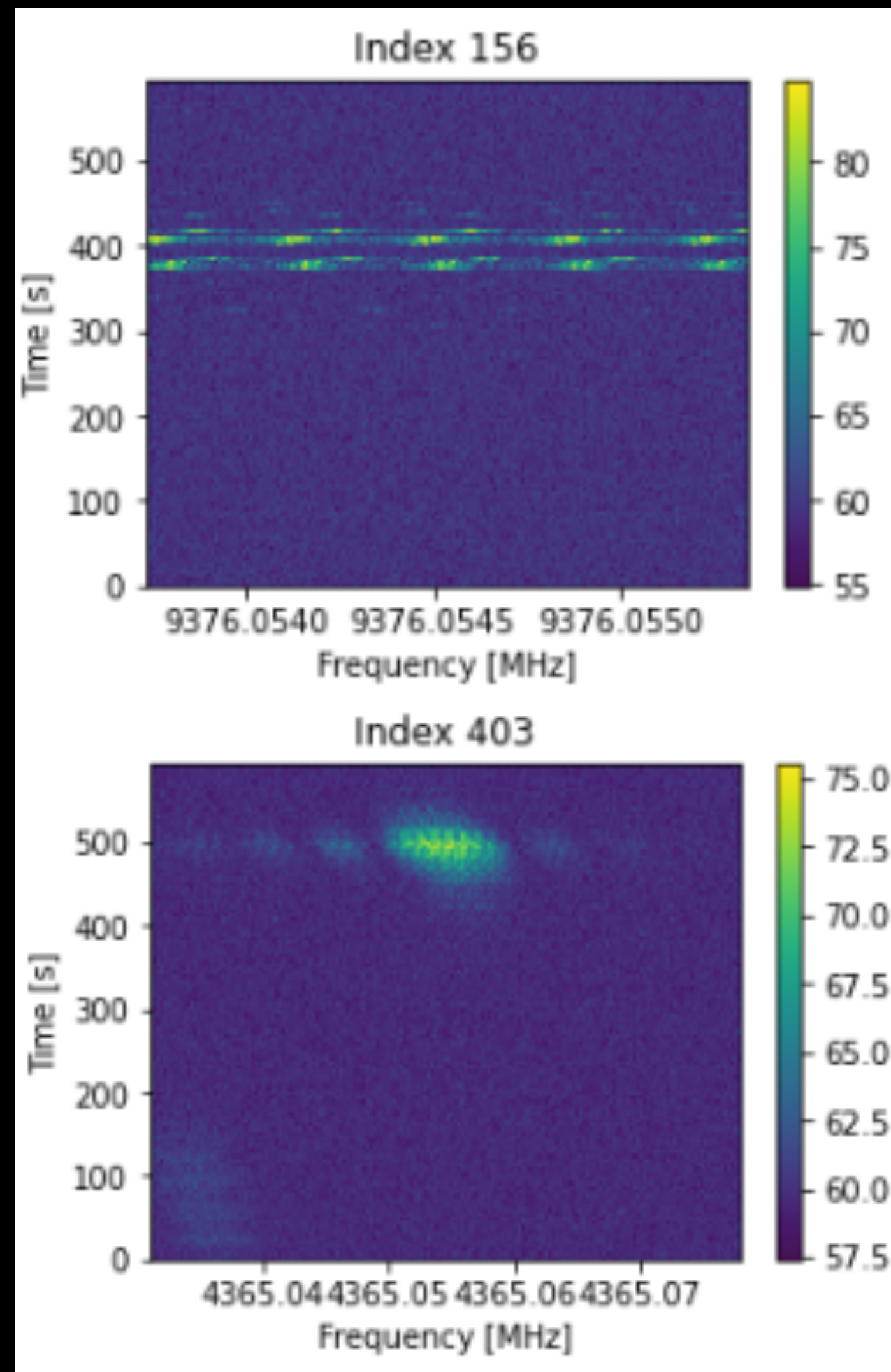


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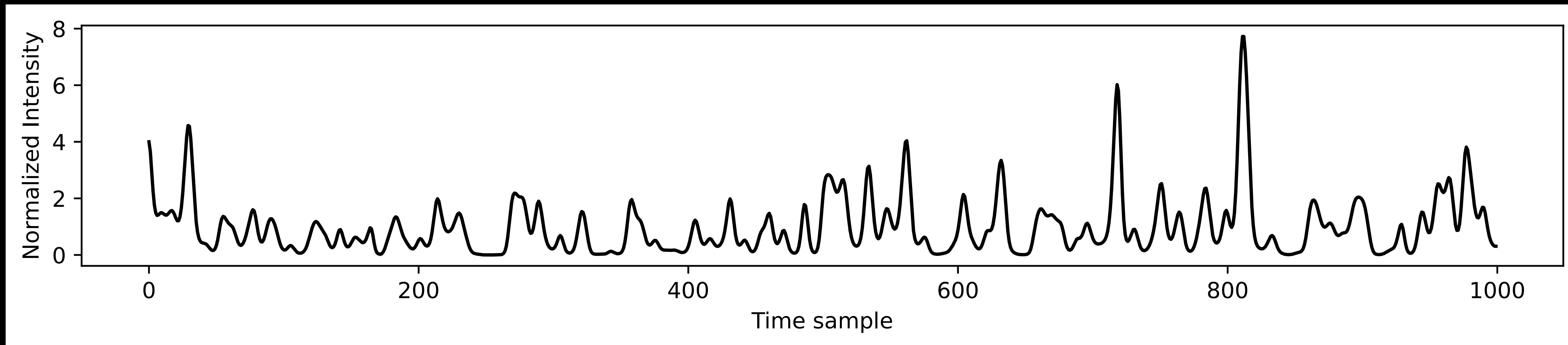


High standard deviation (RMS) signals are pulsed - or broadband



Quick way to produce synthetic data with asymptotic statistics

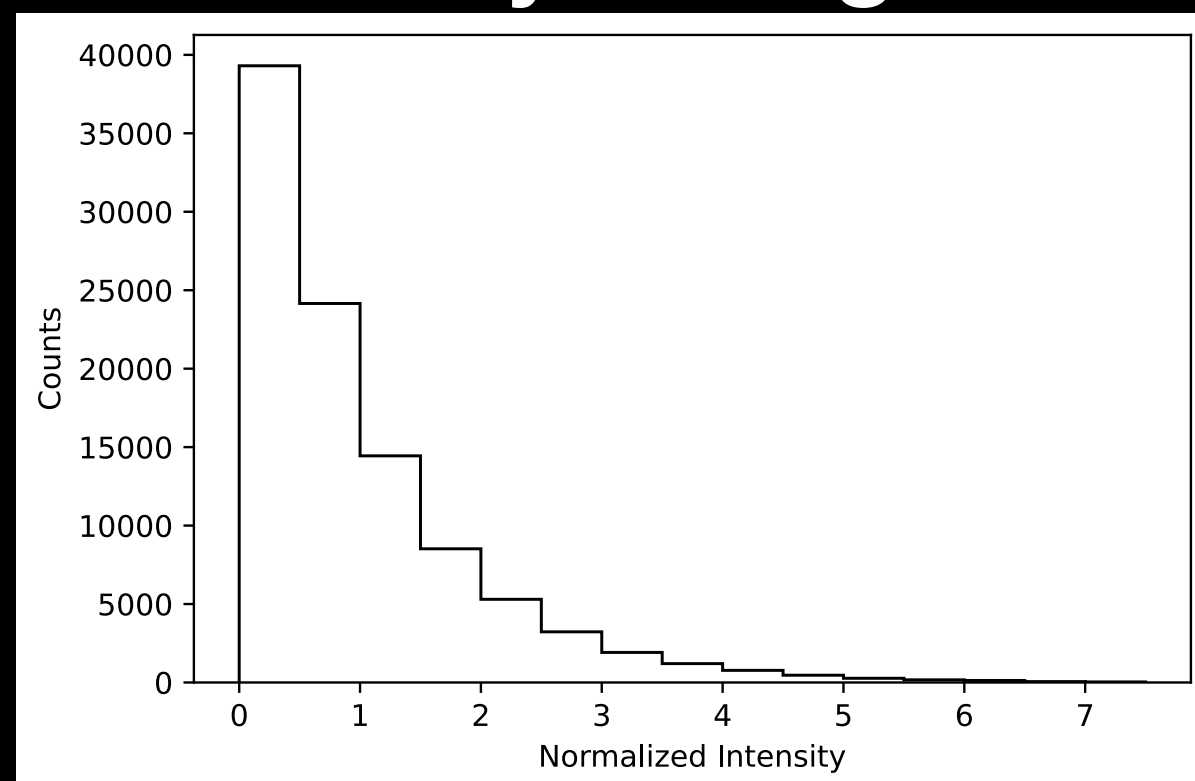
- (Cario & Nelson 1996) The ARTA random process matches:
 - Target intensity distribution
 - Target autocorrelation structure (with custom asymptotic precision)



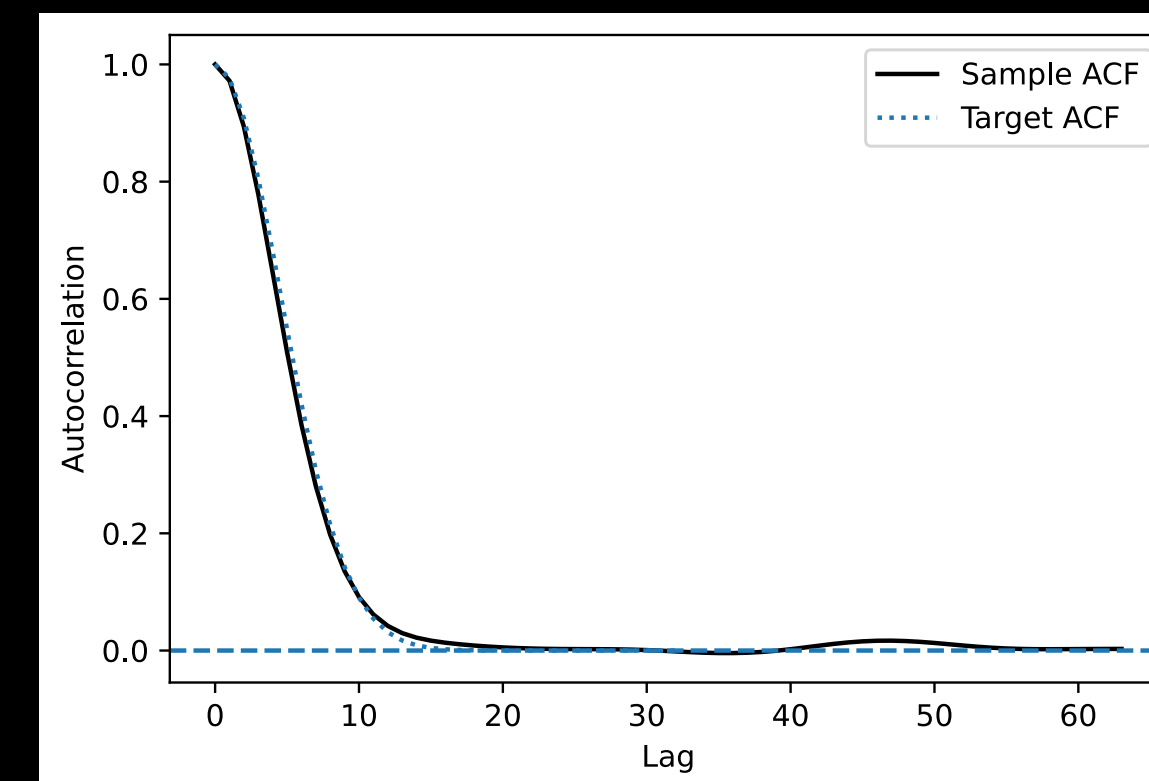
$n = 1e5$

$\Delta t_d = 30 \text{ s}$

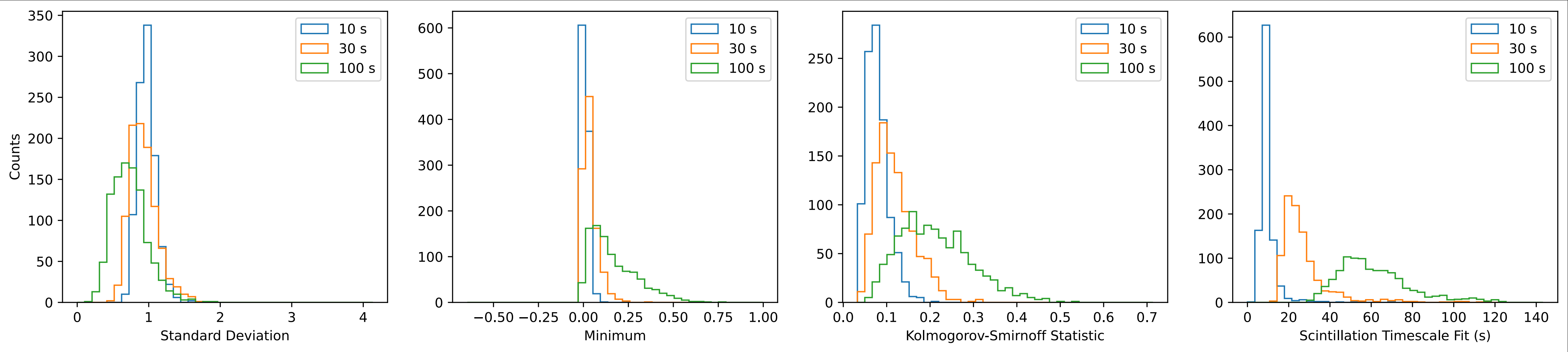
Intensity histogram



Autocorrelation



Statistics using low number of synthetic samples



Std. Dev.

Minimum

KS Statistic

Timescale Fit

10 min “observation”, 4.65 s resolution

Estimating scattering strength

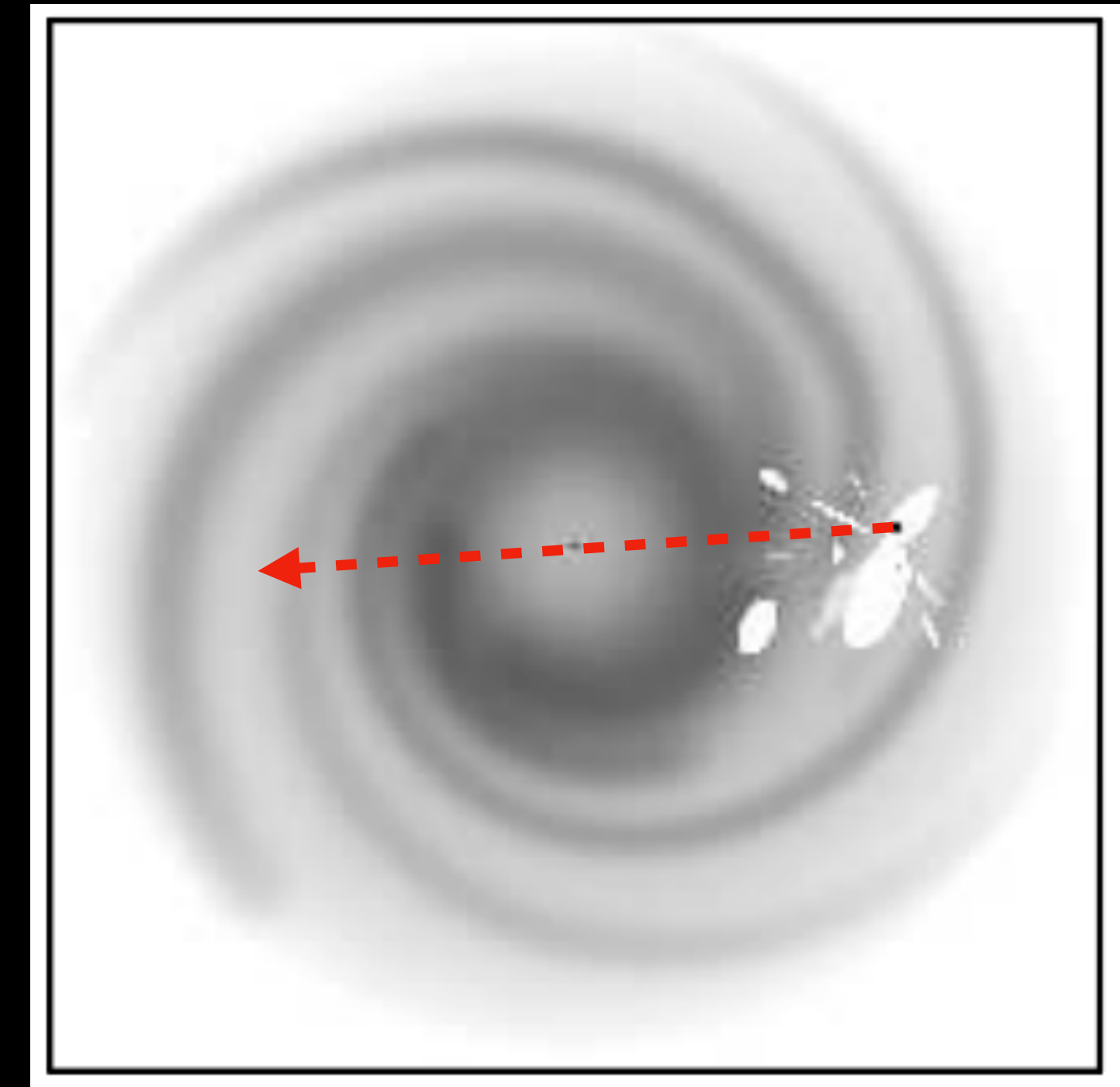
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- We use Monte Carlo sampling for unknown parameters

$$\Delta t_d \propto \nu^{6/5} \nu_T^{-1}$$

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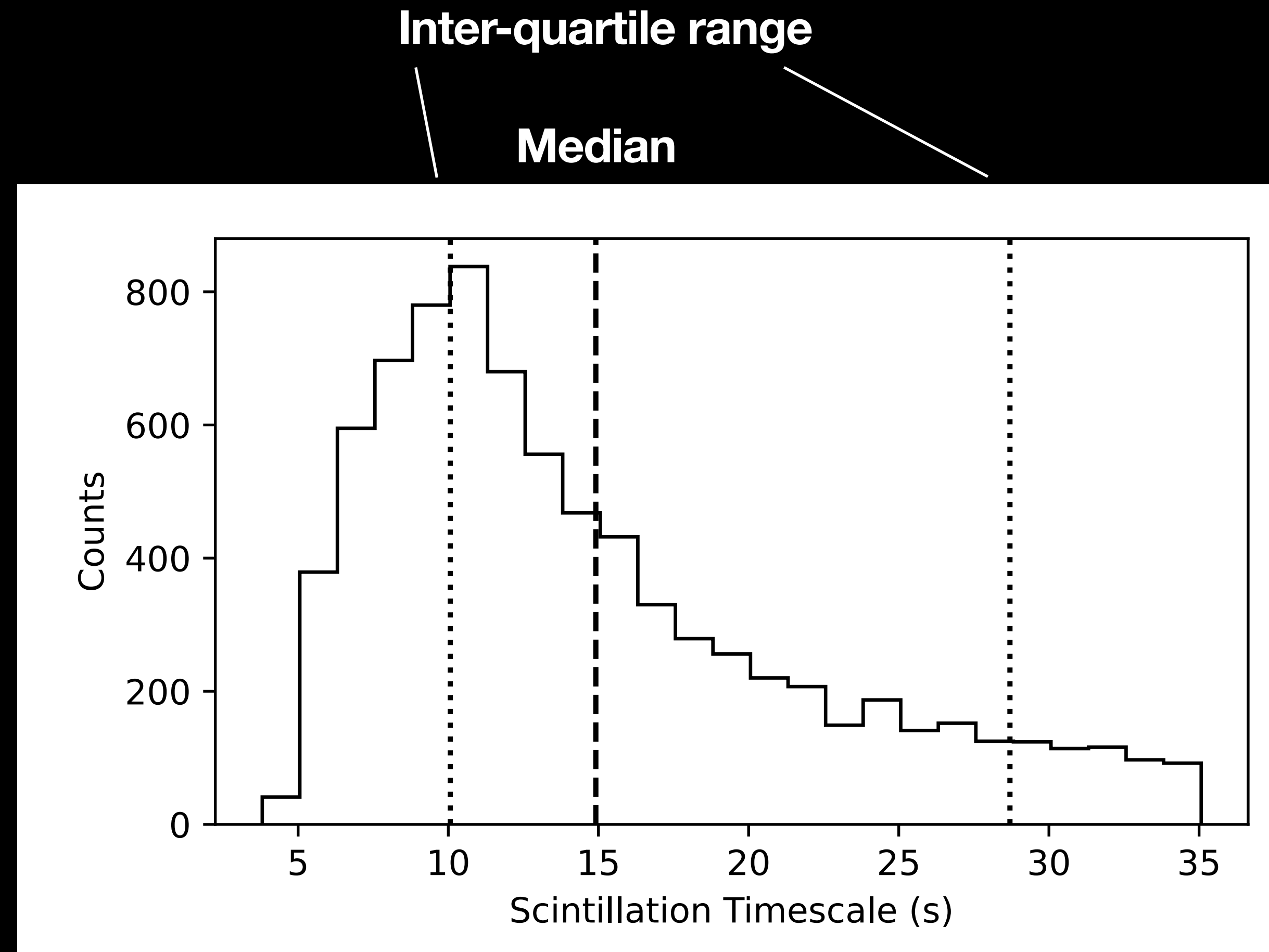
J. M. CORDES
Astronomy Department and NAIC, Cornell University, Ithaca, NY 14853
cordes@spacenet.tn.cornell.edu

T. JOSEPH W. LAZIO
Naval Research Lab, Code 7213, Washington, D.C. 20375-5351
Joseph.Lazio@nrl.navy.mil



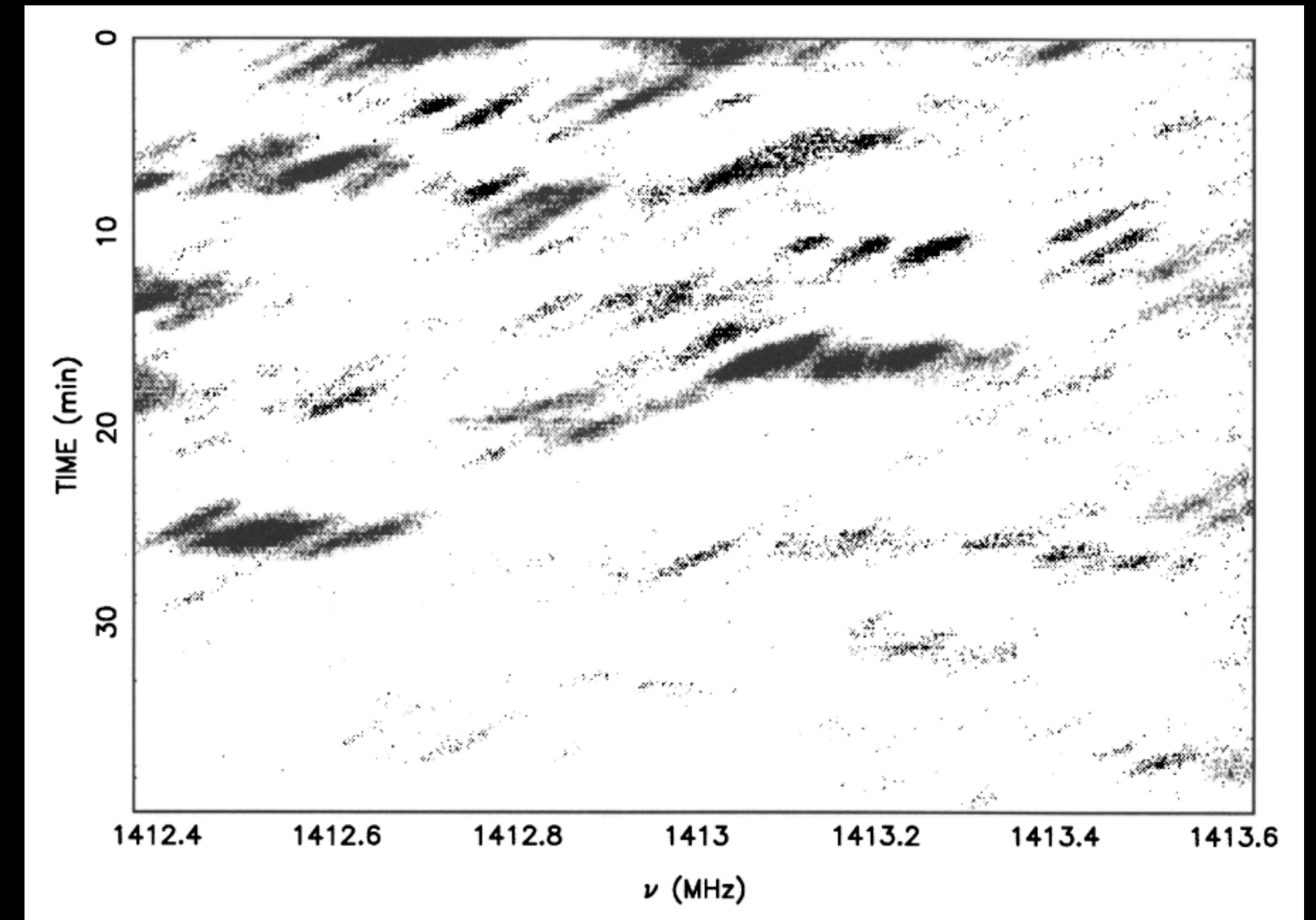
C-band

$(l, b) = (1, 0)$

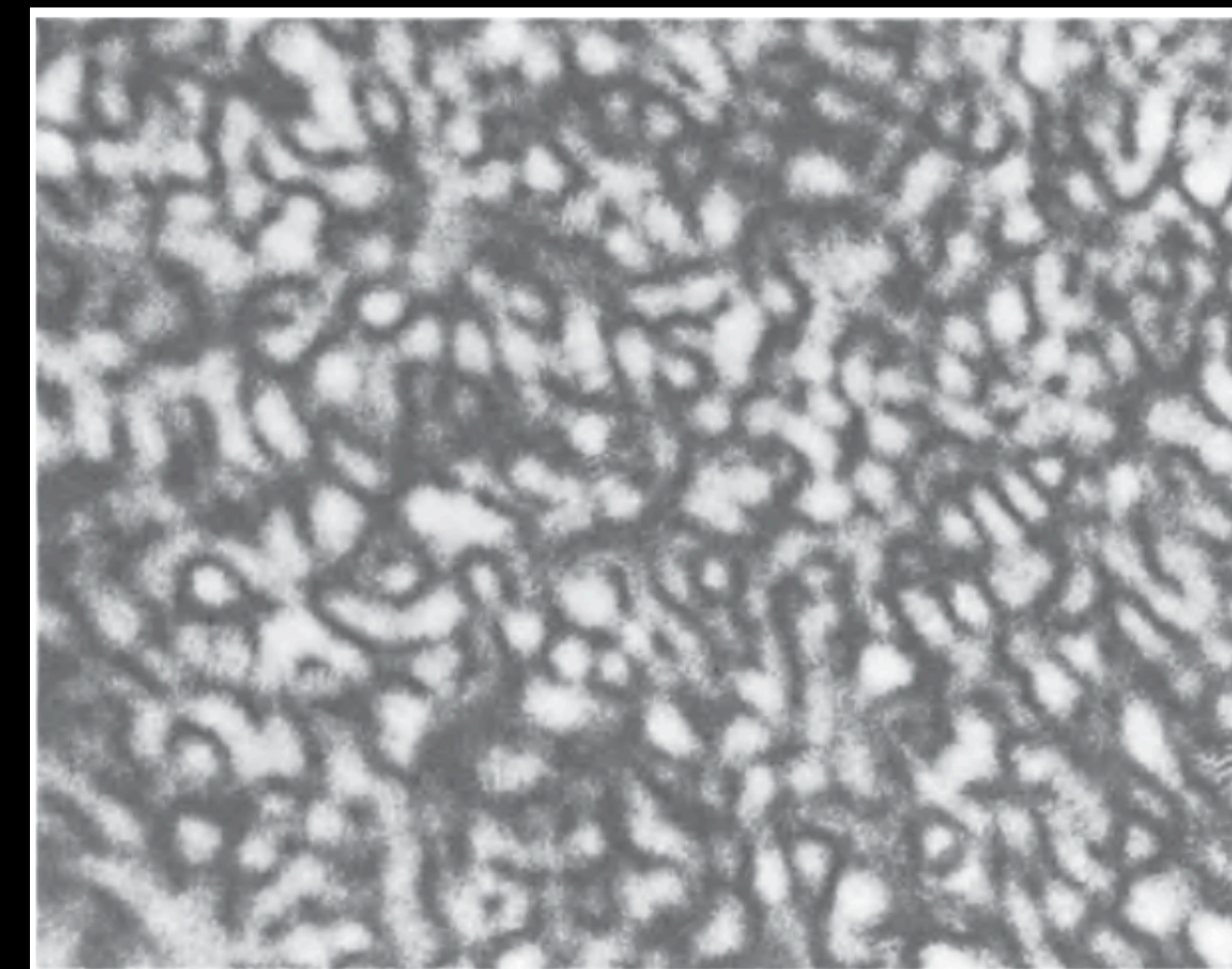


ISM Scattering & Scintillation

- Interaction between radio waves and free electrons in plasma
- Pulsar observations paved the way
- Parallels with laser speckle



Cordes & Lazio 1991



Goodman 1984
BREAKTHROUGH
LISTEN

Scattering and SETI research

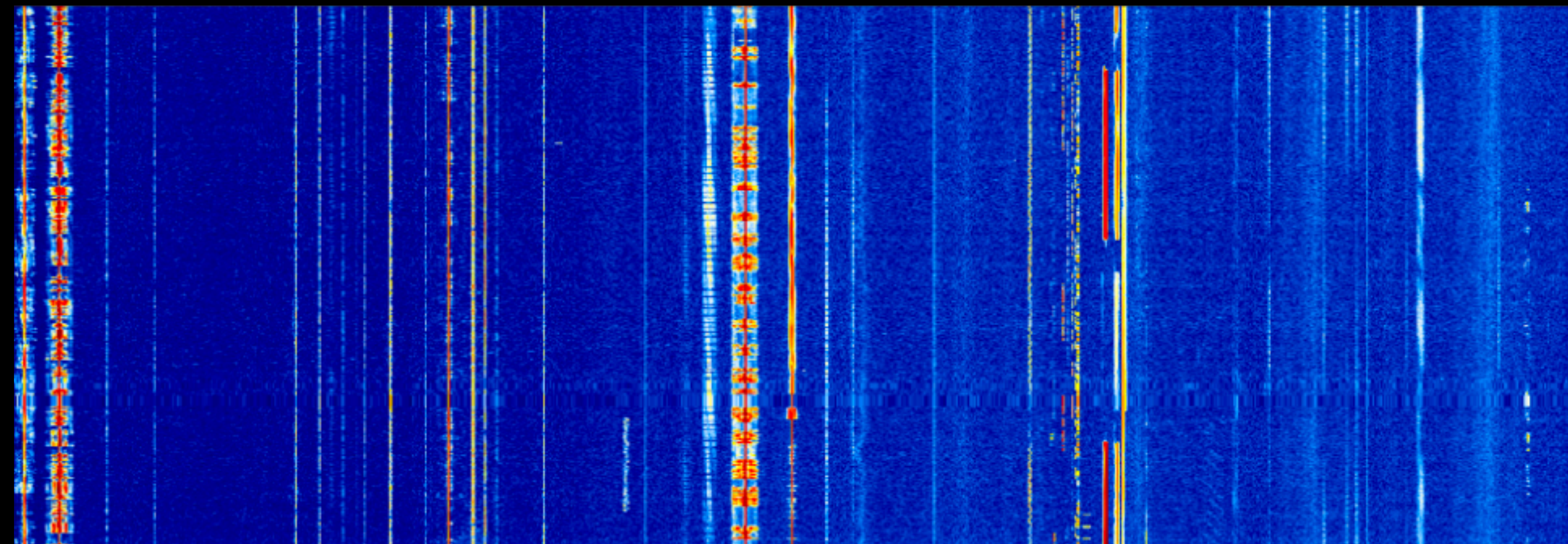
INTERSTELLAR SCATTERING EFFECTS ON THE DETECTION OF NARROW-BAND SIGNALS

JAMES M. CORDES AND T. JOSEPH LAZIO

National Astronomy and Ionosphere Center and Department of Astronomy, Cornell University, Ithaca, NY 14853

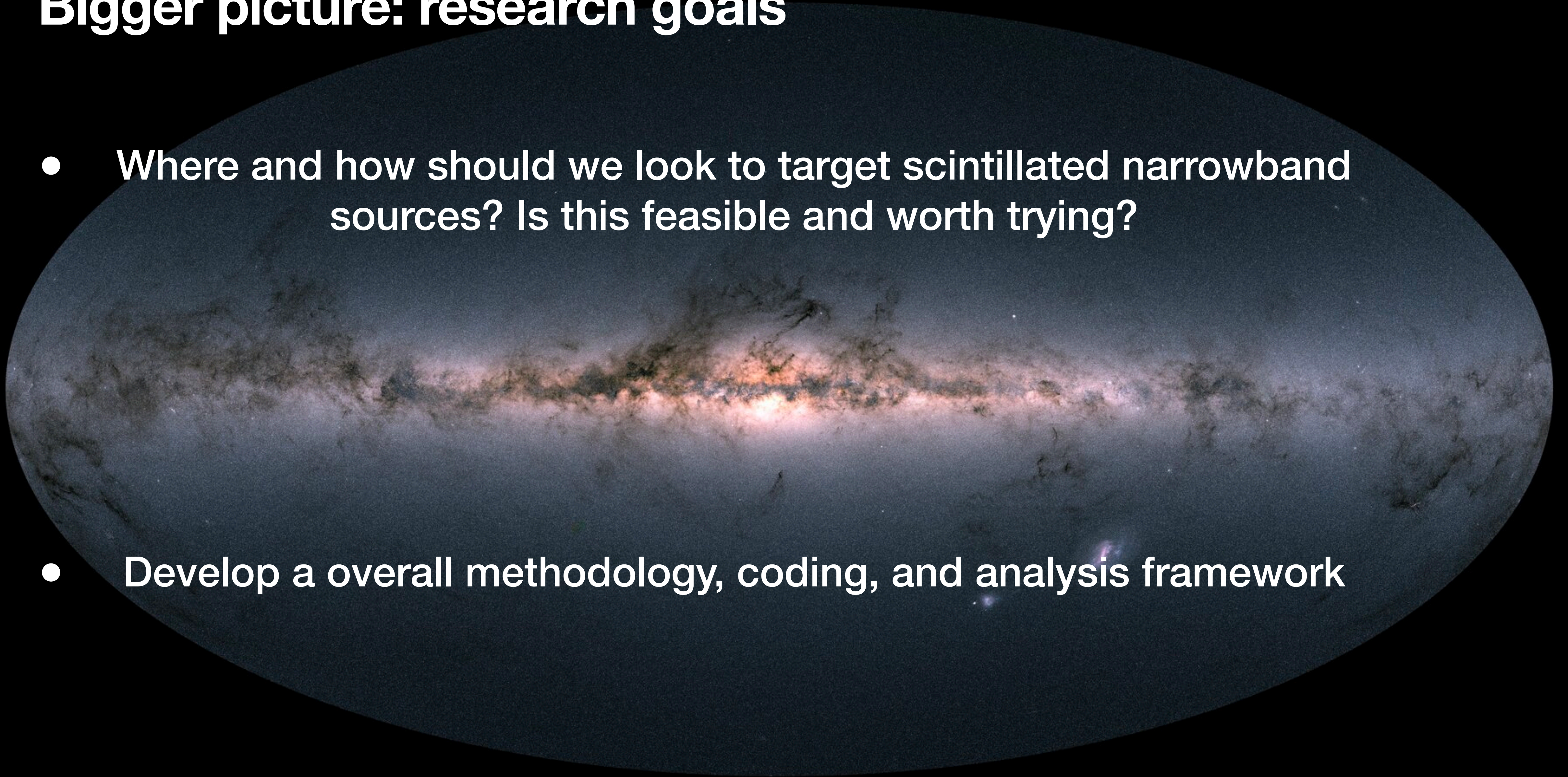
Received 1990 October 4; accepted 1991 January 15

- Many studies acknowledge scattering but attempt to avoid it
- Generally, SETI techniques aren't sensitive to detailed morphology
 - Noise, modulation, S/N
- Stochastic effects are hard to describe



Bigger picture: research goals

- Where and how should we look to target scintillated narrowband sources? Is this feasible and worth trying?
- Develop a overall methodology, coding, and analysis framework



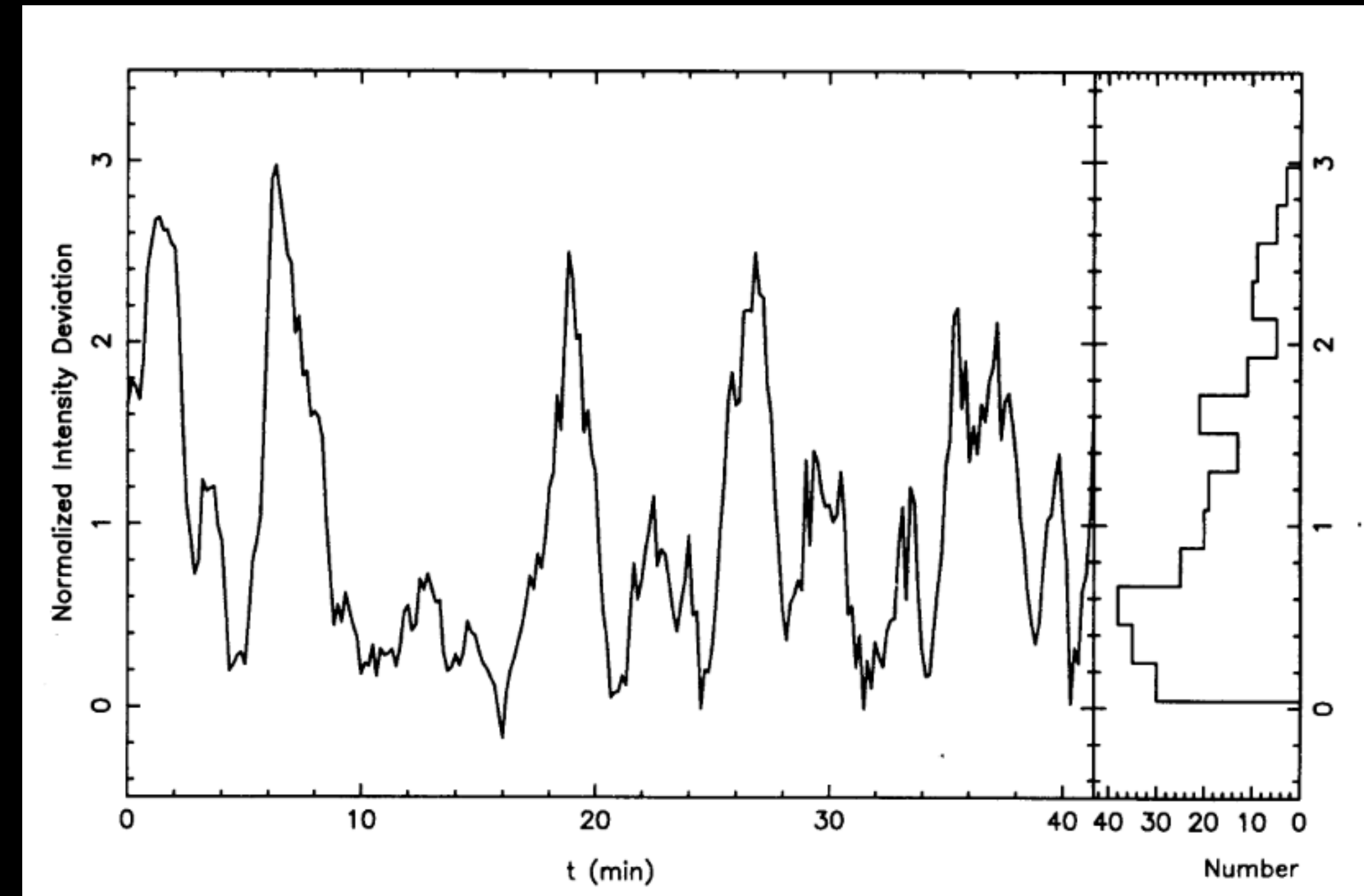
Can we detect scintillated narrowband technosignatures?

1. What scintillation timescales should we expect?
2. How can we probe asymptotic statistics?
3. Can we differentiate scintillated signals from existing RFI?

What would strongly scintillated signals look like?

- Asymptotic behavior:
 - Exponential intensity distribution
 - Approximately Gaussian autocorrelation, with characteristic timescale

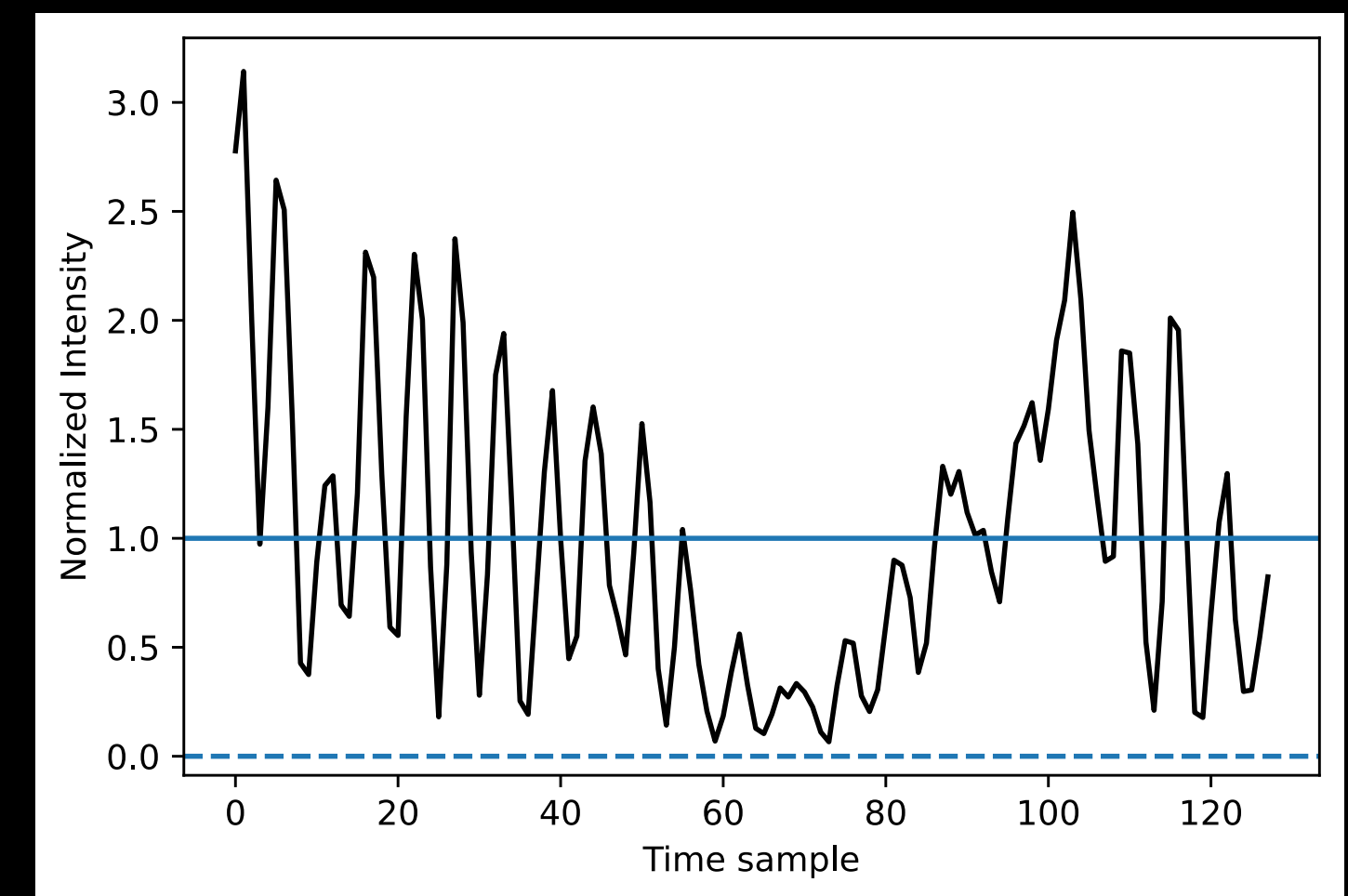
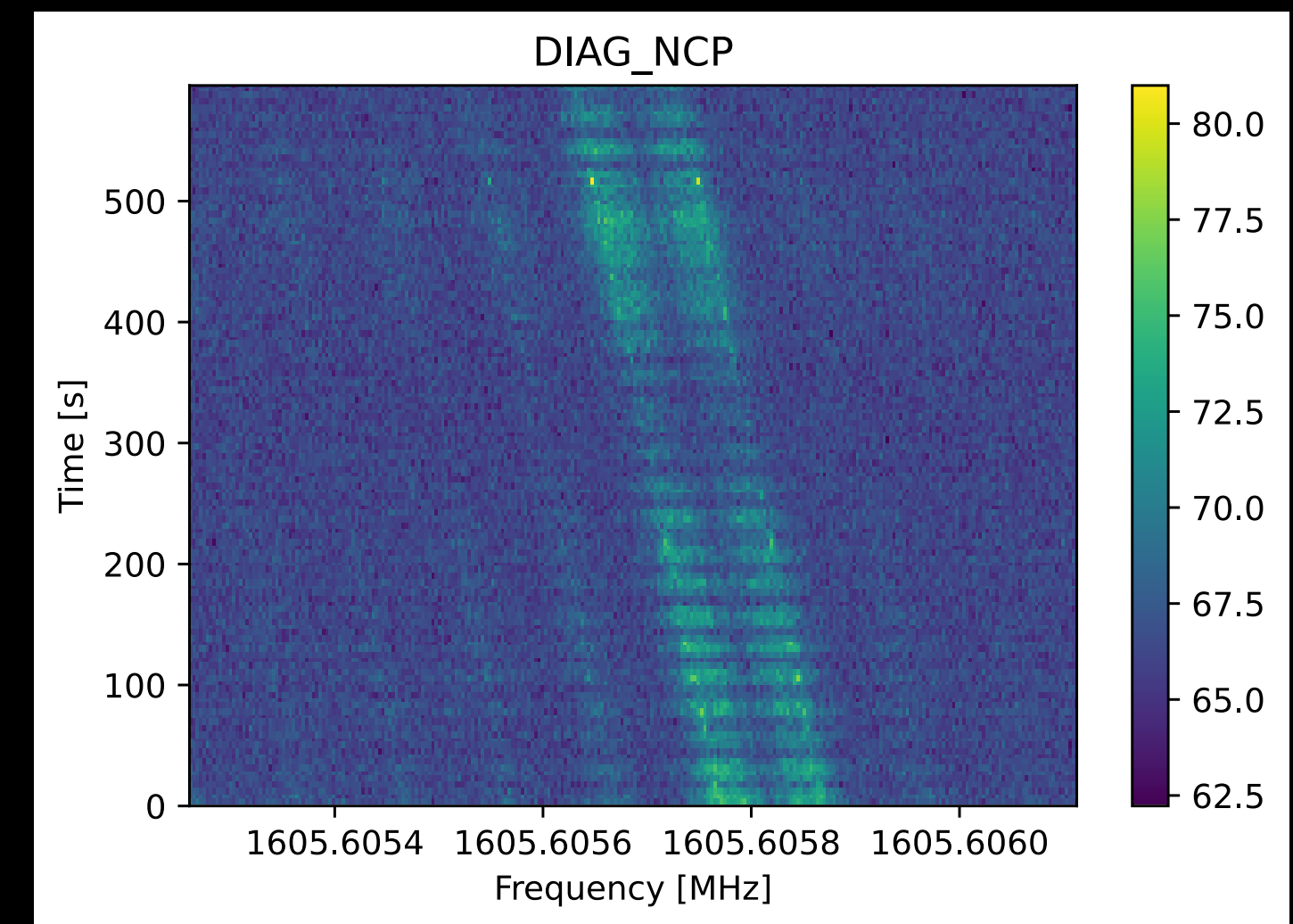
Assuming 100% duty-cycle narrowband emission



Cordes & Lazio 1991;
Cordes, Lazio, Sagan 1997

Given a signal... is it scintillated?

- Create bounding box around narrowband signal
- Estimate noise-subtracted intensity time series, normalized to mean 1
- Compute “diagnostic statistics” that pertain to asymptotic behavior
 - E.g. standard deviation, Kolmogorov-Smirnoff statistic, fit to autocorrelation function



What would strongly scintillated signals look like?

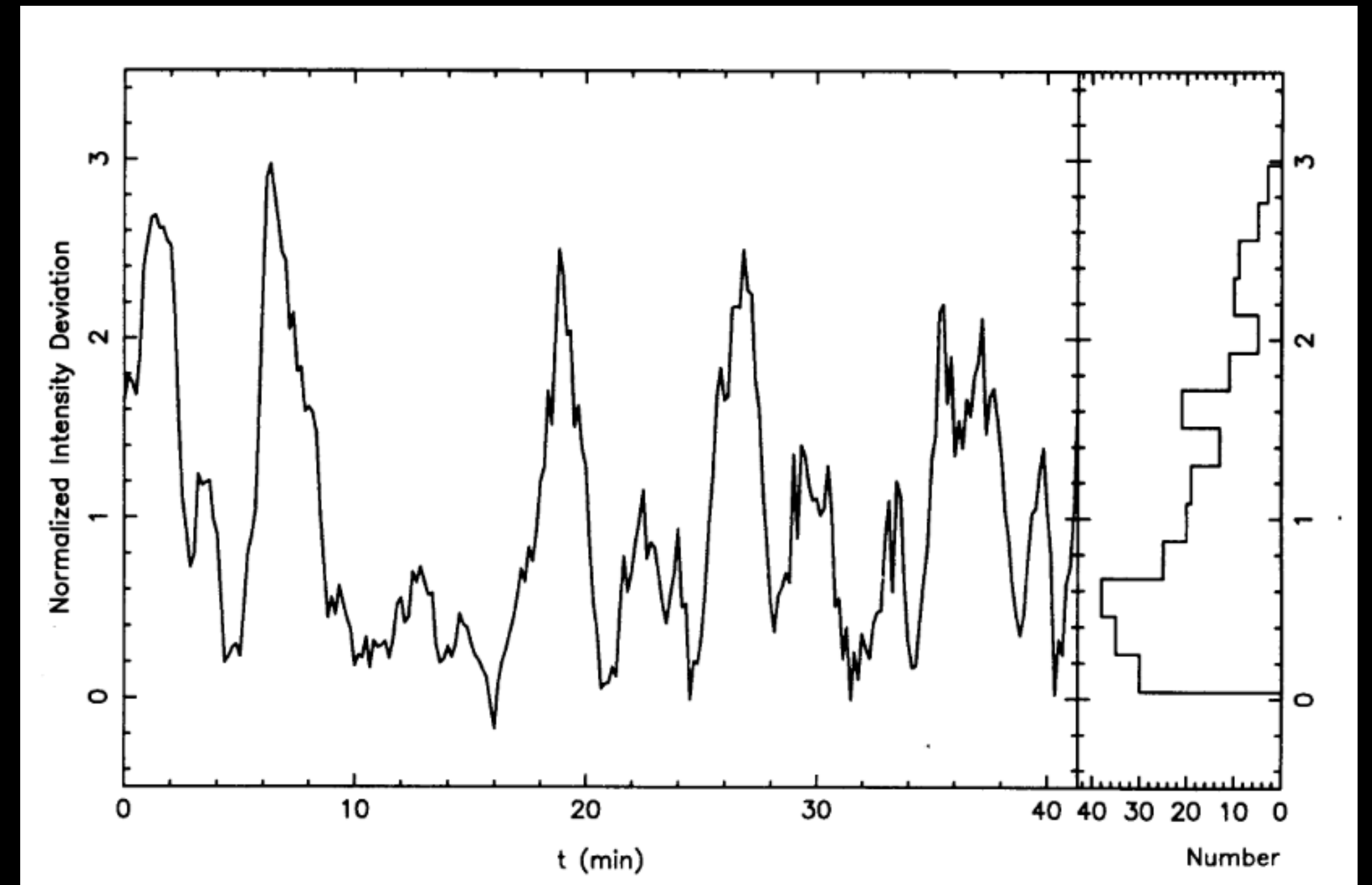
- Expected asymptotic behavior:
- Exponential intensity distribution

$$p(I) \propto e^{-I/\langle I \rangle}$$

- Near Gaussian autocorrelation, with characteristic timescale

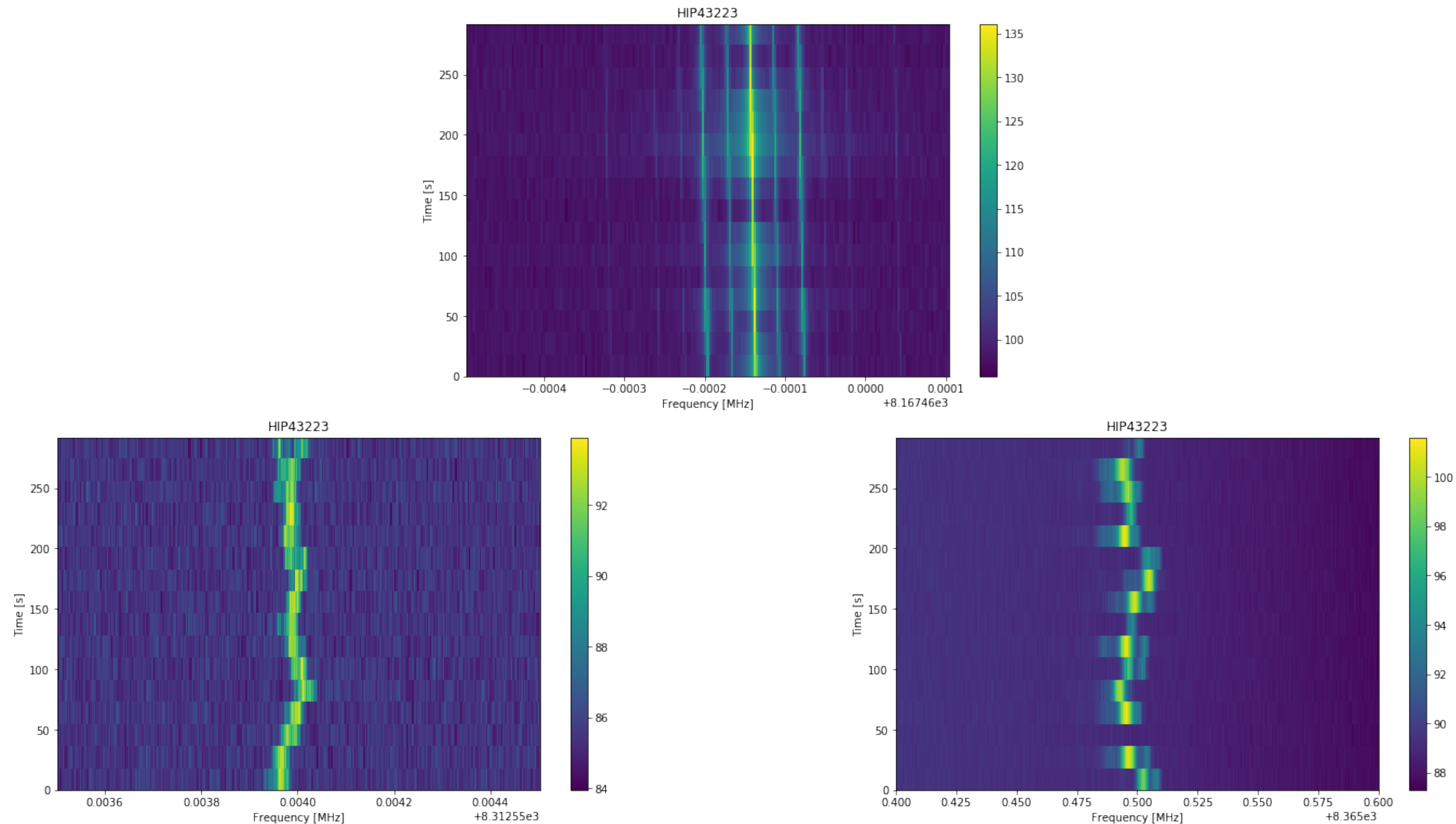
$$\rho(\tau) \sim e^{-(\tau/\Delta t_d)^2}$$

Assuming 100% duty-cycle narrowband emission



Cordes & Lazio 1991;
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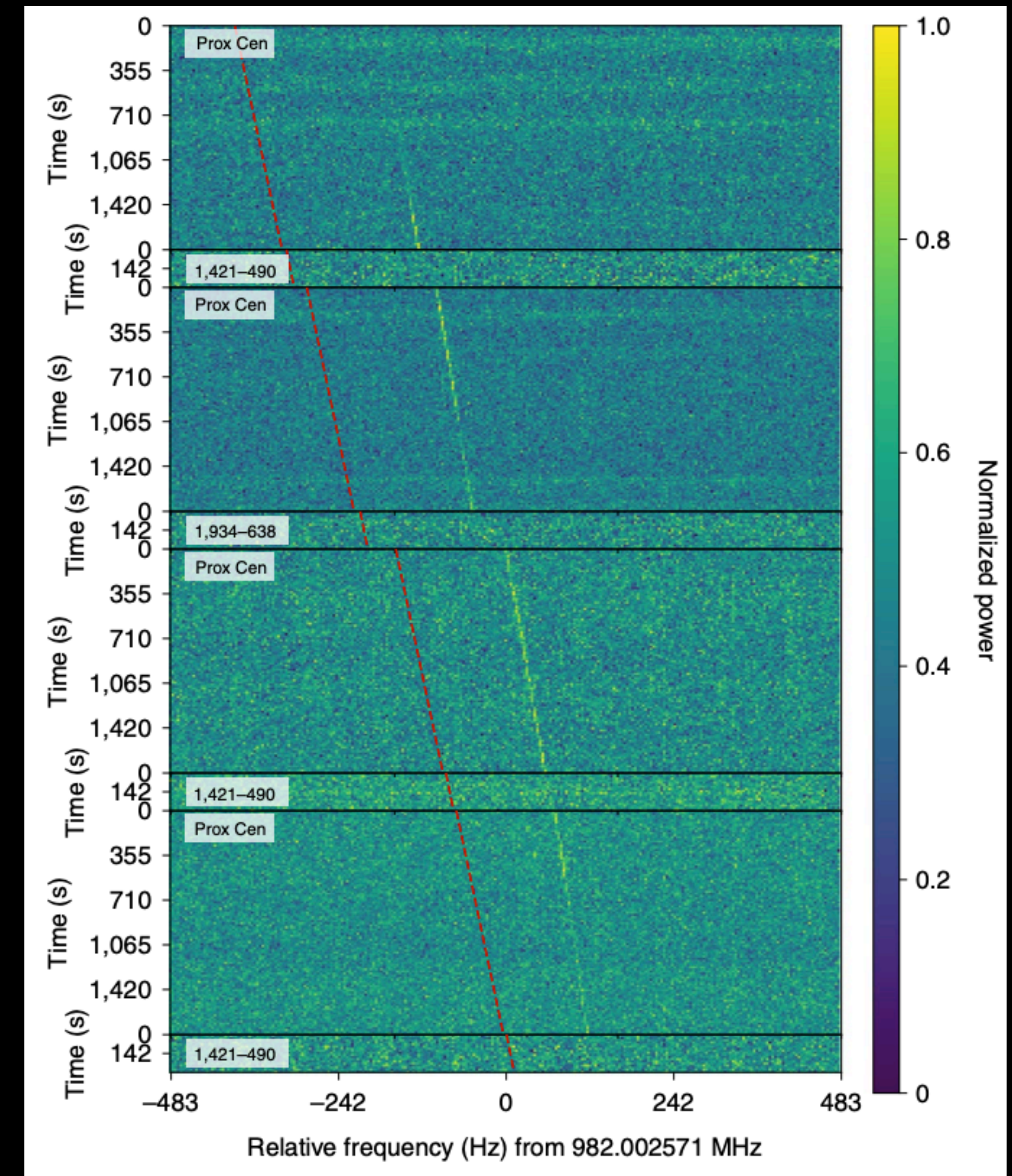
But what does the RFI environment look like?



Extra Slides x2

Plasma effects as a search filter for SETI

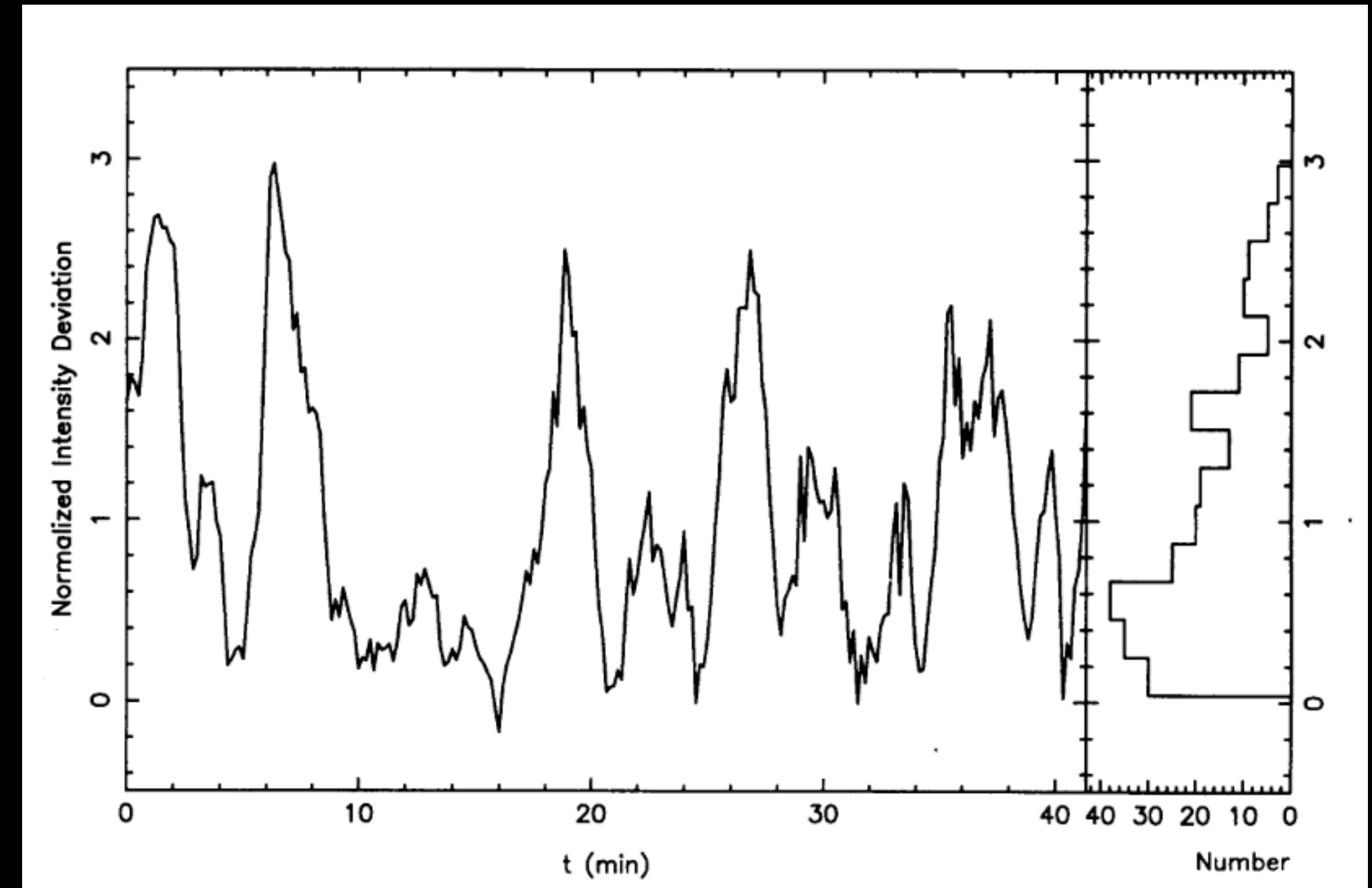
- Modern radio SETI involves detecting a vast number of signals and filtering likely candidates
- For a few reasons, most filters do not involve the effects on the signal itself
- We propose that in some cases, we can detect scintillation from the ISM in narrowband signals, which would heavily imply extrasolar origin



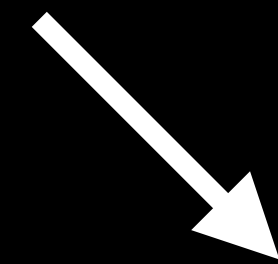
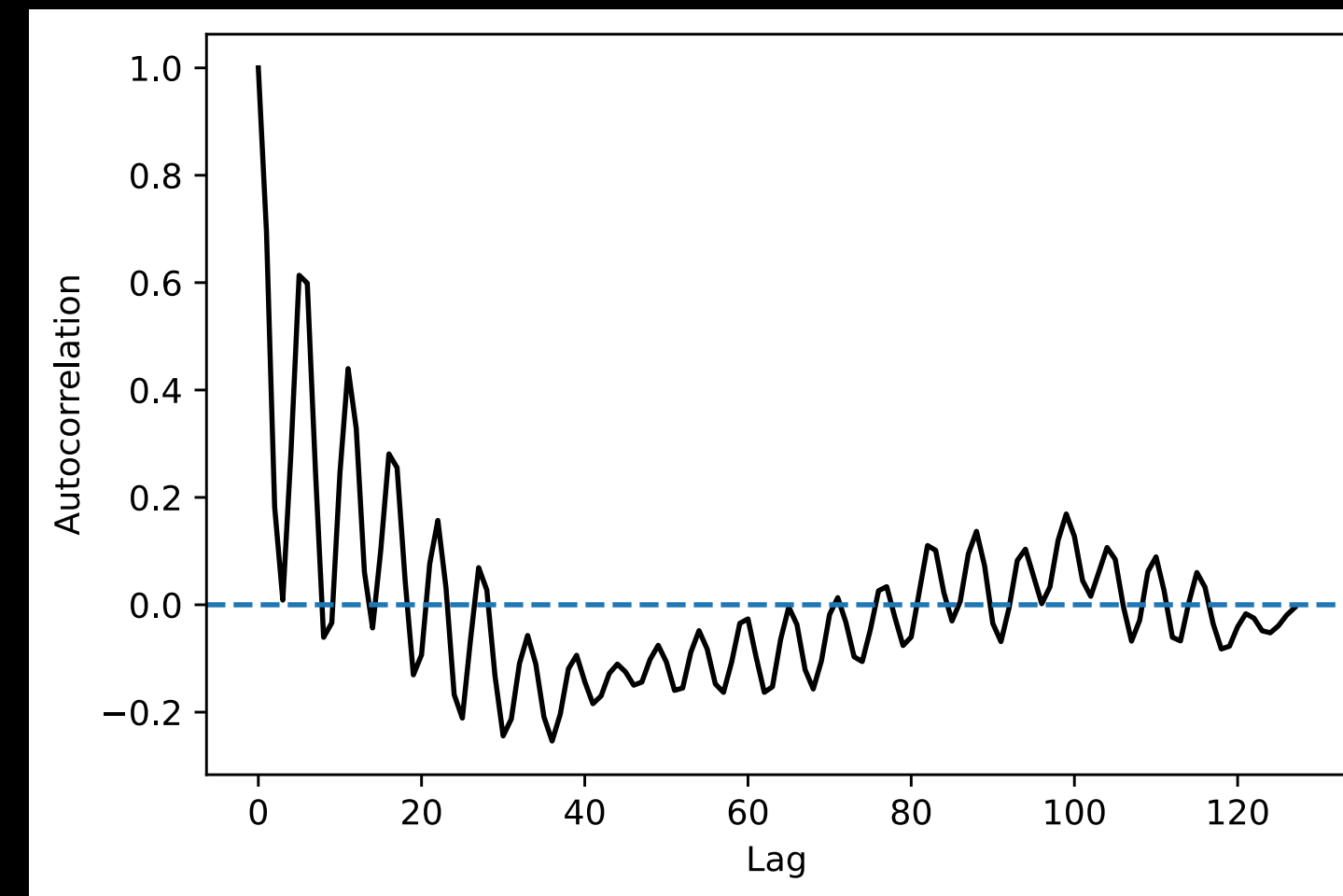
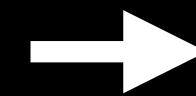
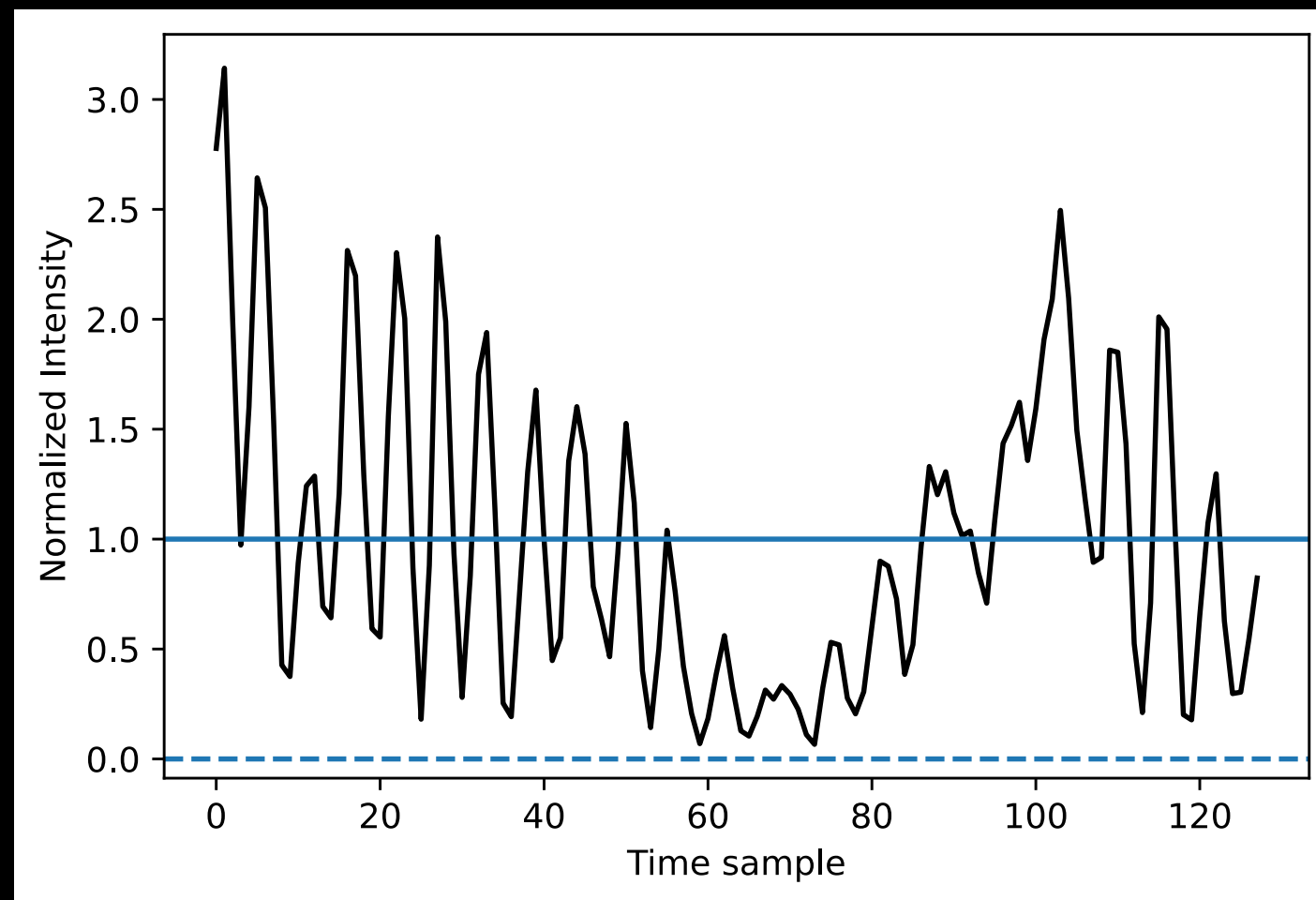
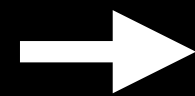
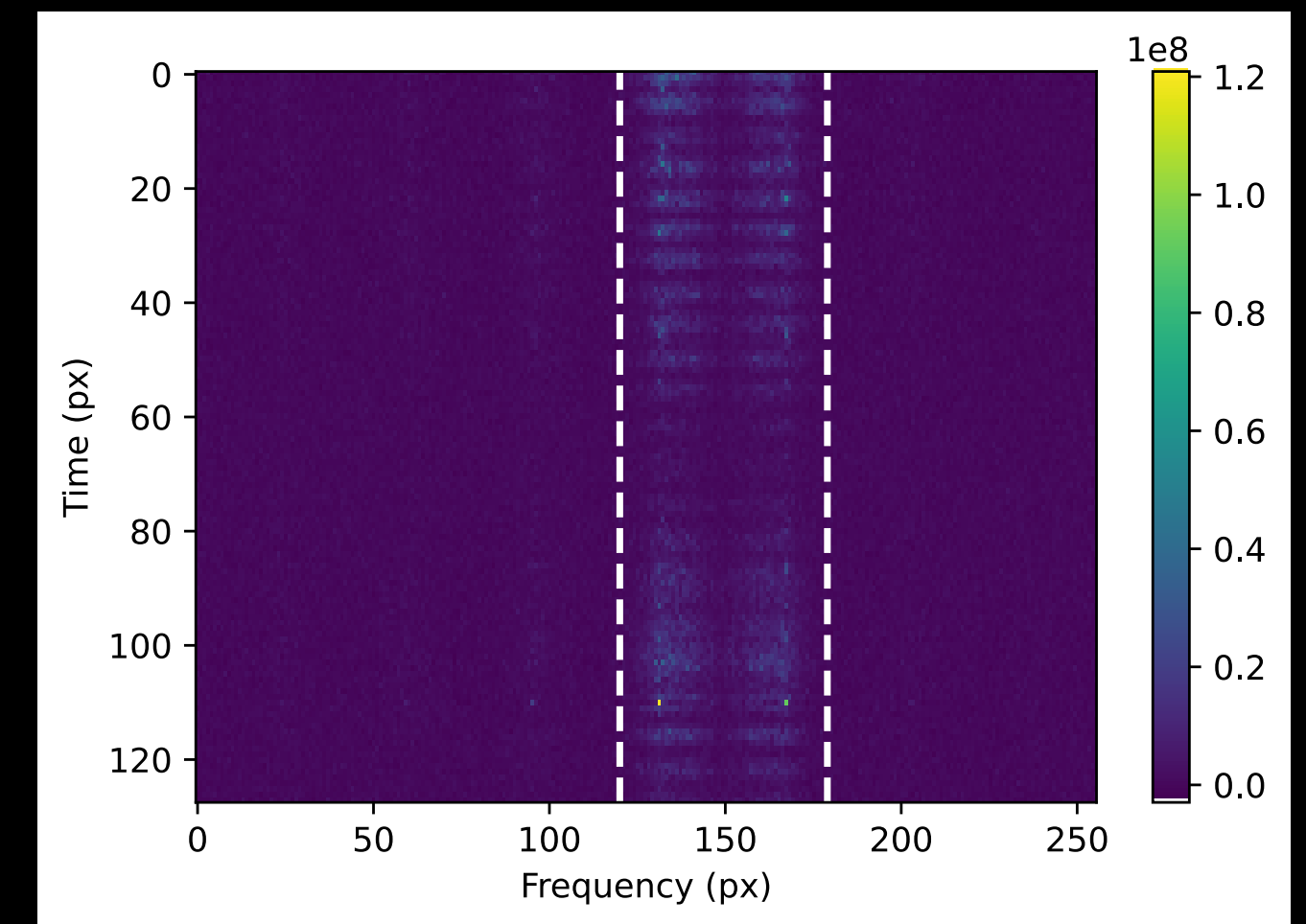
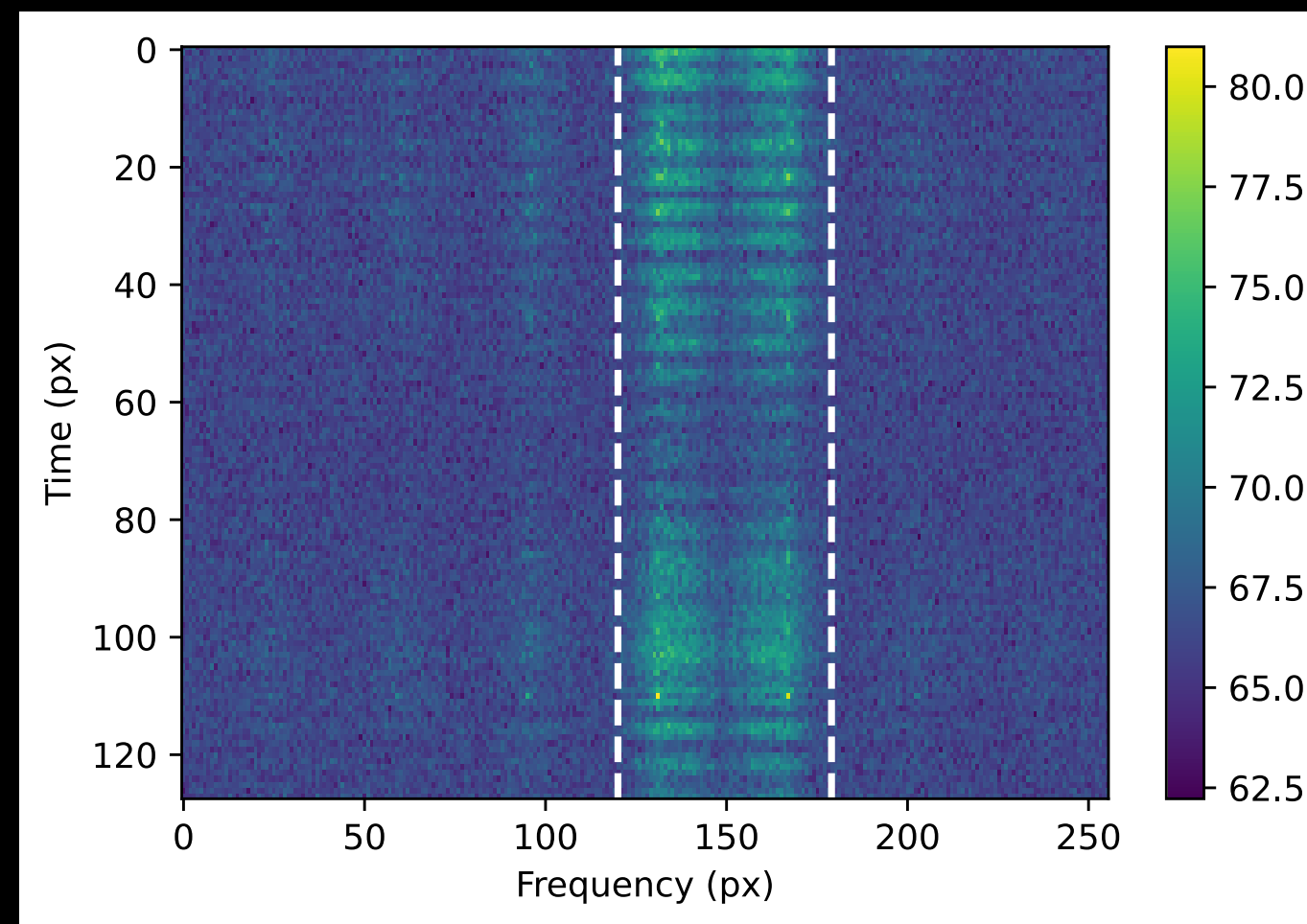
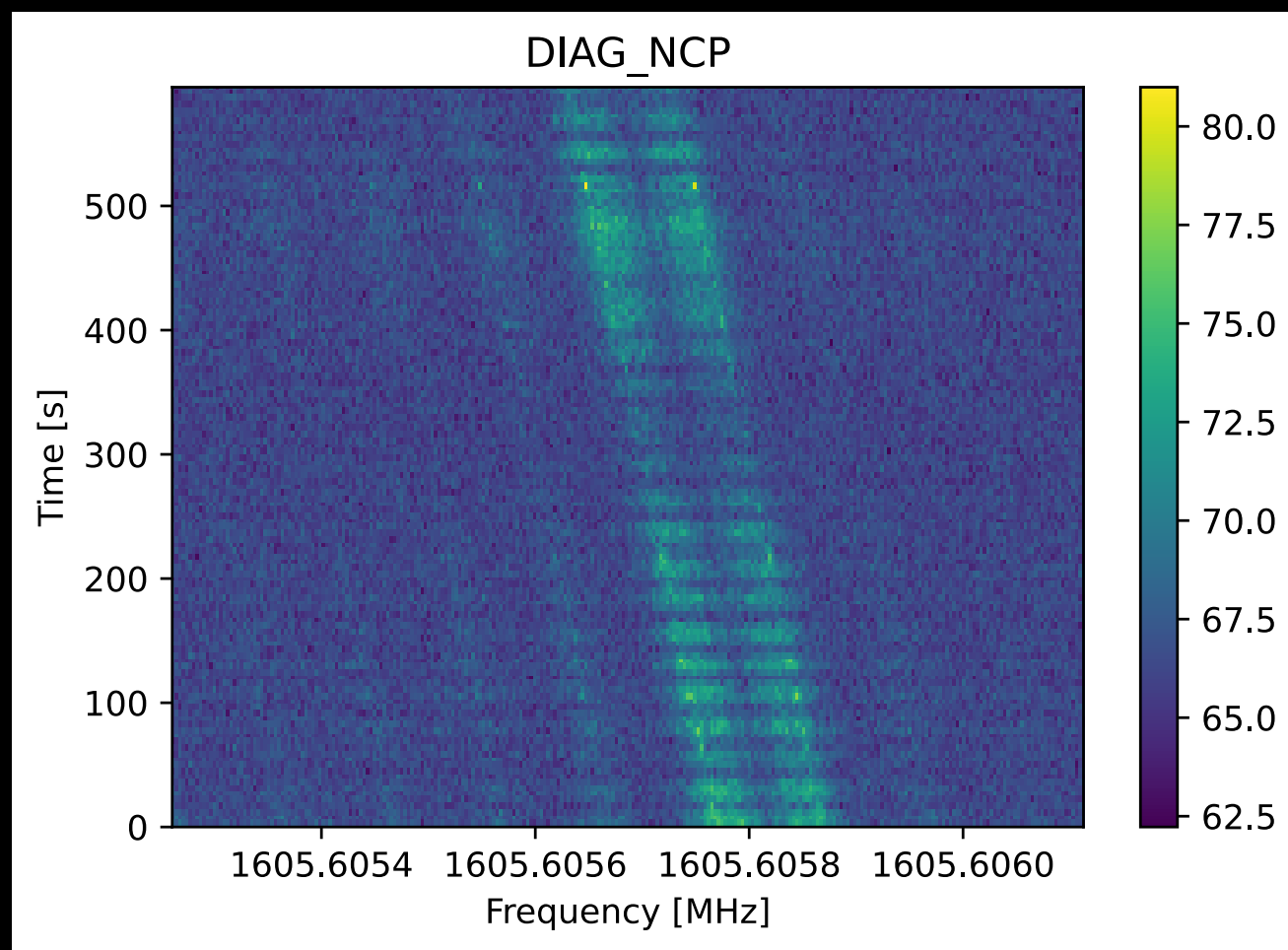
Smith et al. 2021

Methods

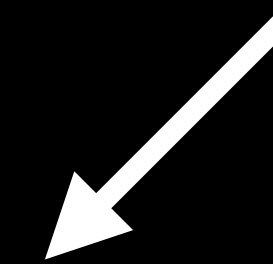
- Target 100% duty-cycle, narrowband transmitters
- Since scintillation is stochastic by definition, identify measurable statistics
- Estimate intensity time series from detected signals for analysis
- Use procedure on RFI in unlikely directions to probe the interference environment



Cordes & Lazio 1991;
Cordes, Lazio, Sagan 1997



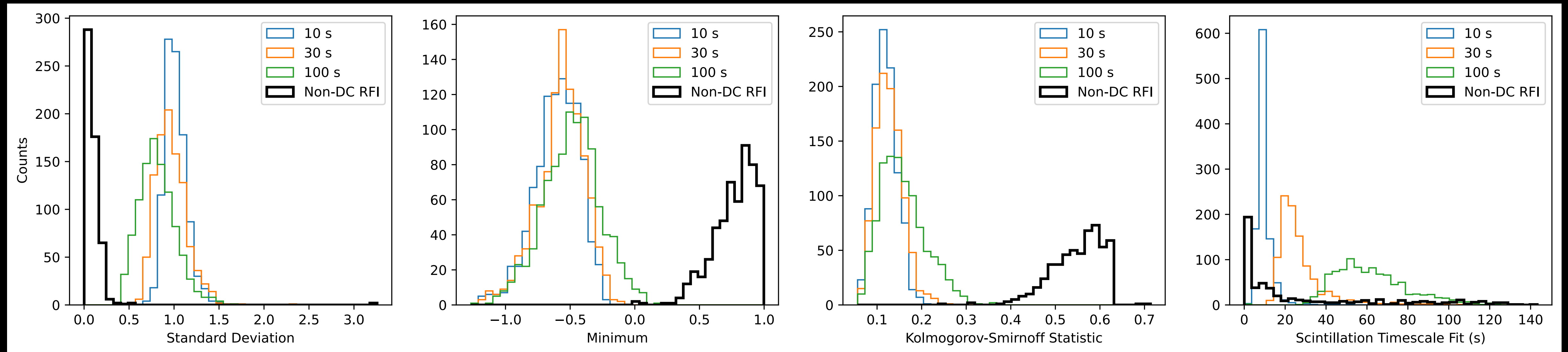
Diagnostic statistics



RFI Analysis

C band

S/N = 25



Standard Deviation

Minimum

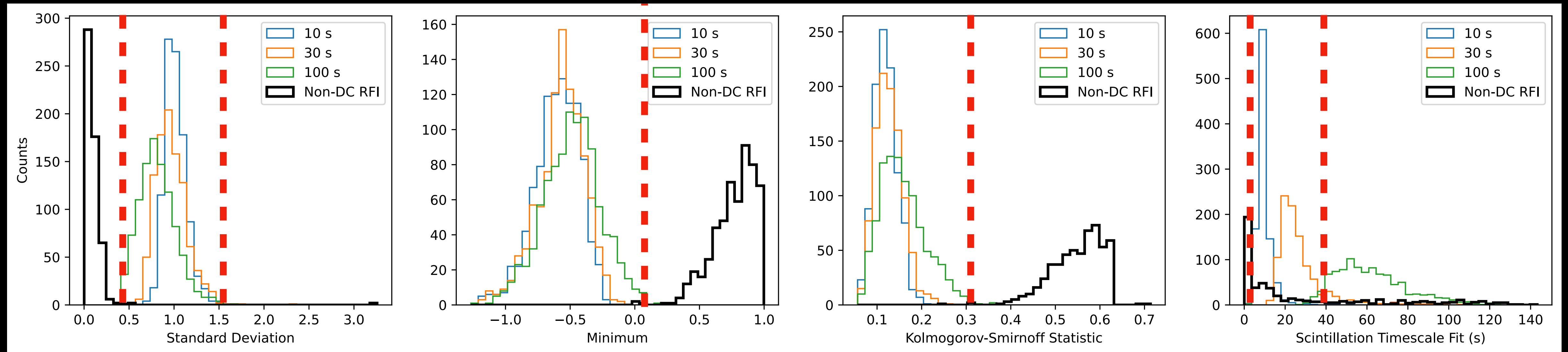
Kolmogorov-
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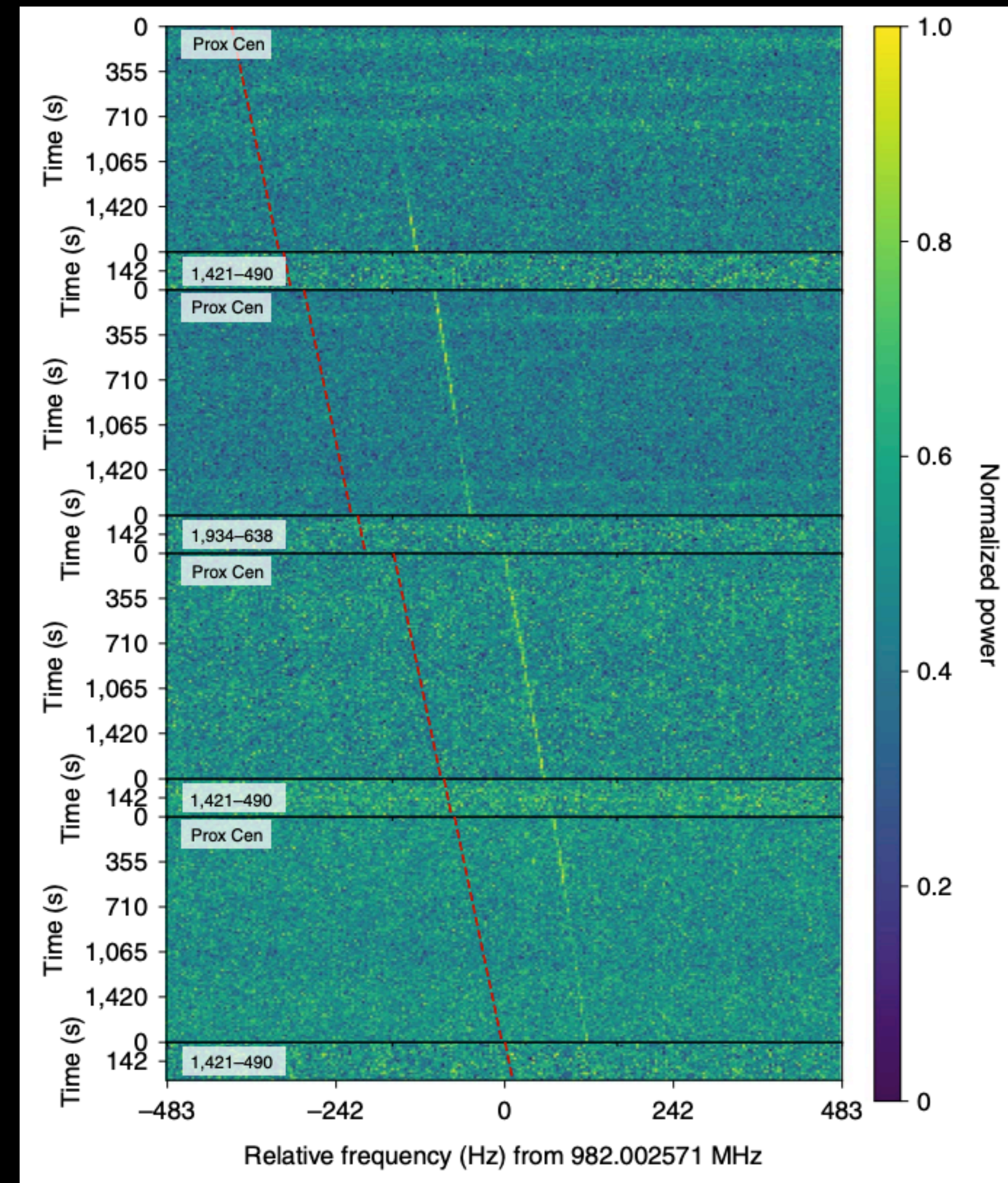
Summary

- Developed a framework for scintillation analysis, with accompanying code
- Because of RFI environment, higher frequencies are more amenable
- Looking forward: dedicated survey with custom resolution to search near the Galactic Center
- Better extraction / classification methods may lead to improvements

Extra Slides

Candidate identification and differentiation

- Narrowband (vs. astrophysical sources)
- Non-zero drift rate (vs. RFI)
- Sky localization (vs. RFI)



Smith et al. 2021

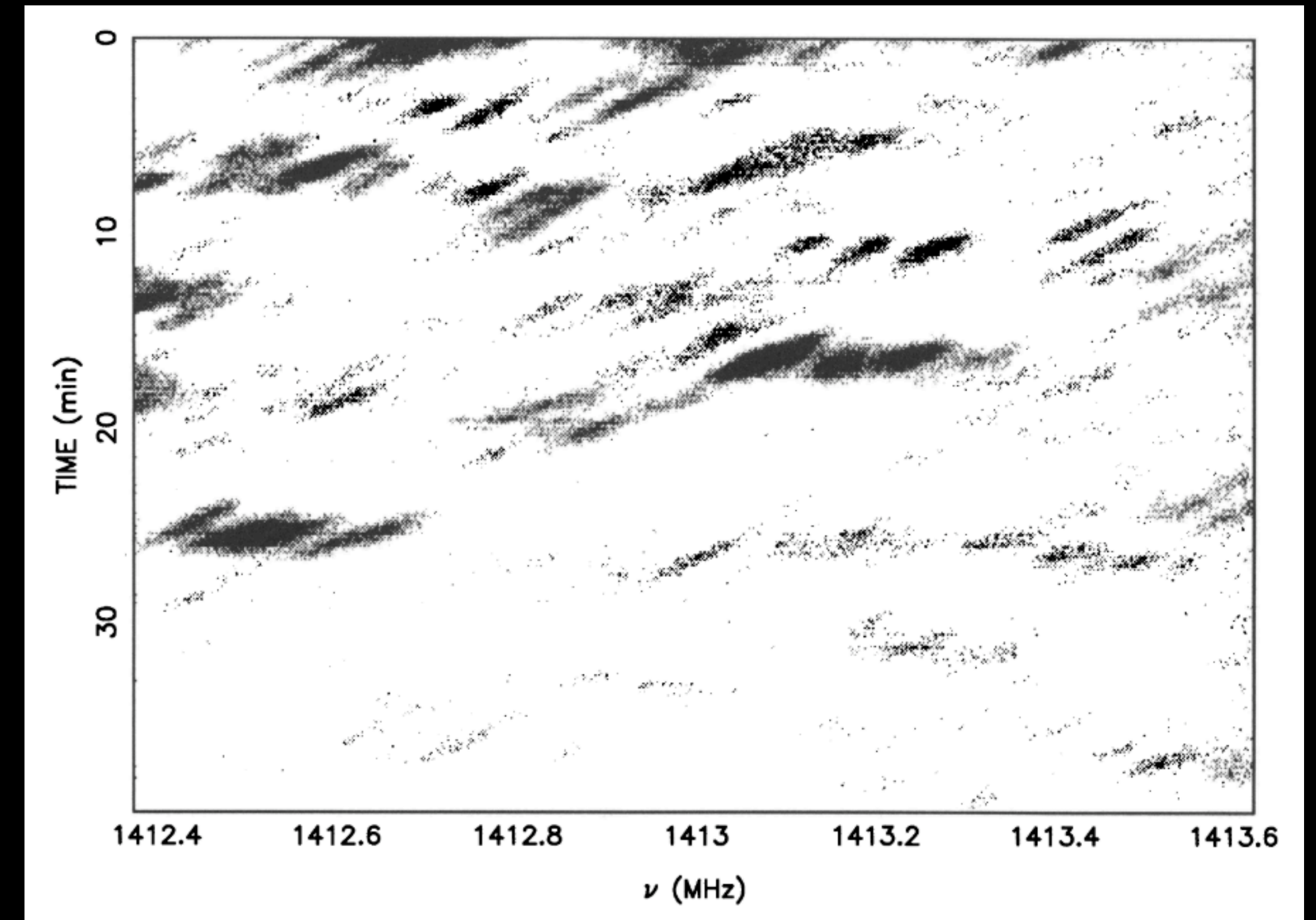
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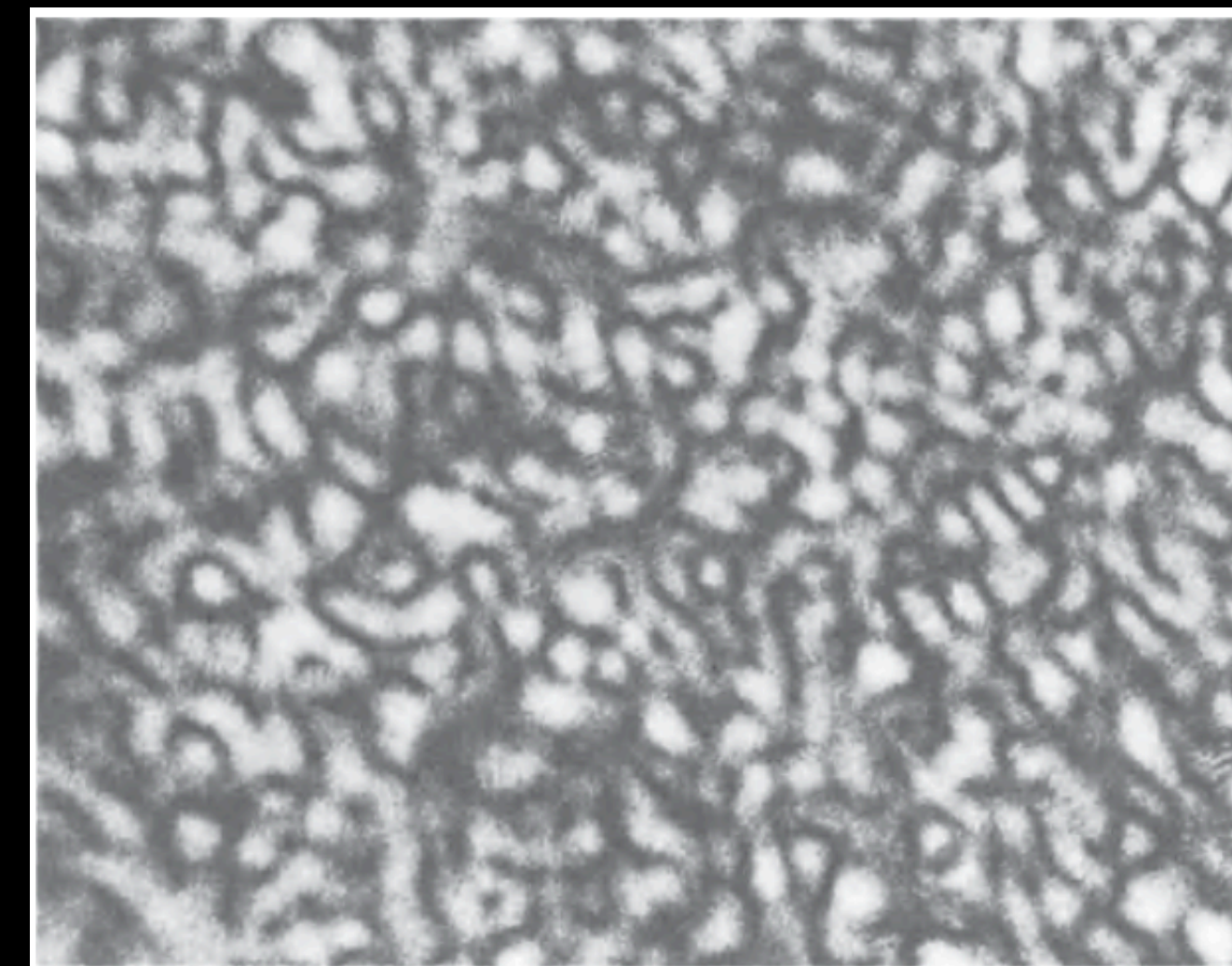
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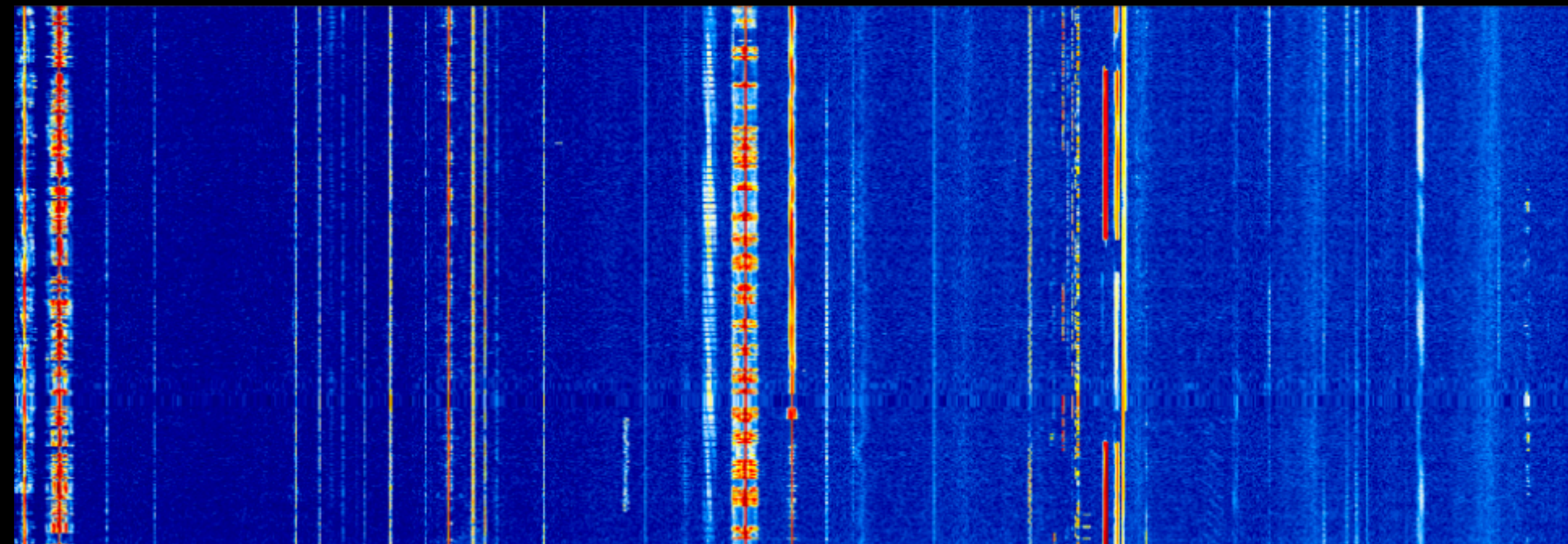
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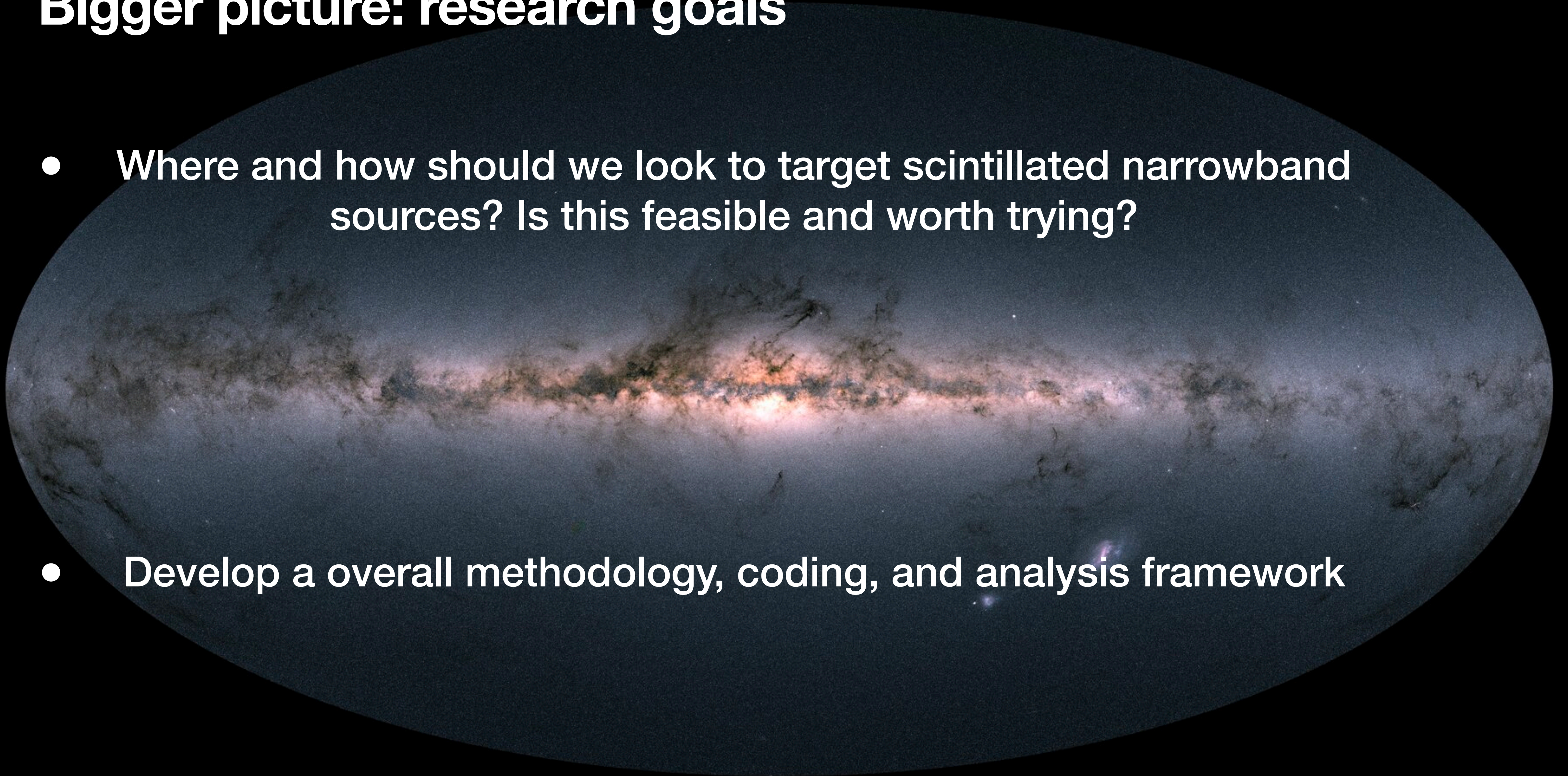
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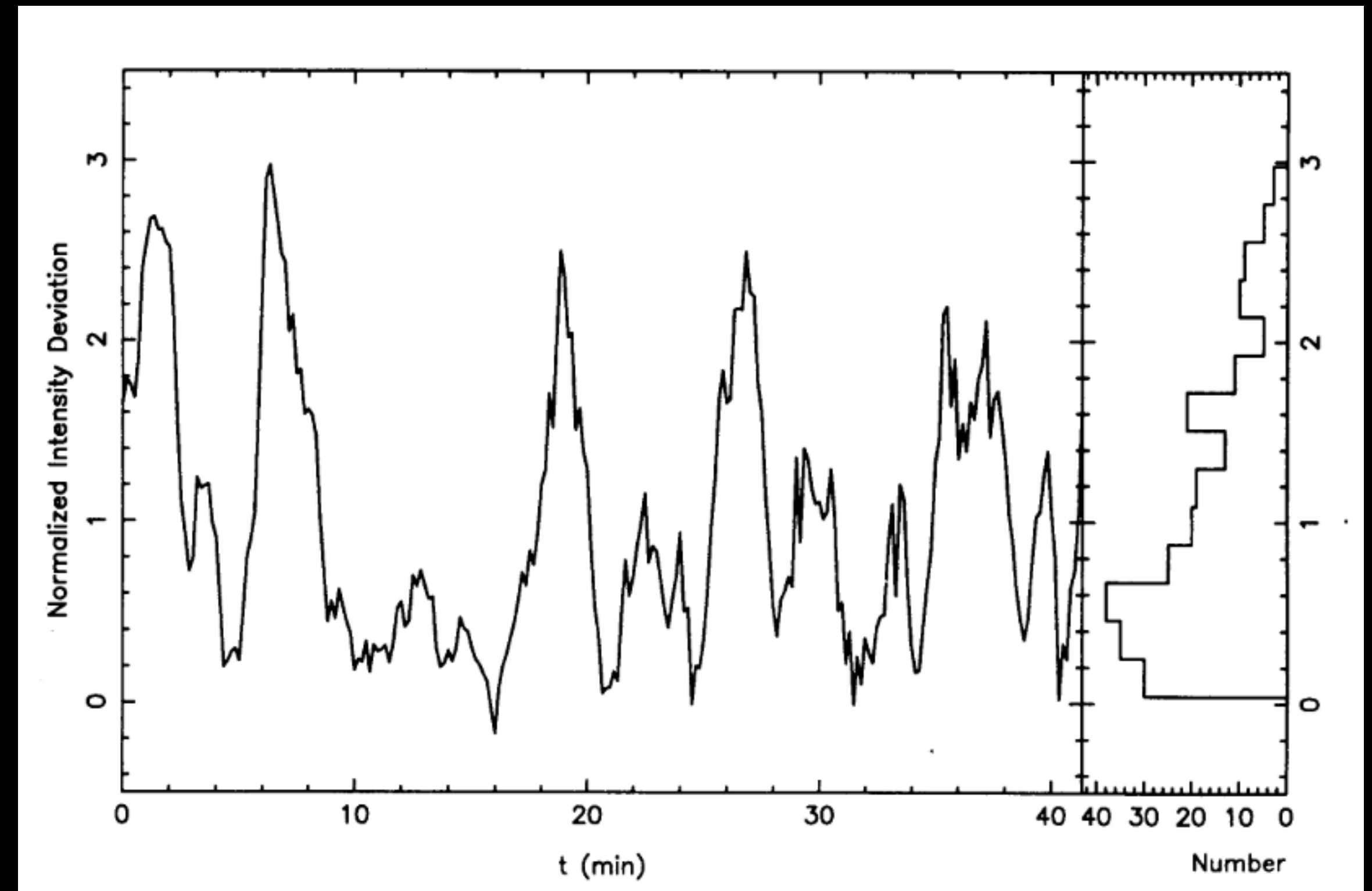
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Estimating scattering strength

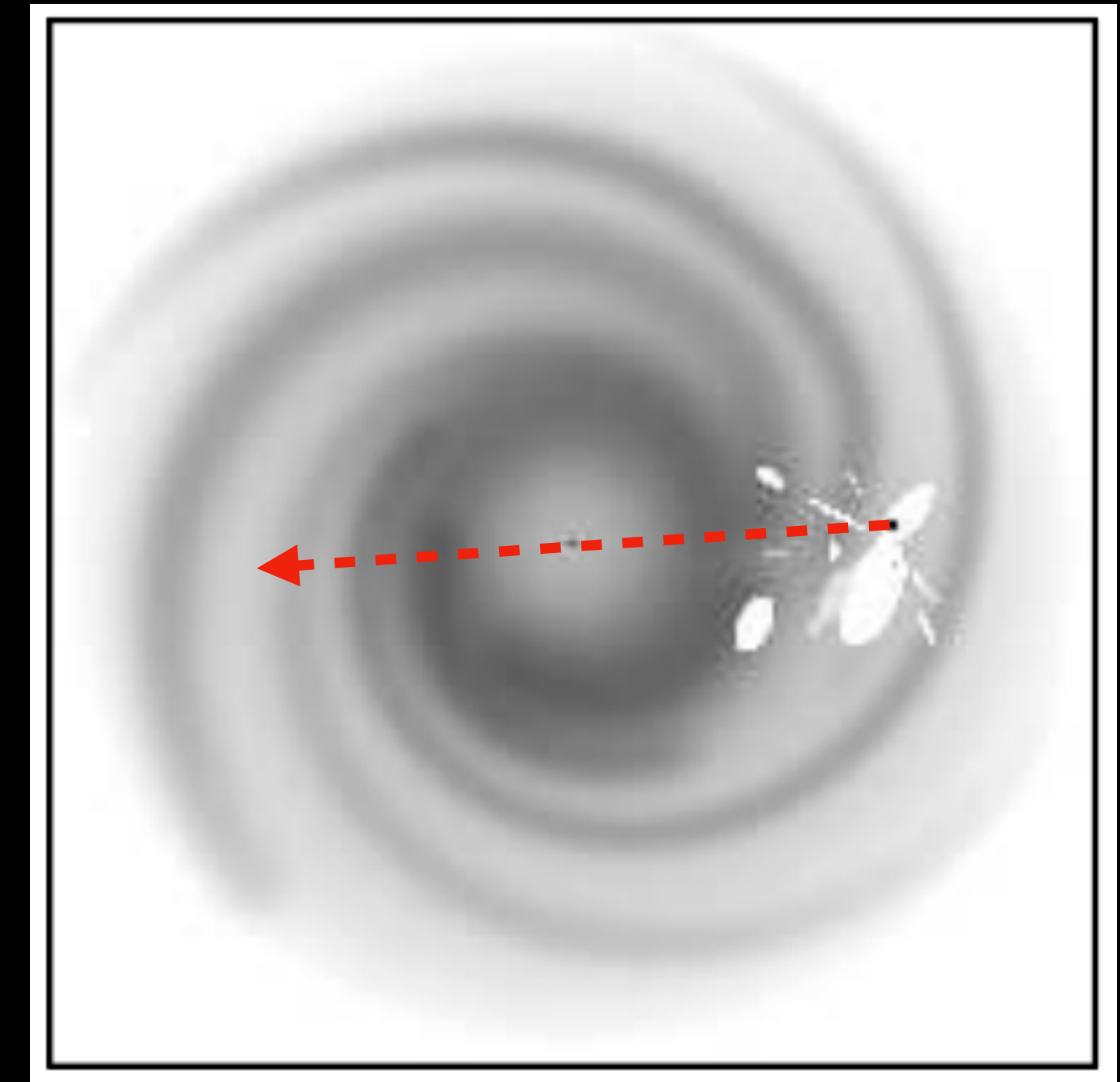
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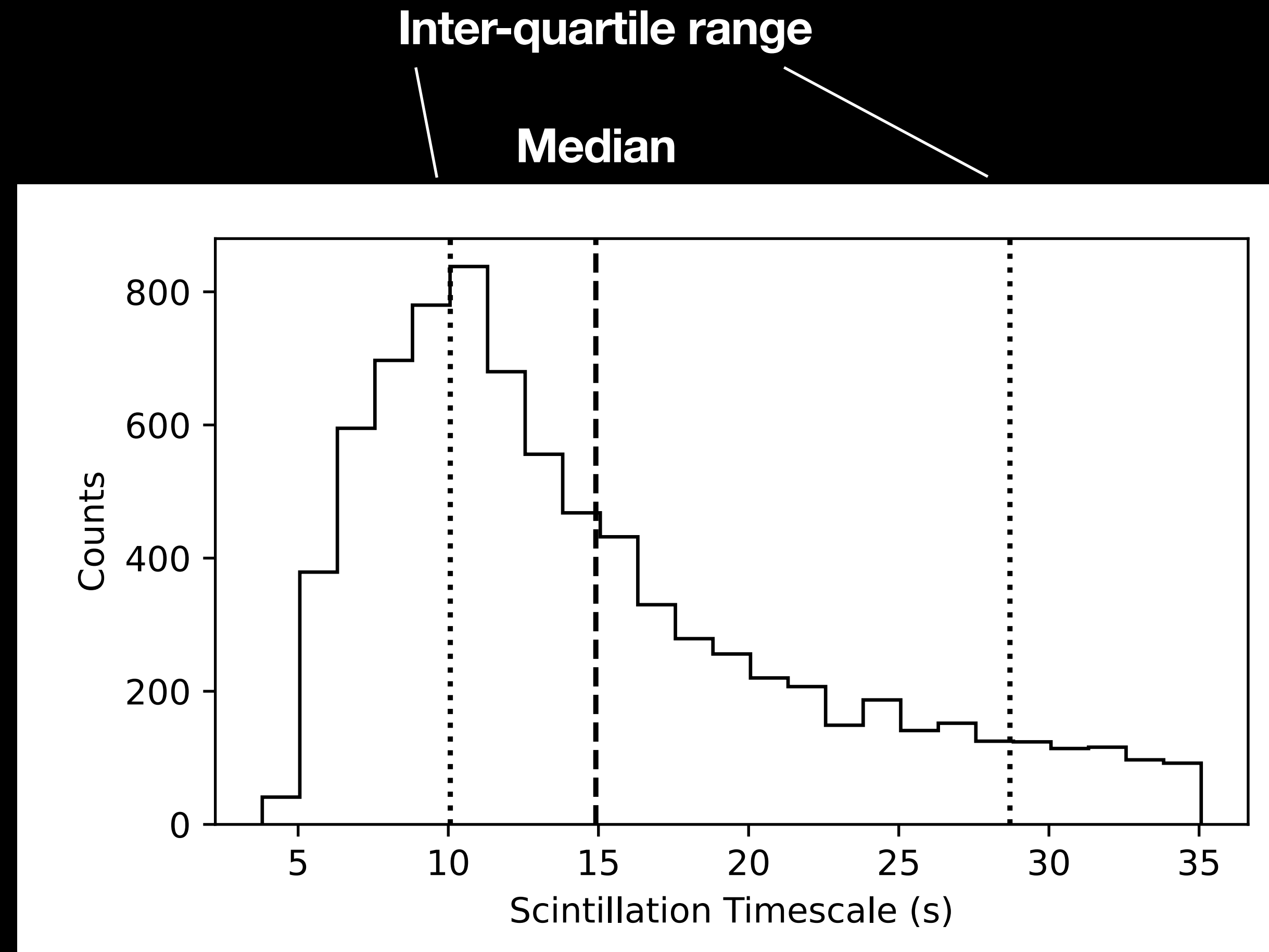
J. M. CORDES
Astronomy Department and NAIC, Cornell University, Ithaca, NY 14853
cordes@spacenet.tn.cornell.edu

T. JOSEPH W. LAZIO
Naval Research Lab, Code 7213, Washington, D.C. 20375-5351
Joseph.Lazio@nrl.navy.mil



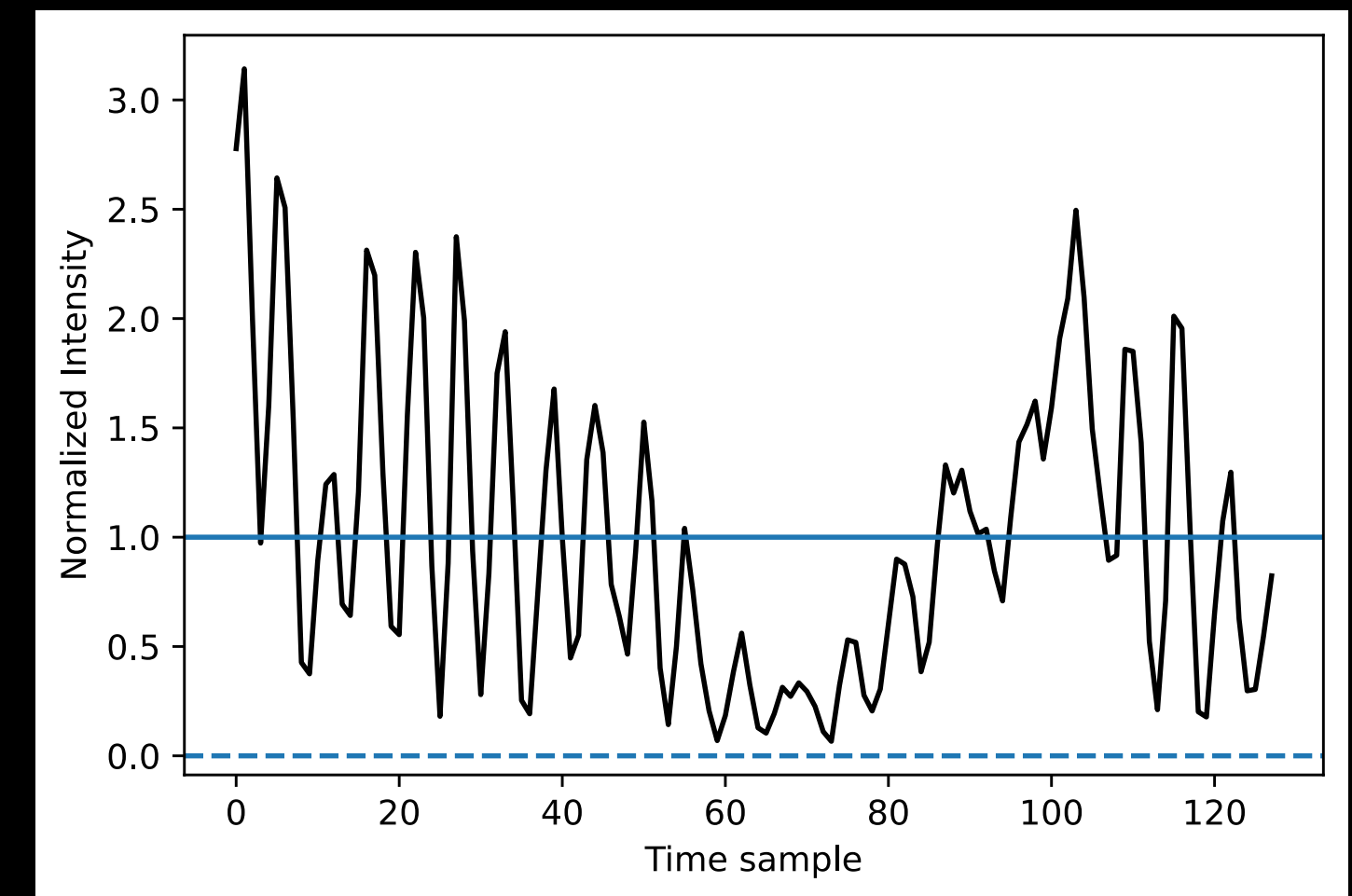
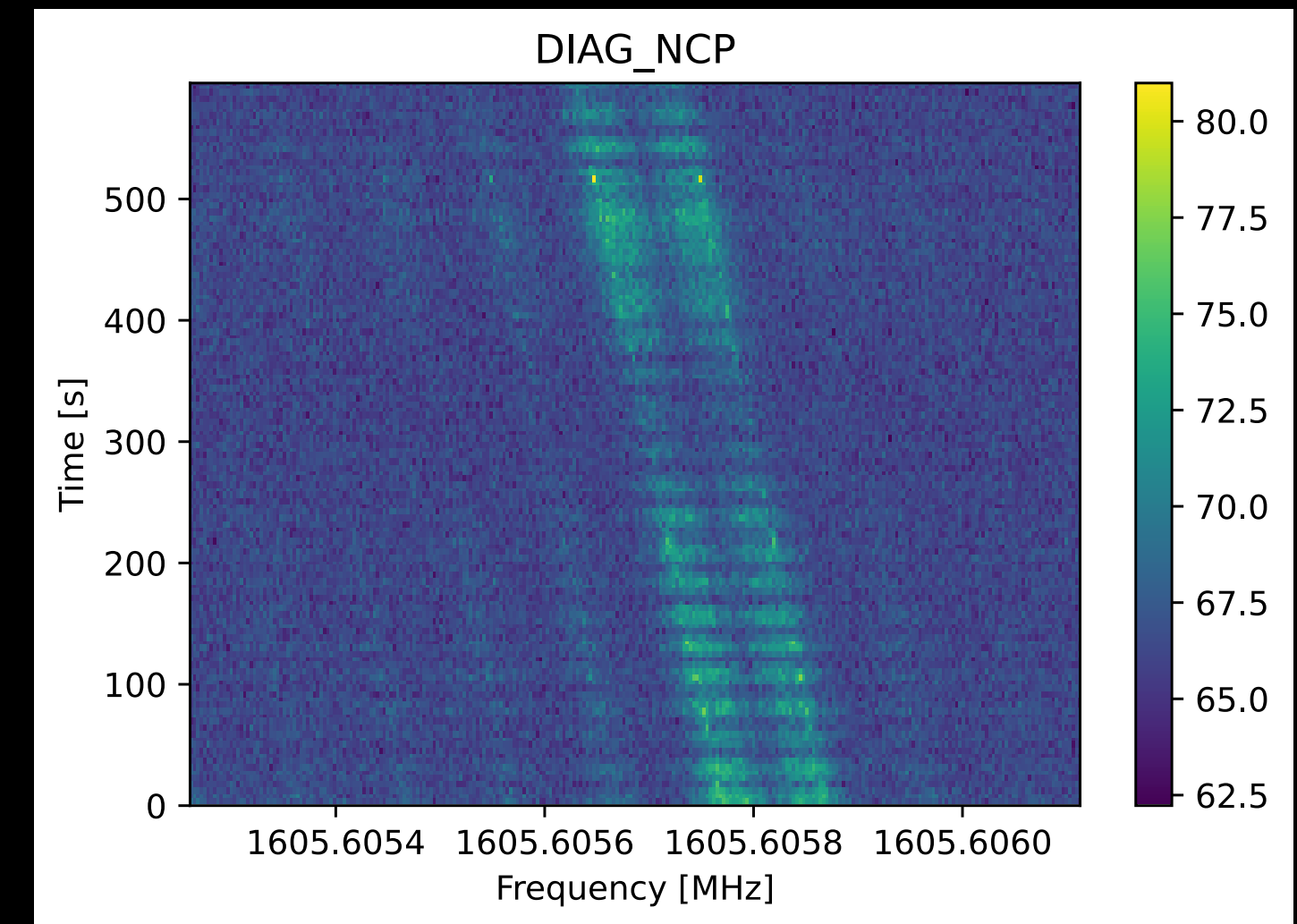
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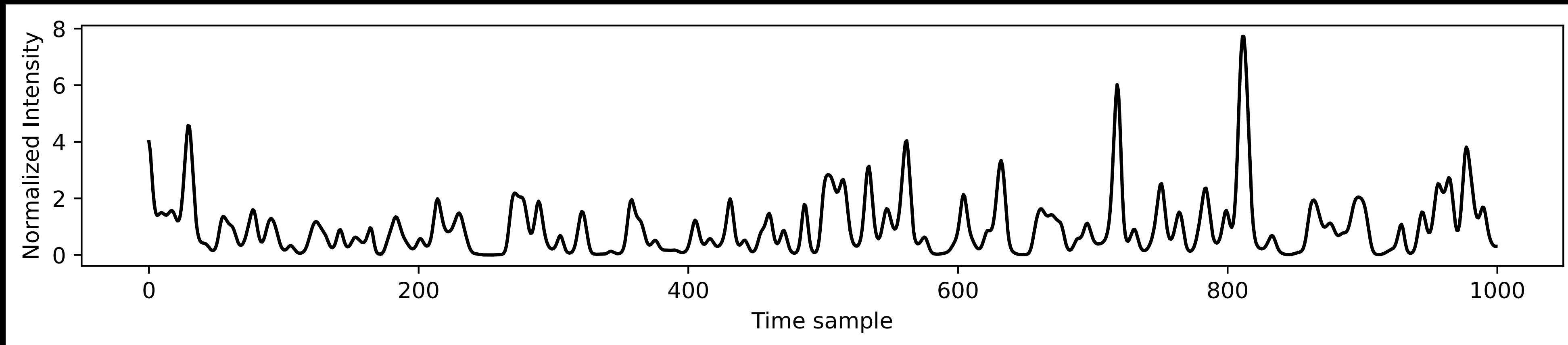
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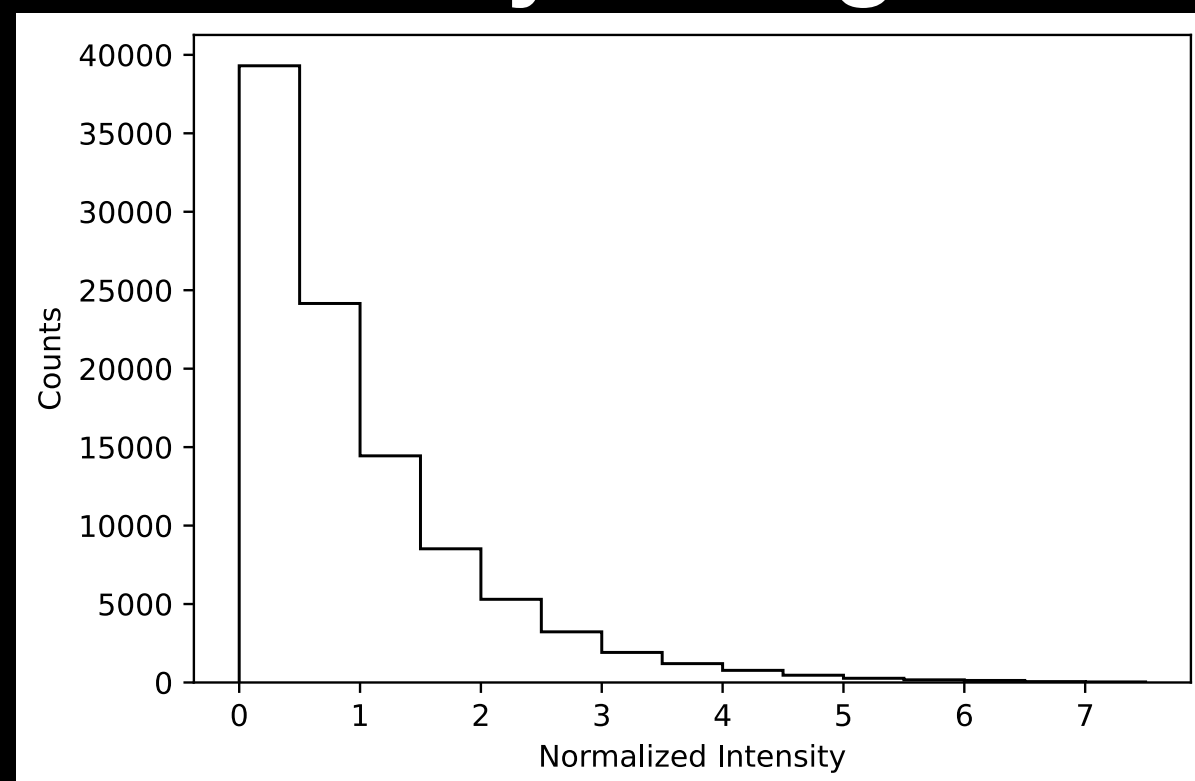
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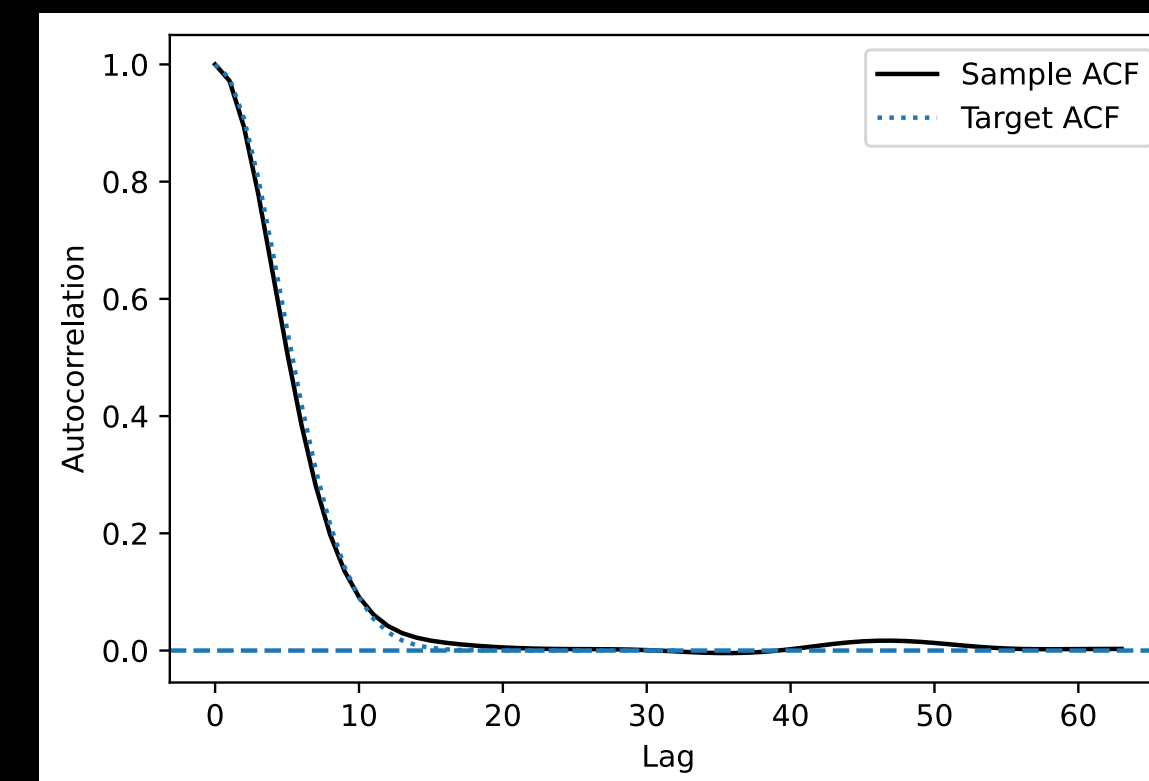
$n = 1e5$

$\Delta t_d = 30 \text{ s}$

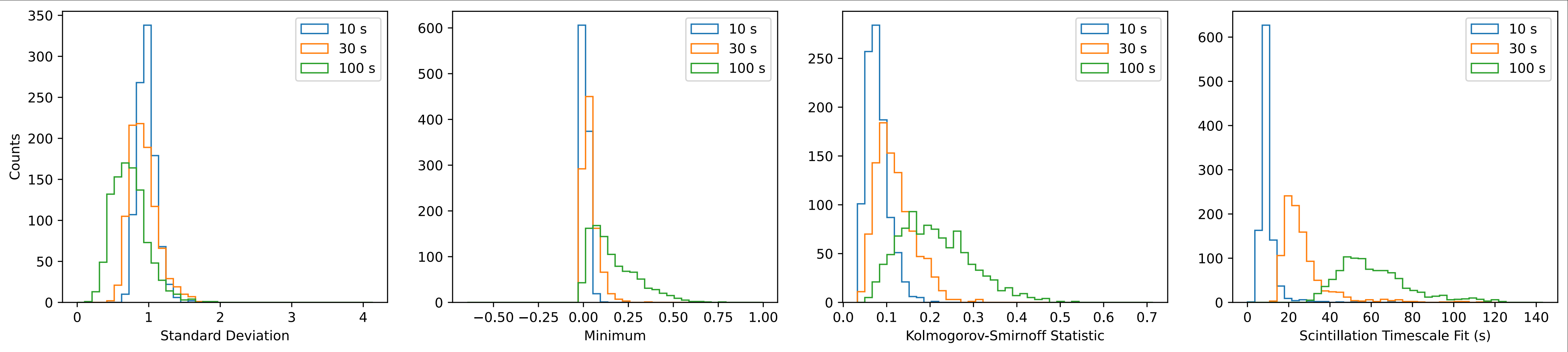
Intensity histogram



Autocorrelation



Statistics using low number of synthetic samples



Std. Dev.

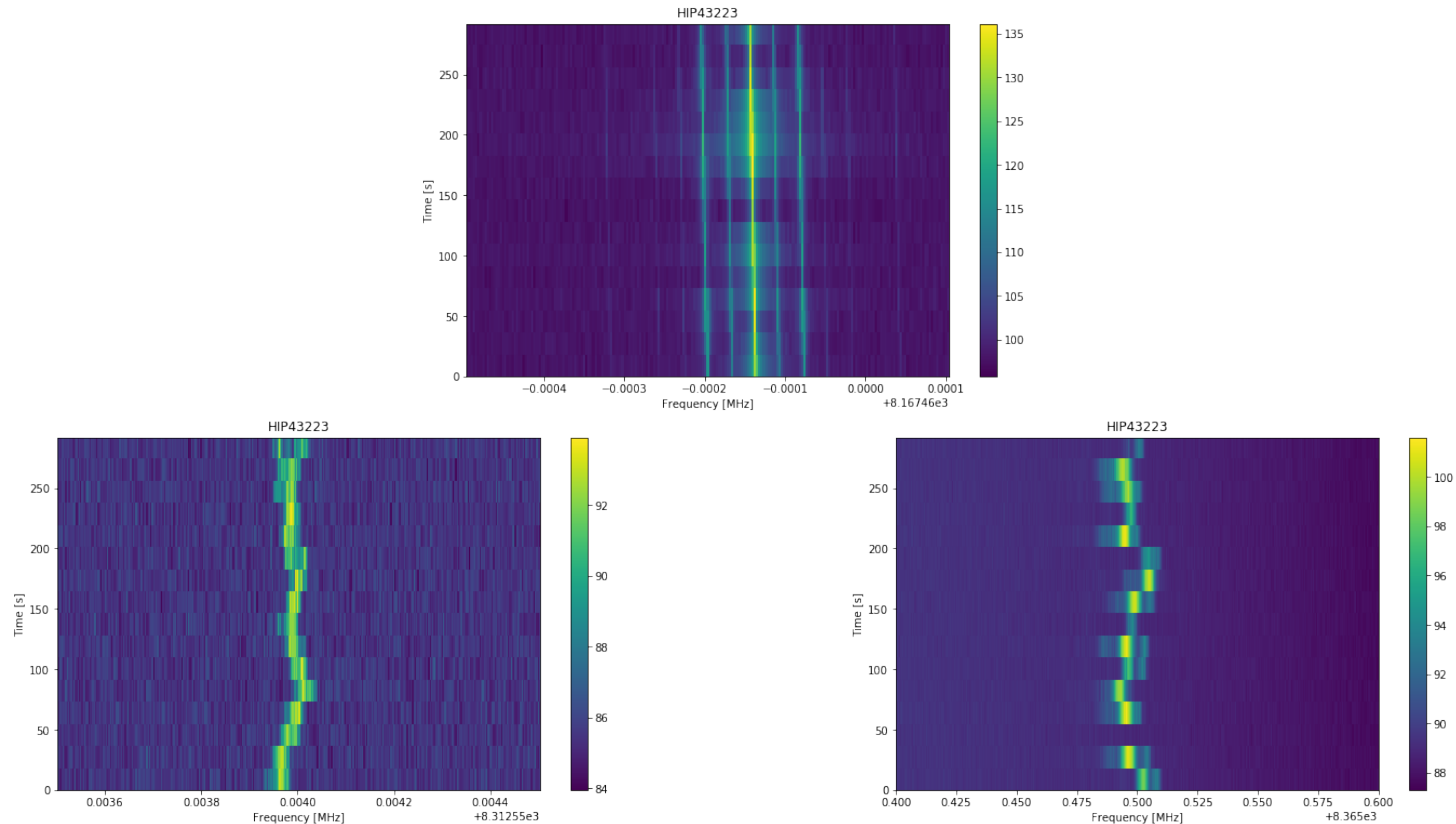
Minimum

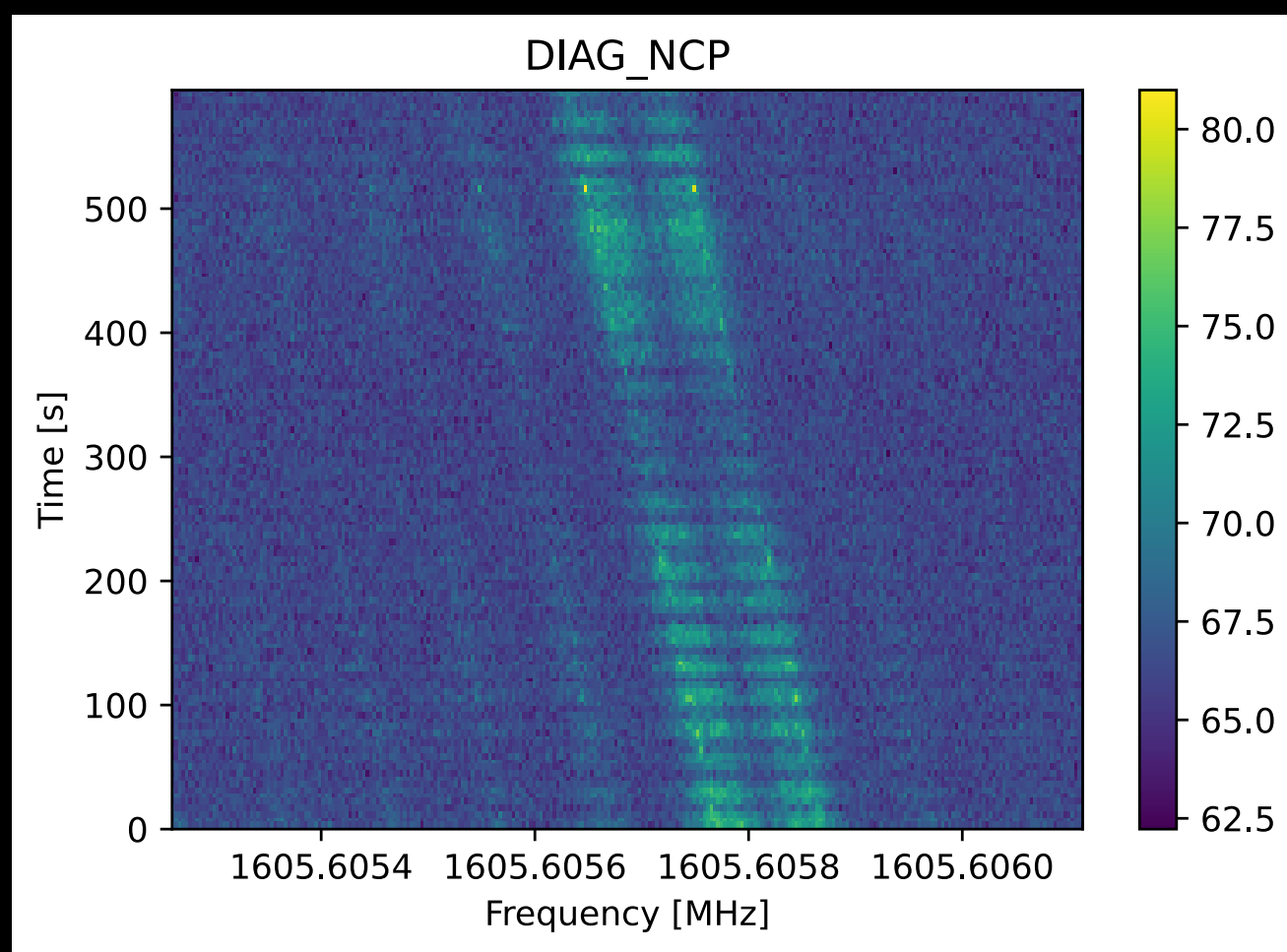
KS Statistic

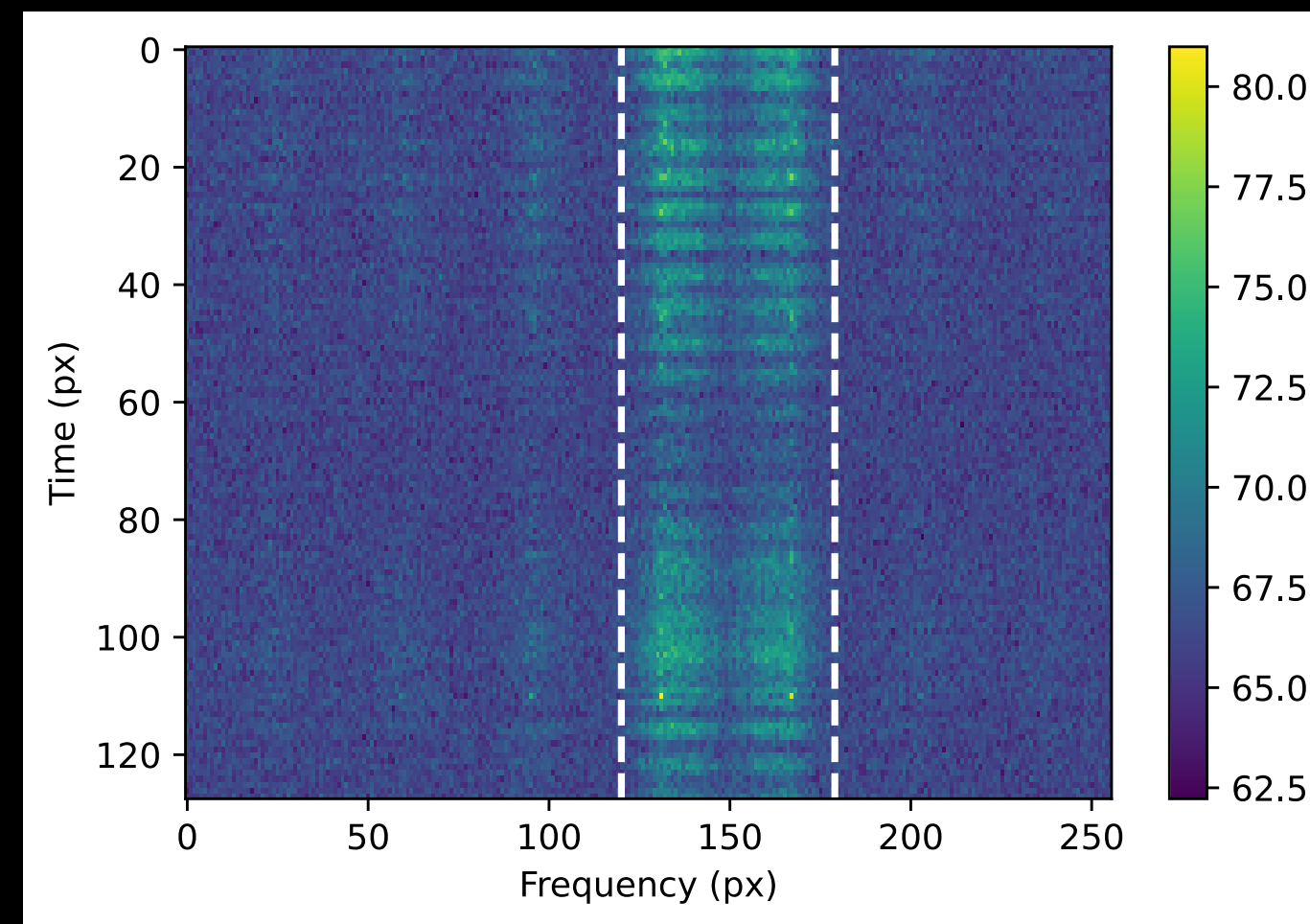
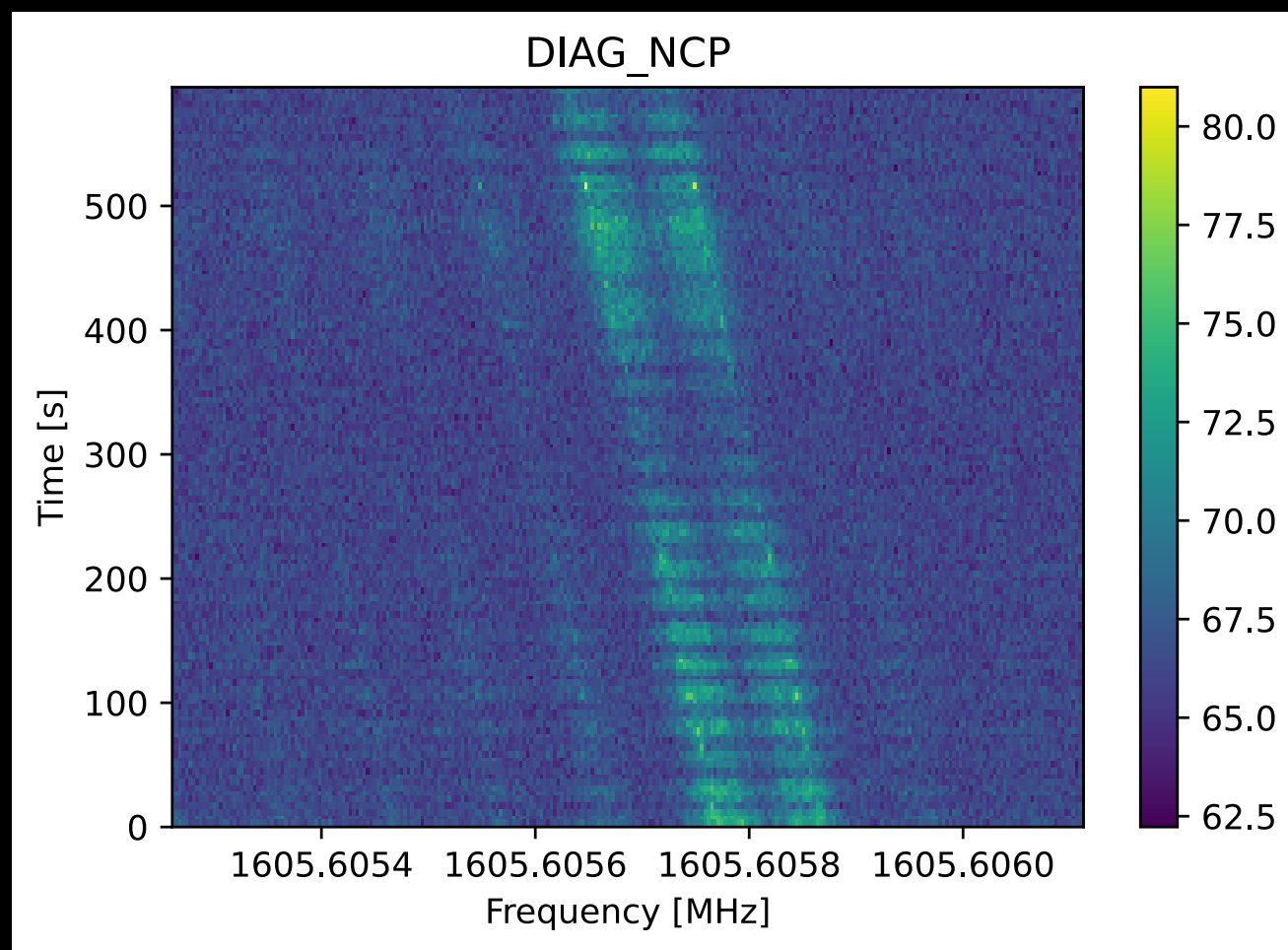
Timescale Fit

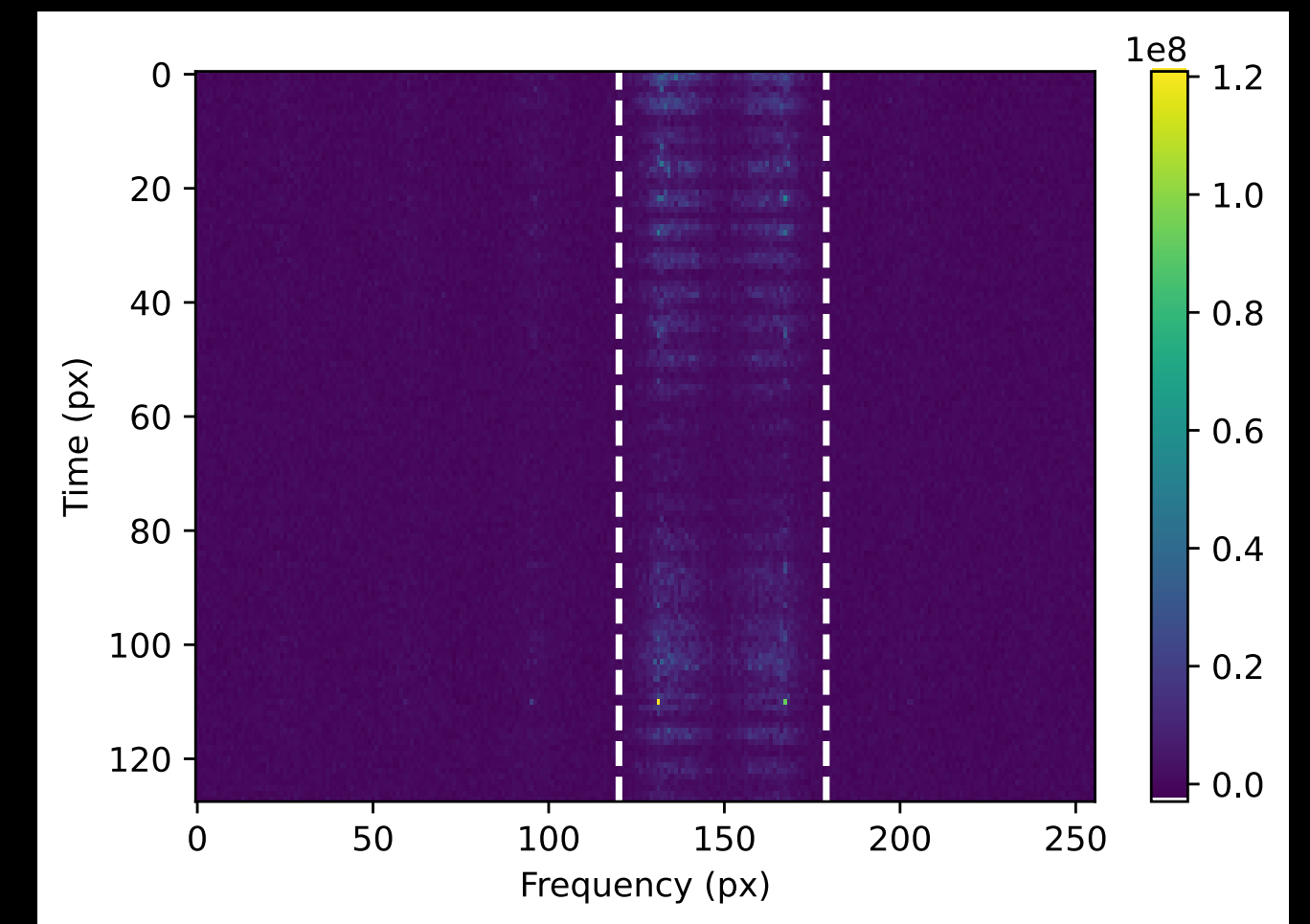
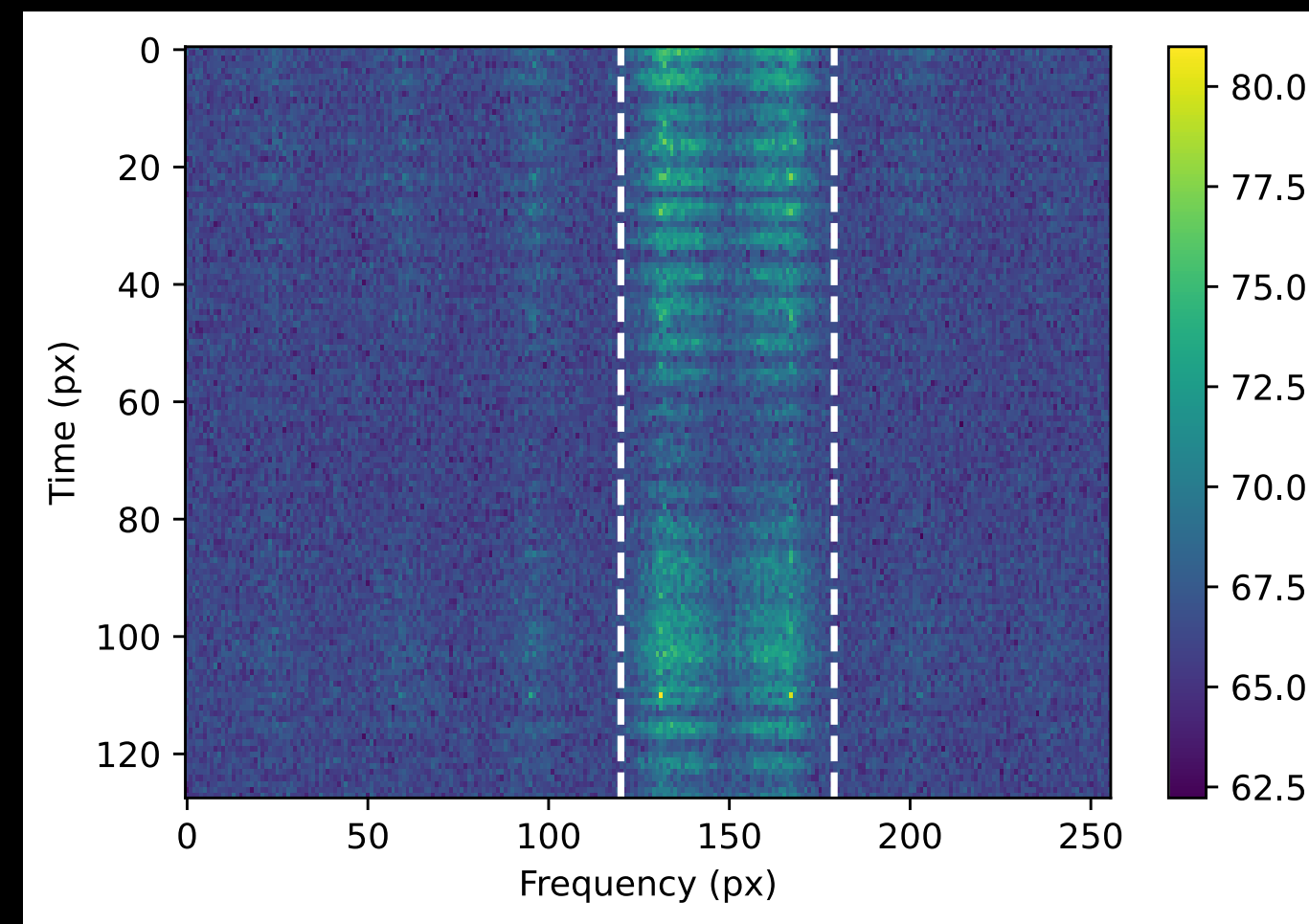
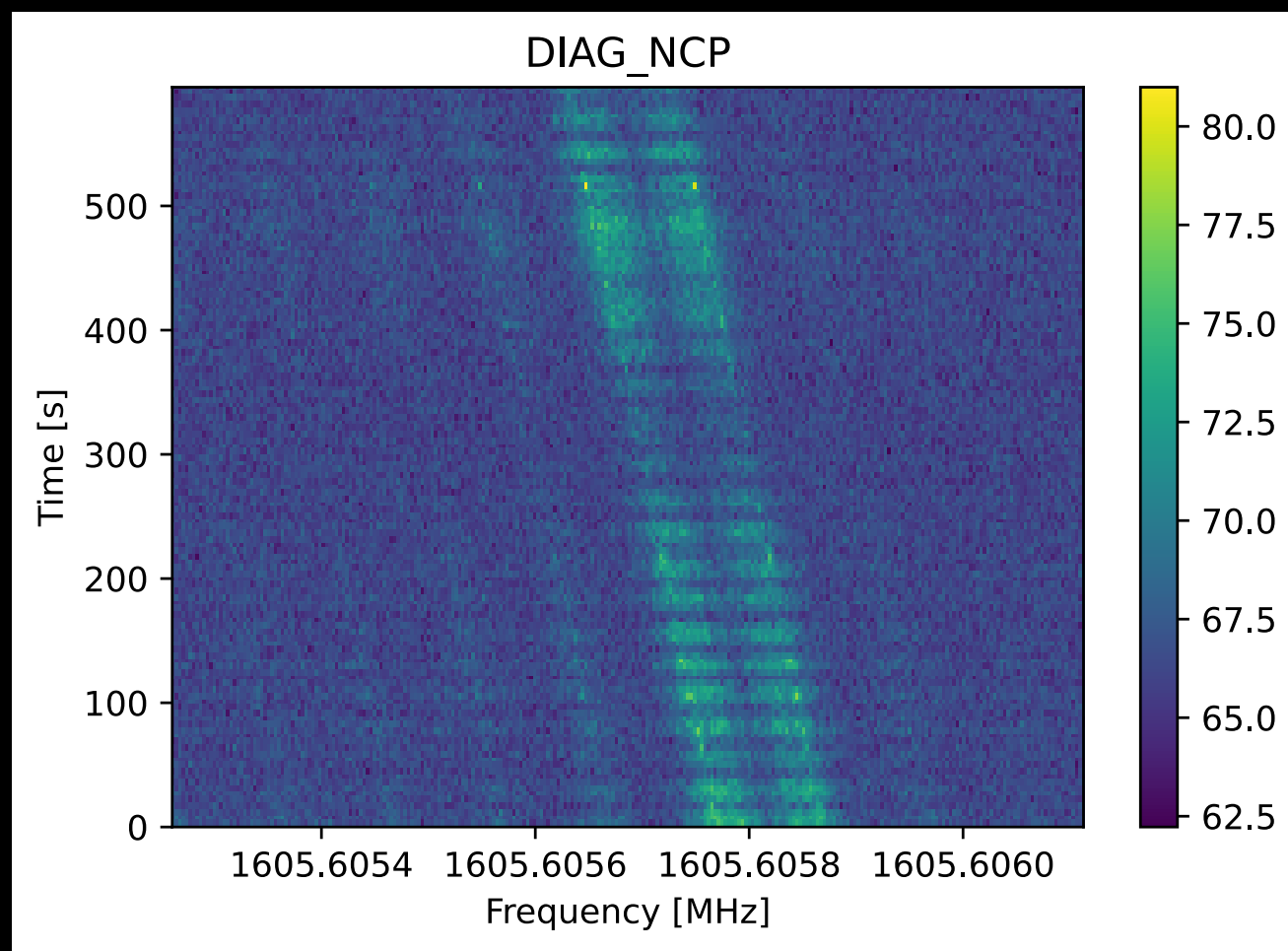
10 min “observation”, 4.65 s resolution

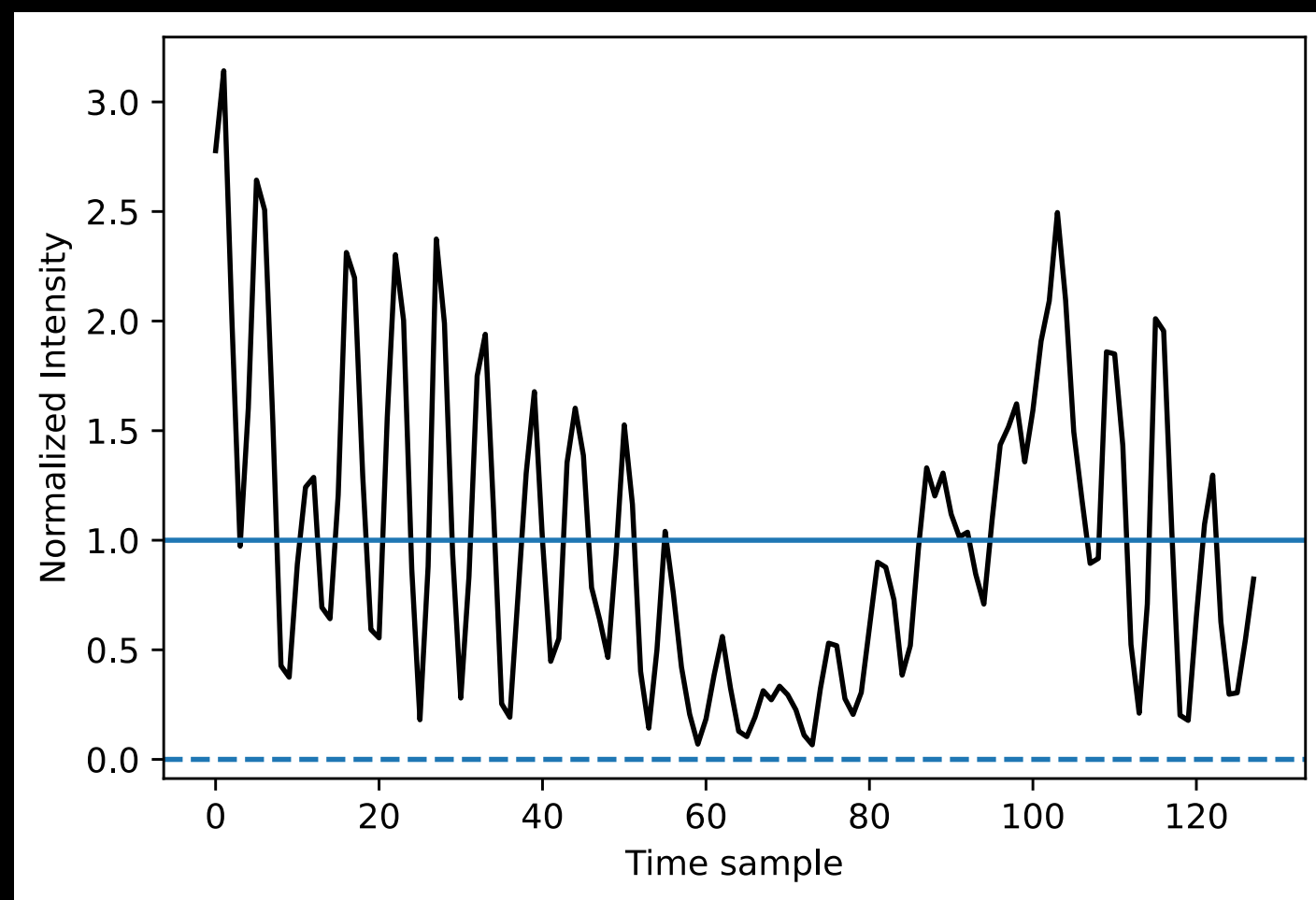
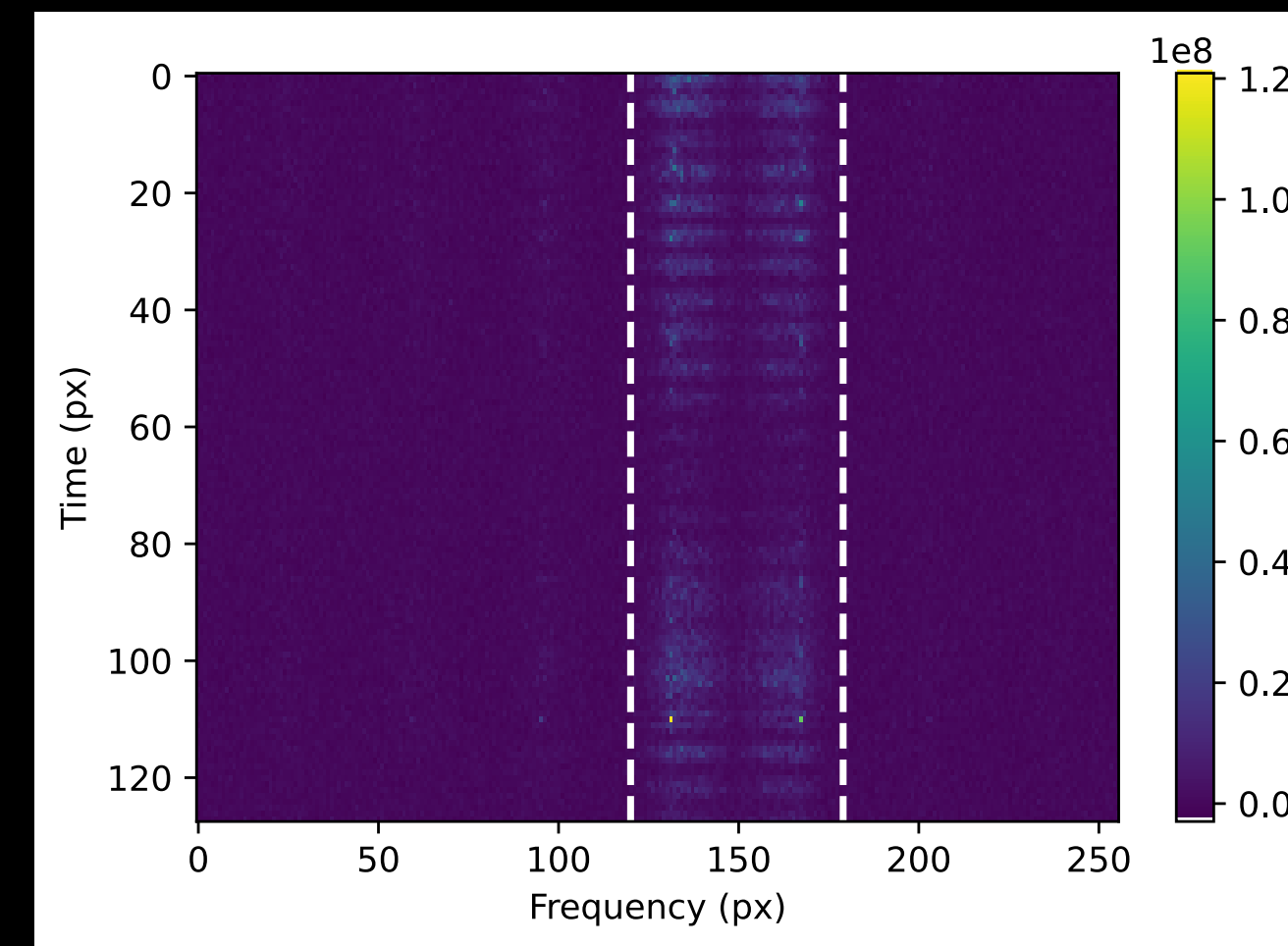
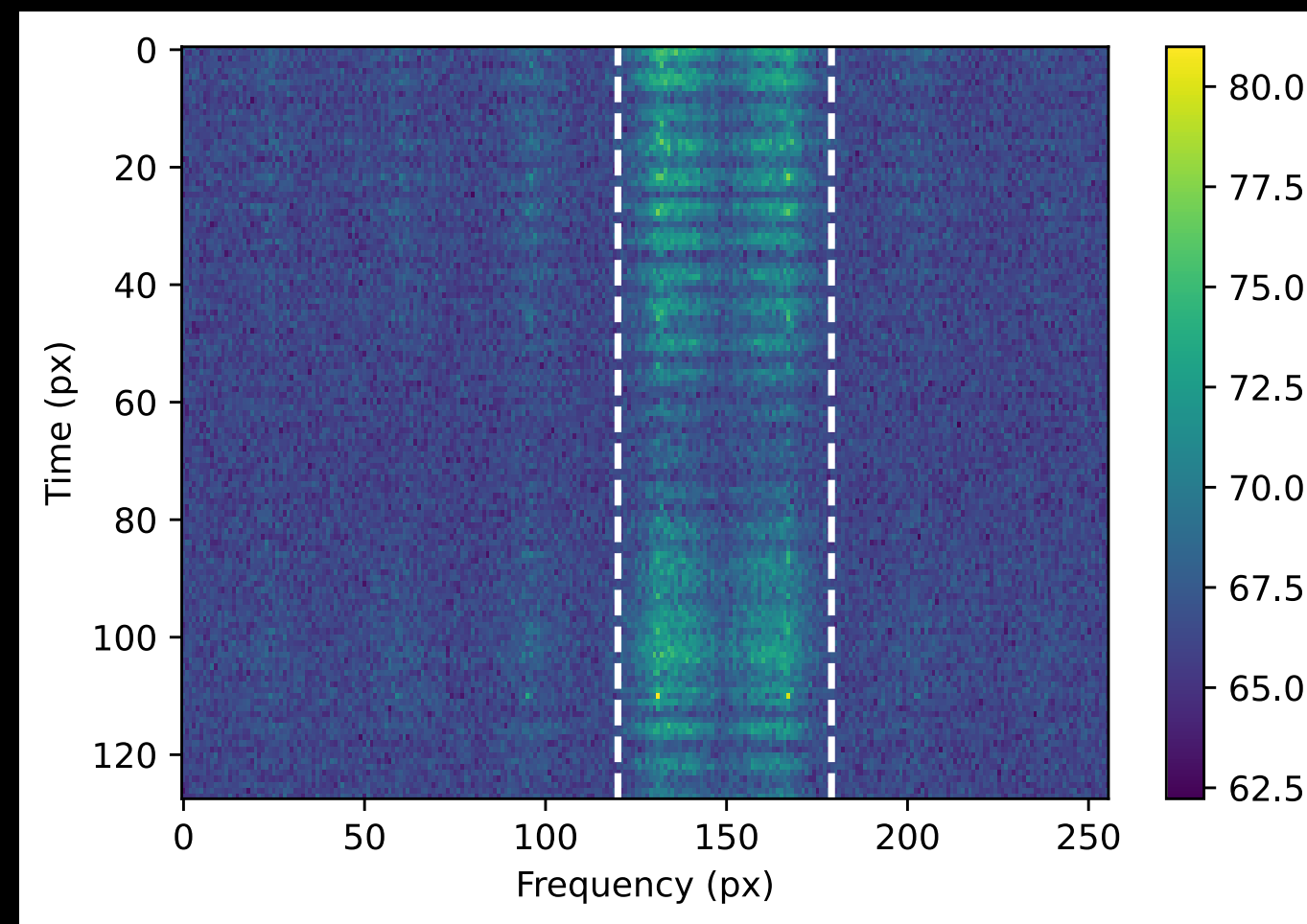
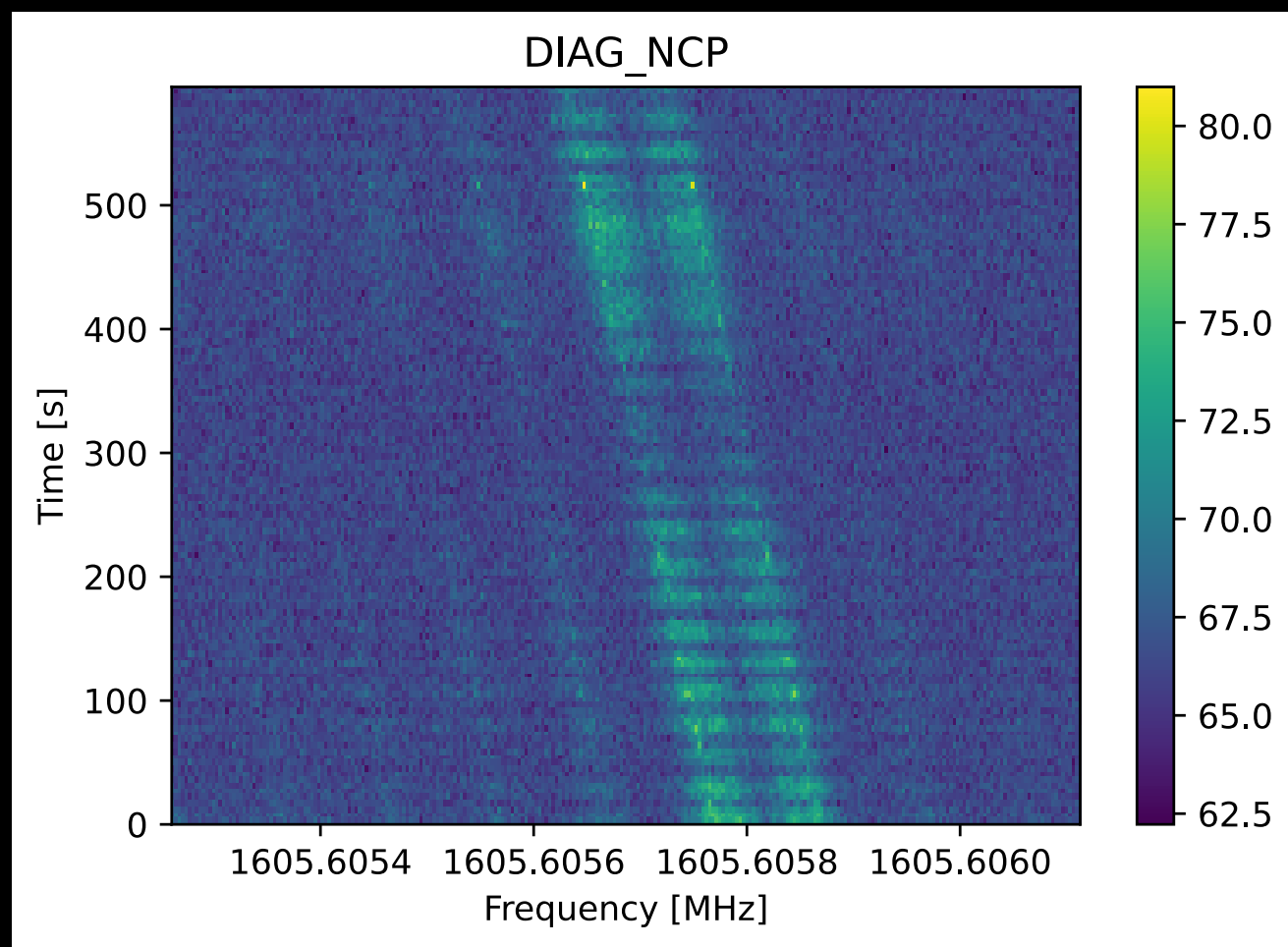
But what does the RFI environment look like?

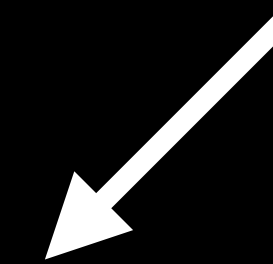
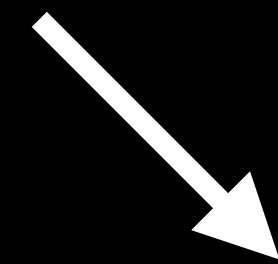
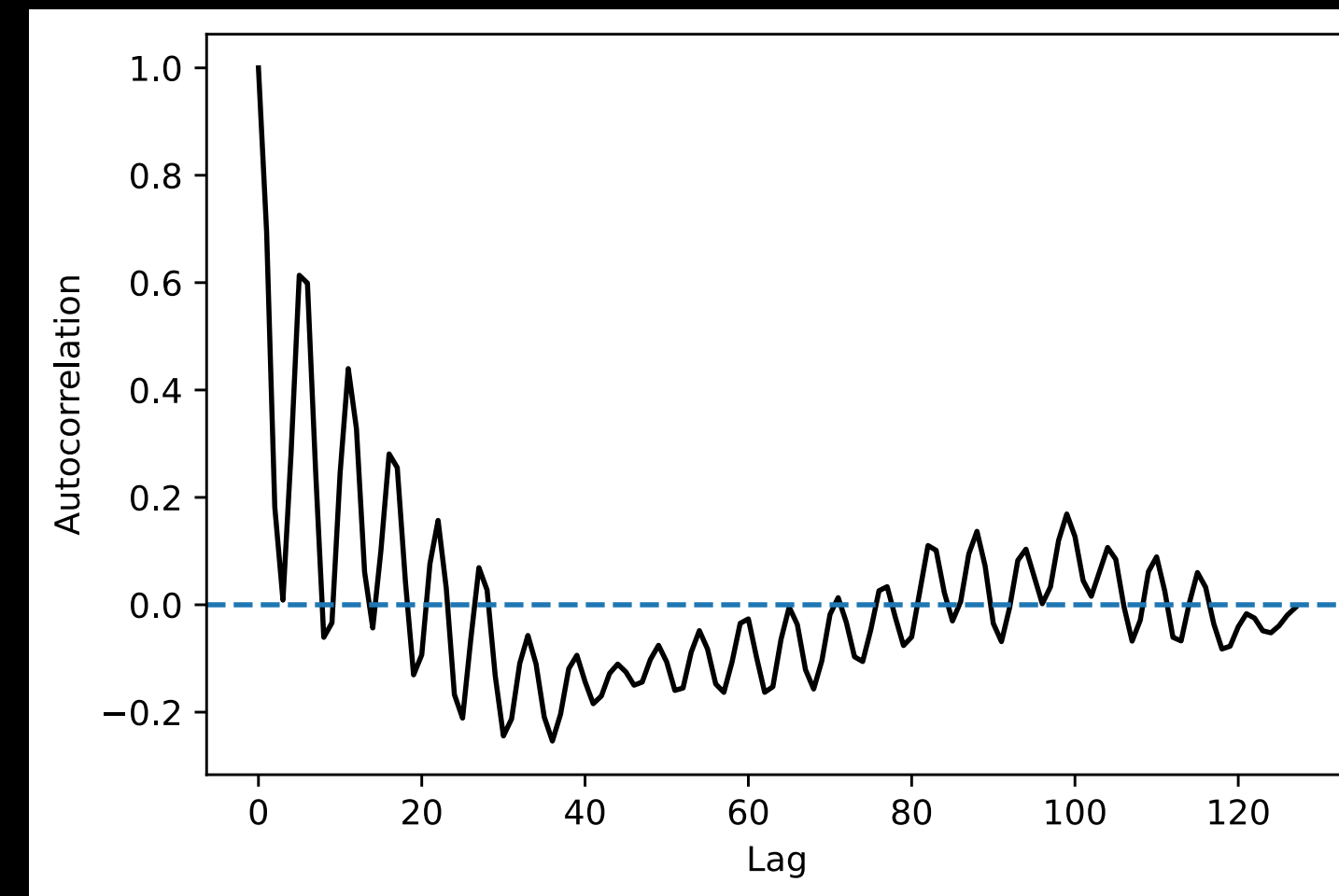
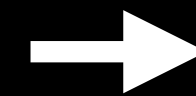
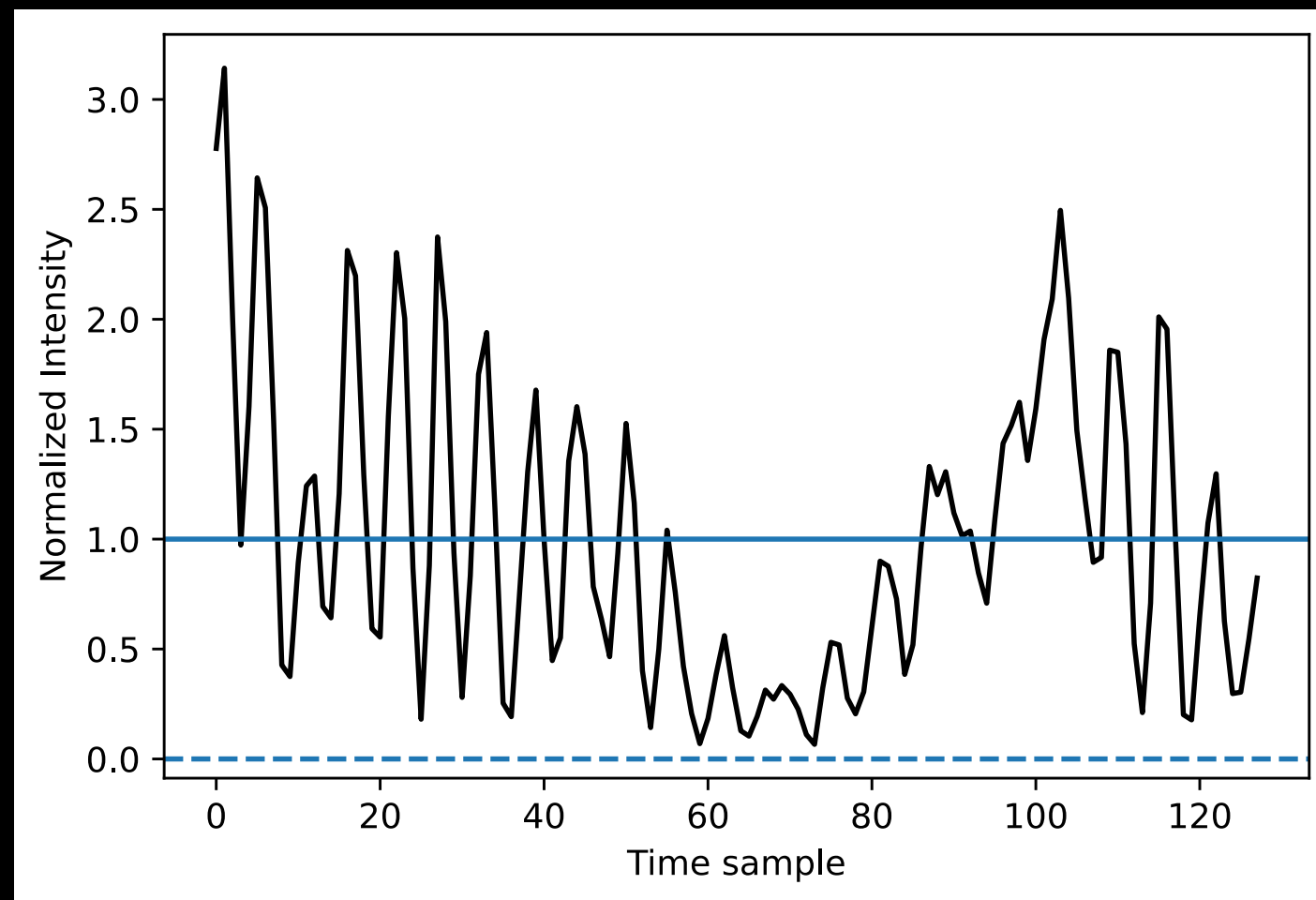
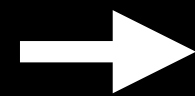
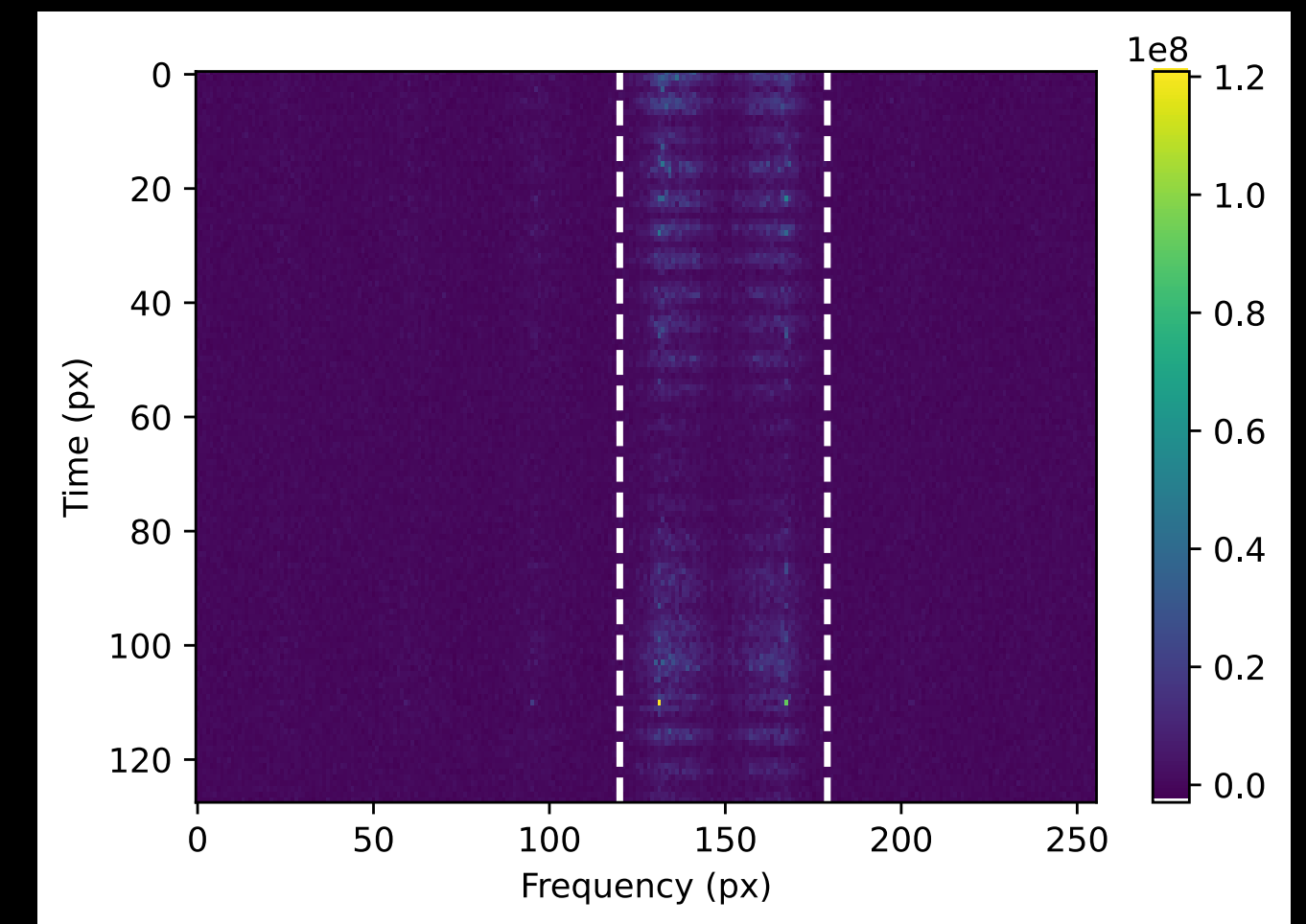
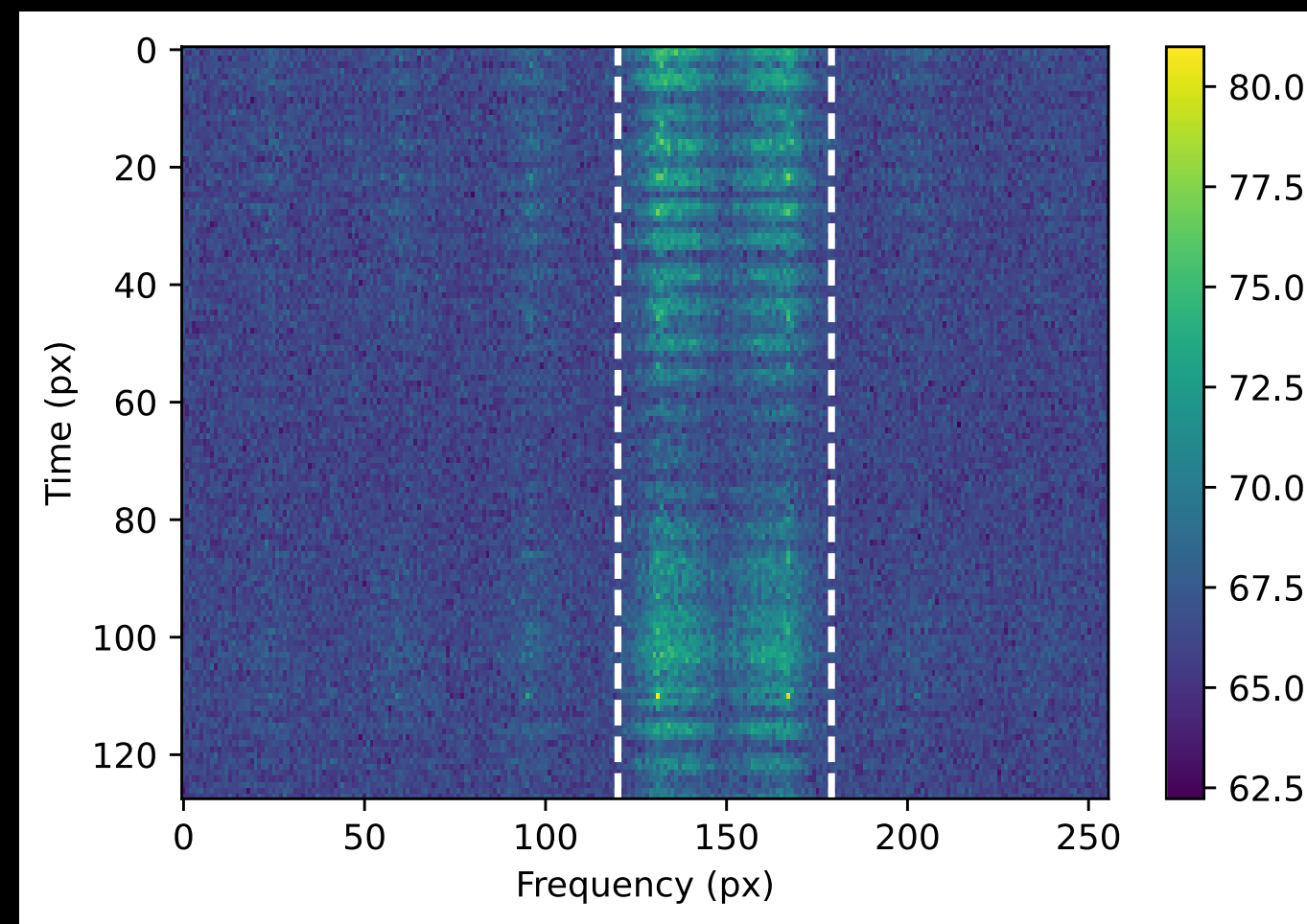
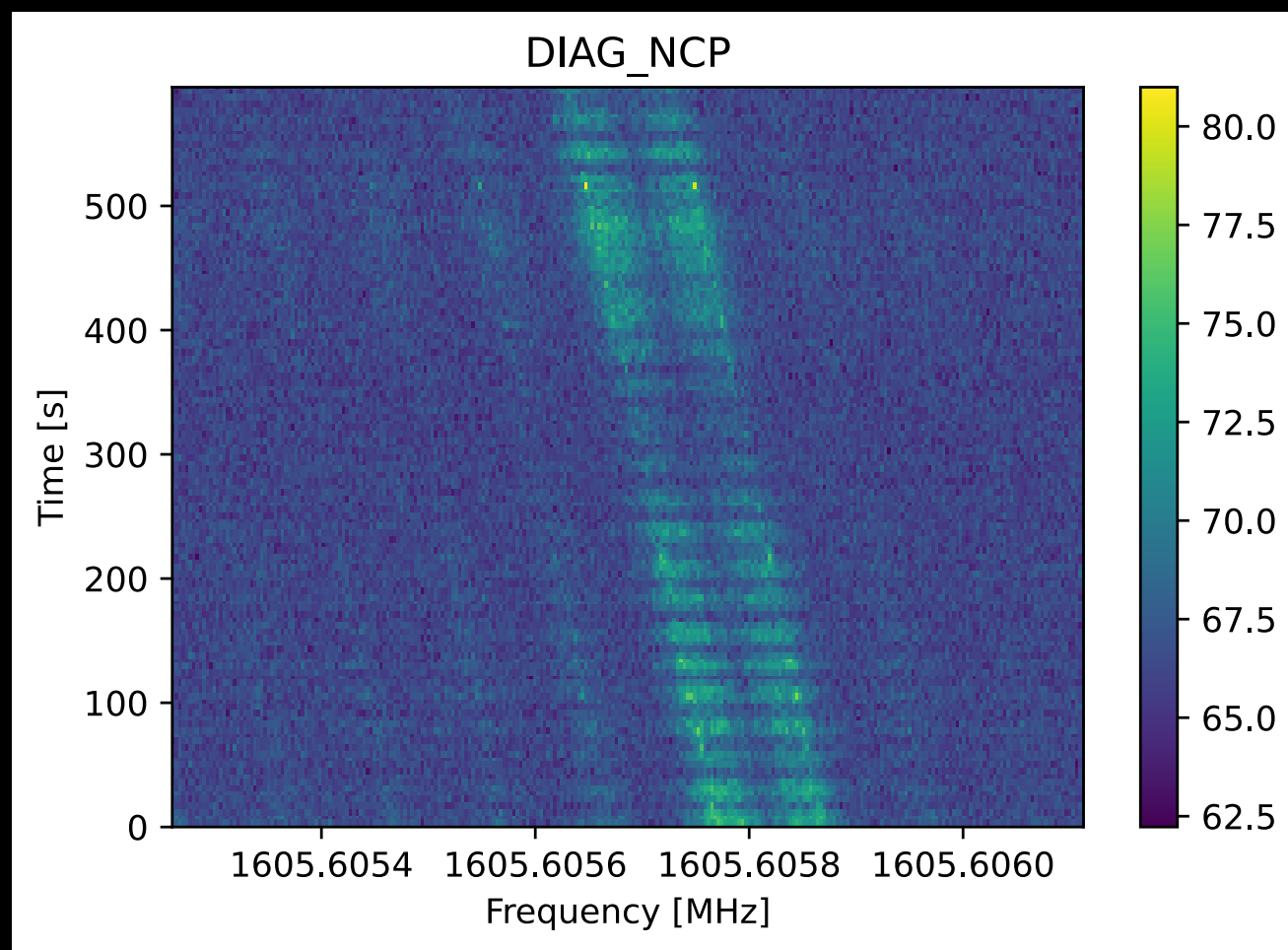








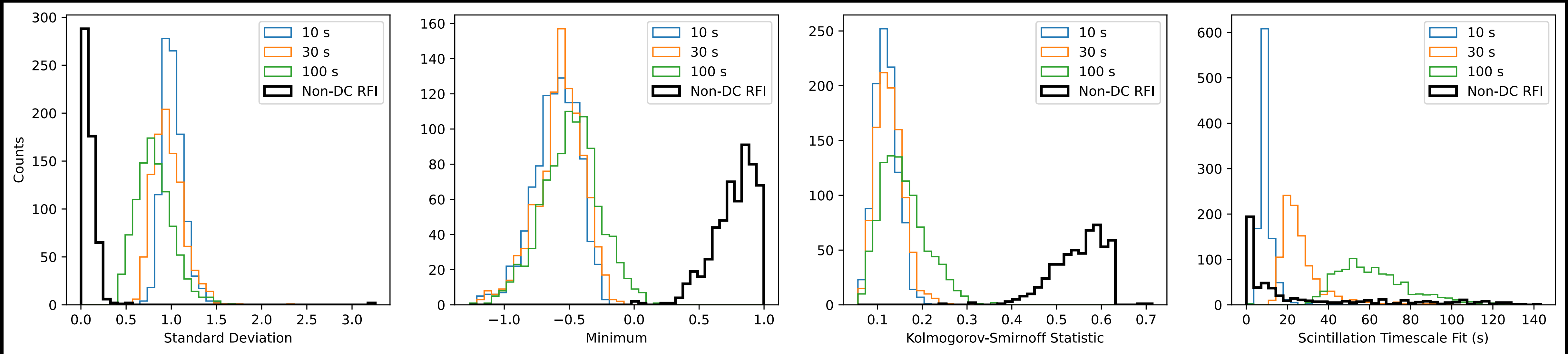




Diagnostic statistics

C band

S/N = 25



Std. Dev.

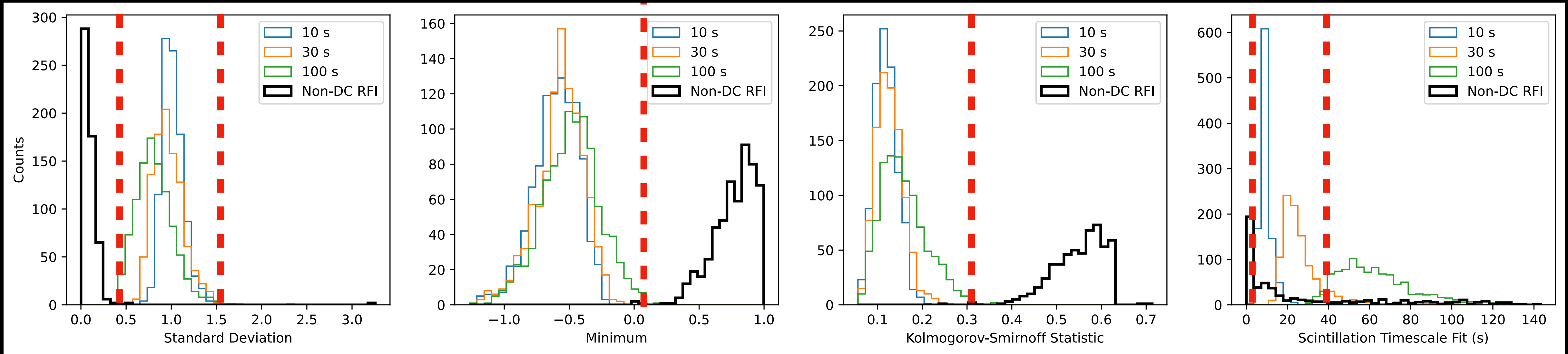
Minimum

KS Statistic

Timescale Fit

C band

S/N = 25



Std. Dev.

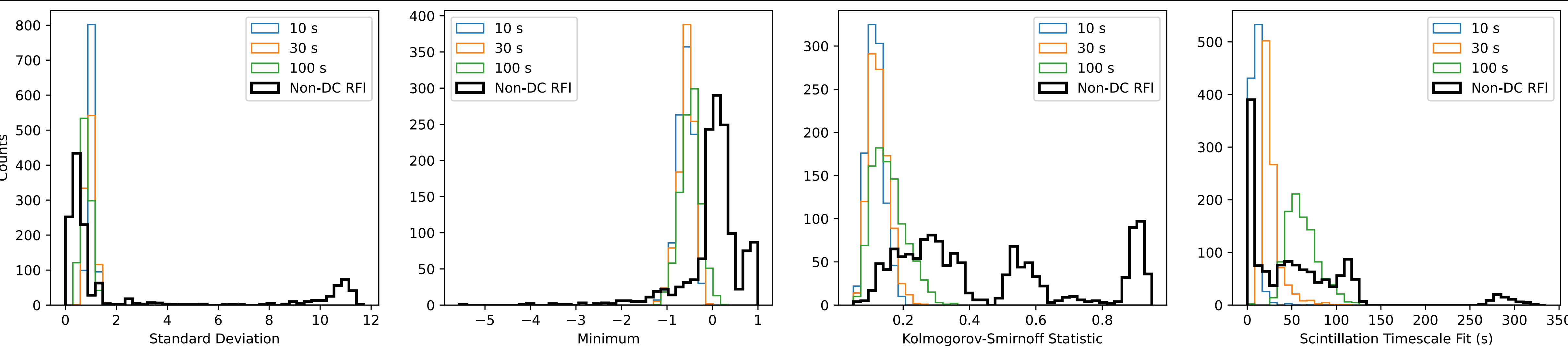
Minimum

KS Statistic

Timescale Fit

L band

S/N = 25



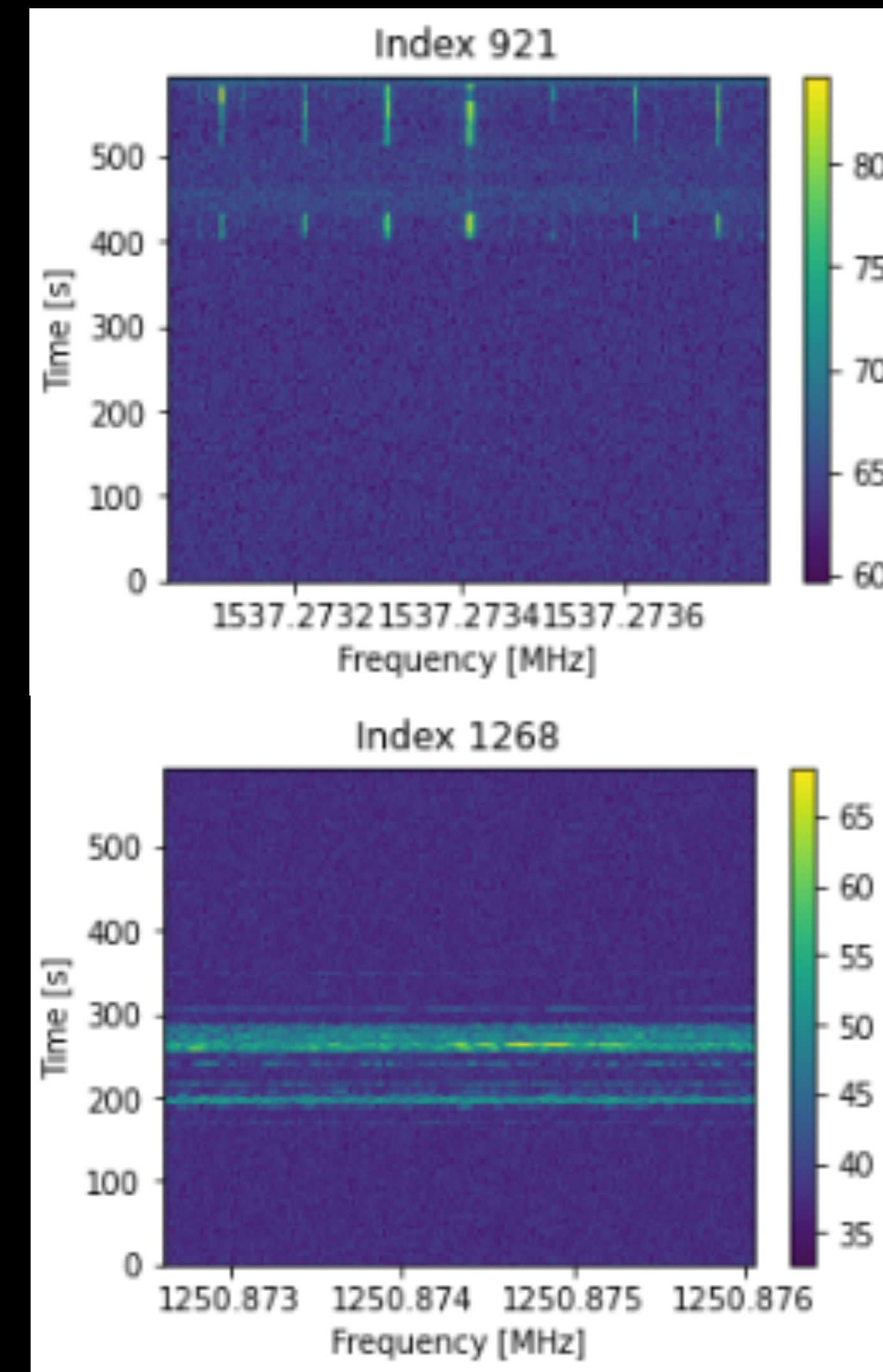
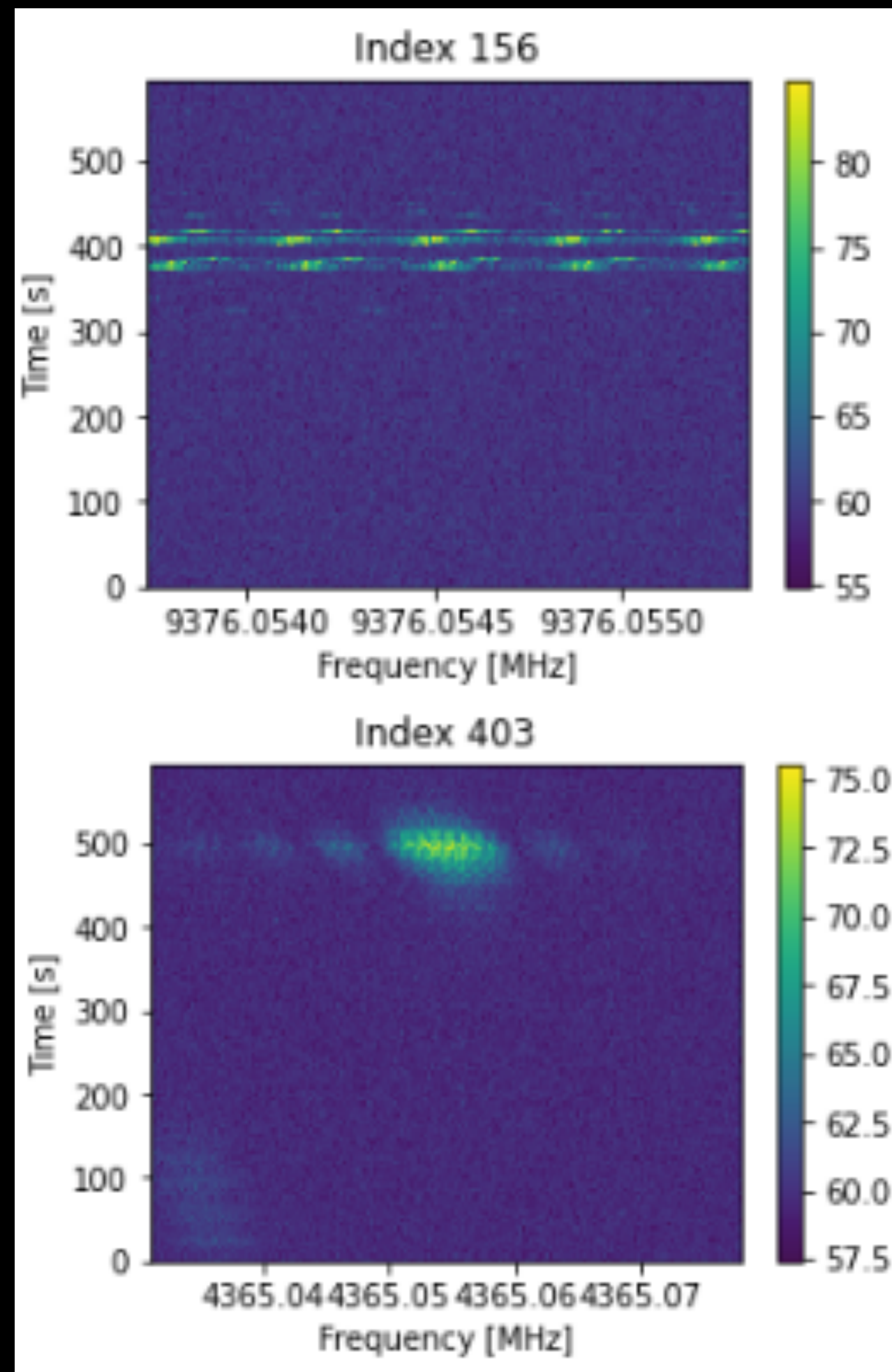
Std. Dev.

Minimum

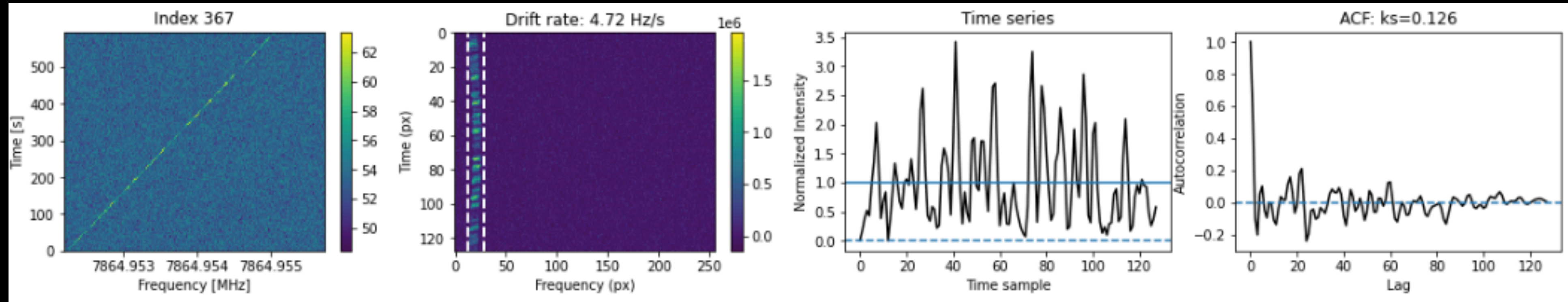
KS Statistic

Timescale Fit

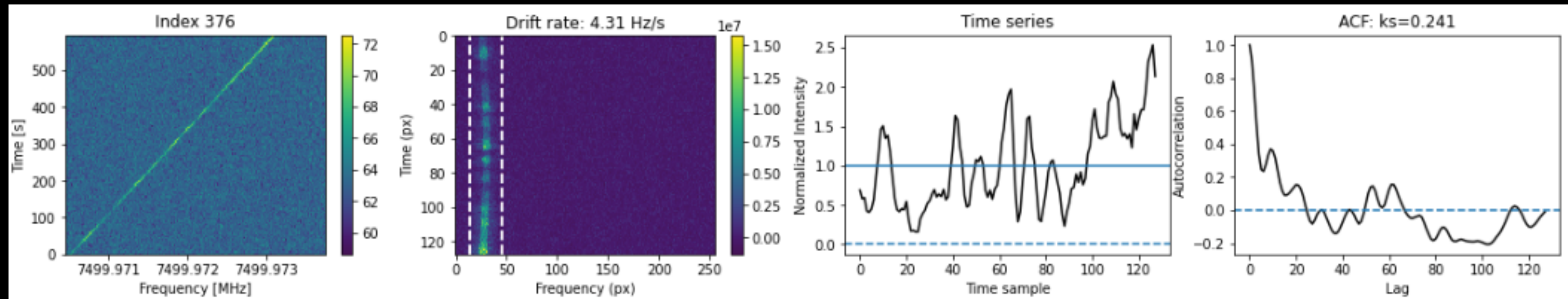
High standard deviation (RMS) signals are pulsed - or broadband



What signals pass the threshold?



Timescale fit ~ 2 s

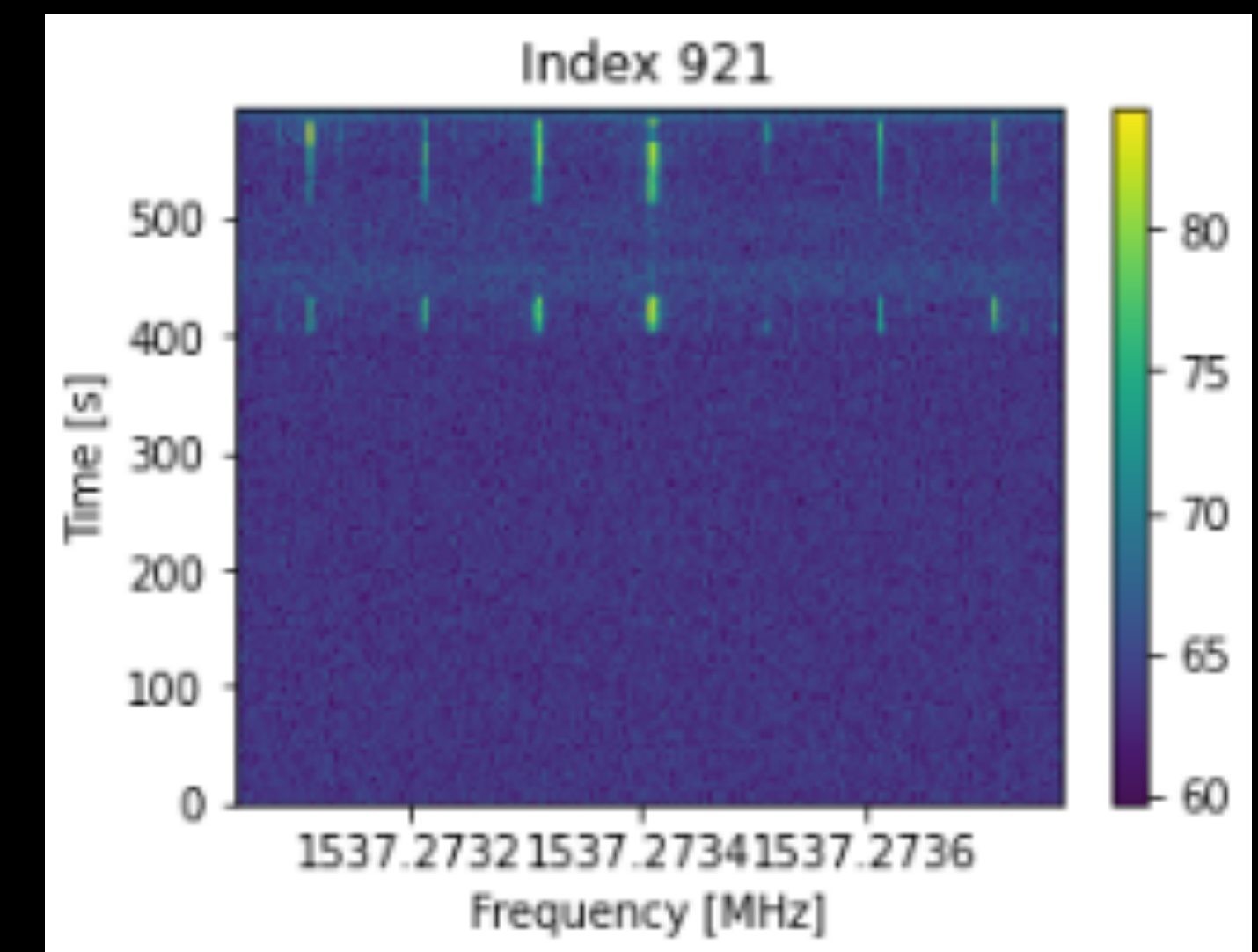
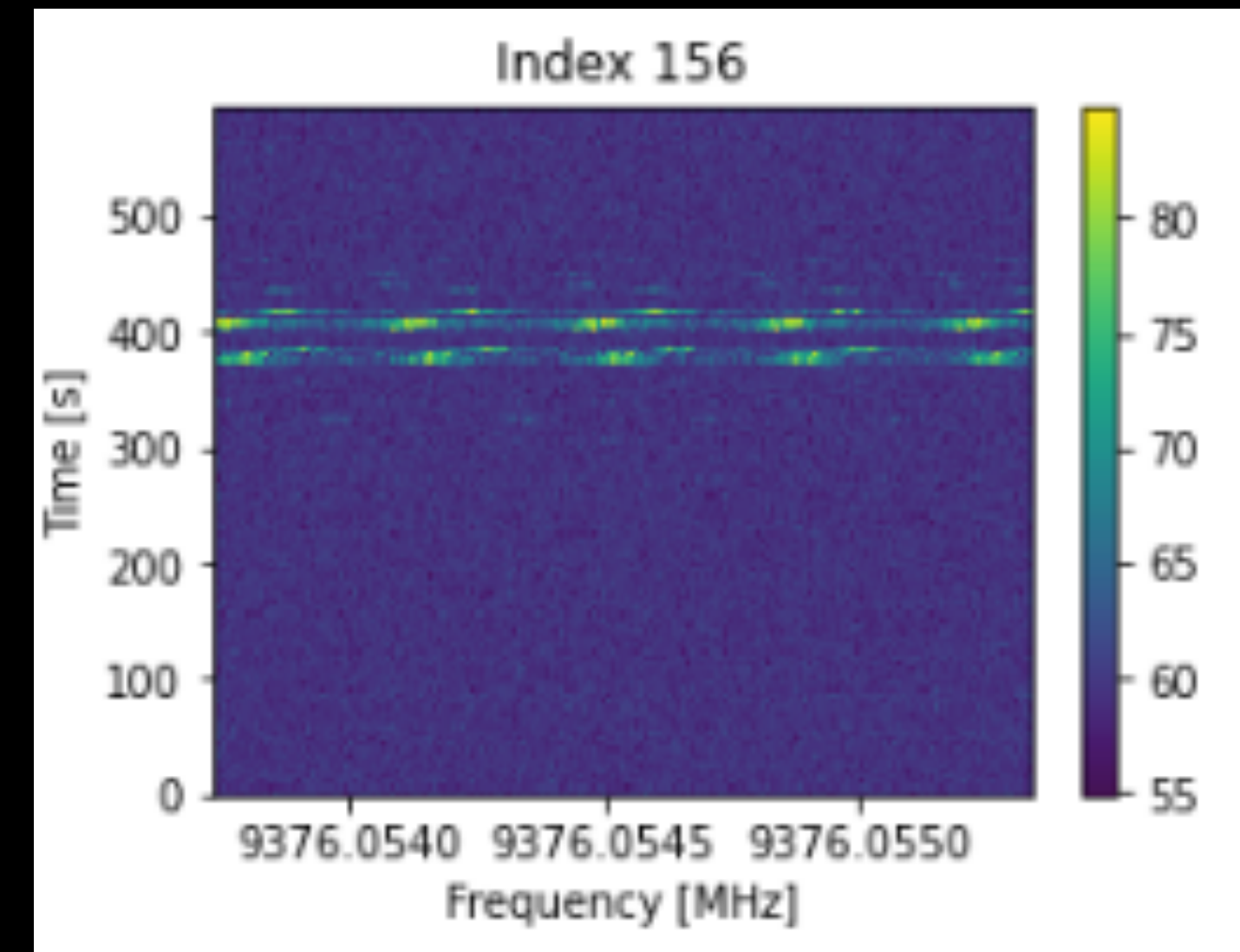


Timescale fit ~ 60 s

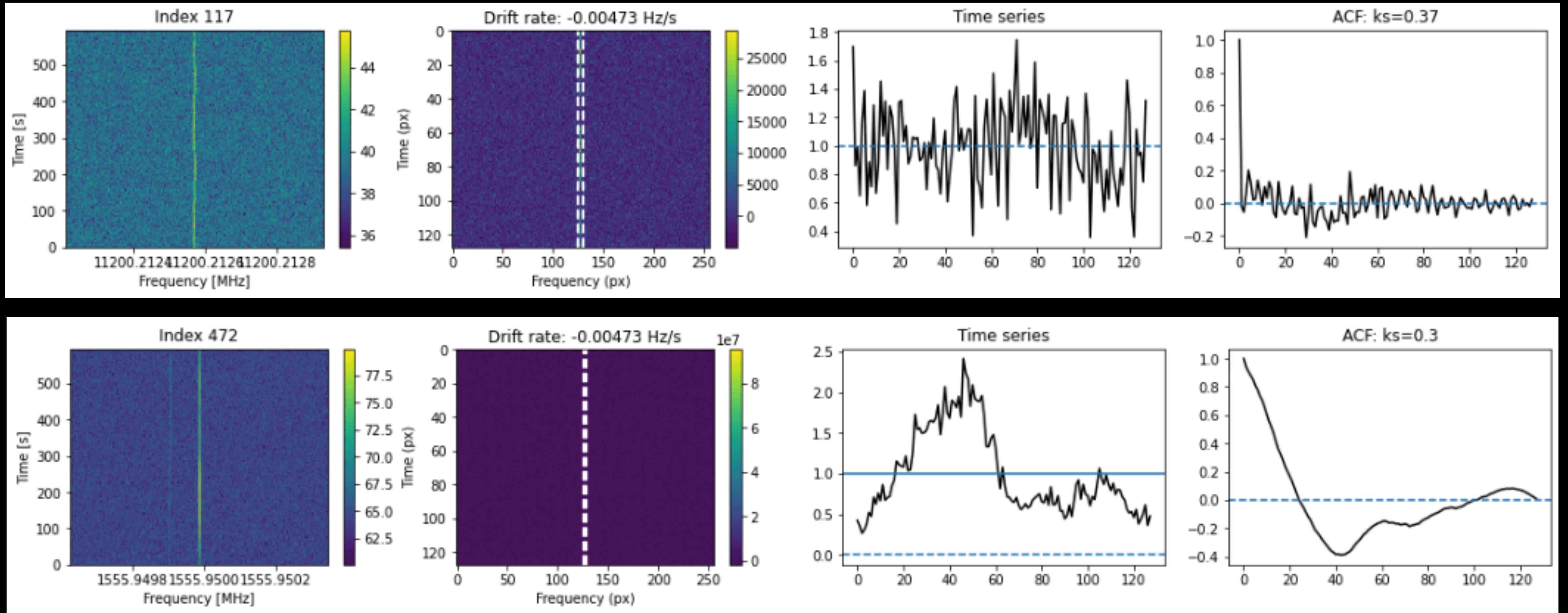
- At C-band, $S/N > 25$, 3 out of 1102

Limitations from RFI analysis?

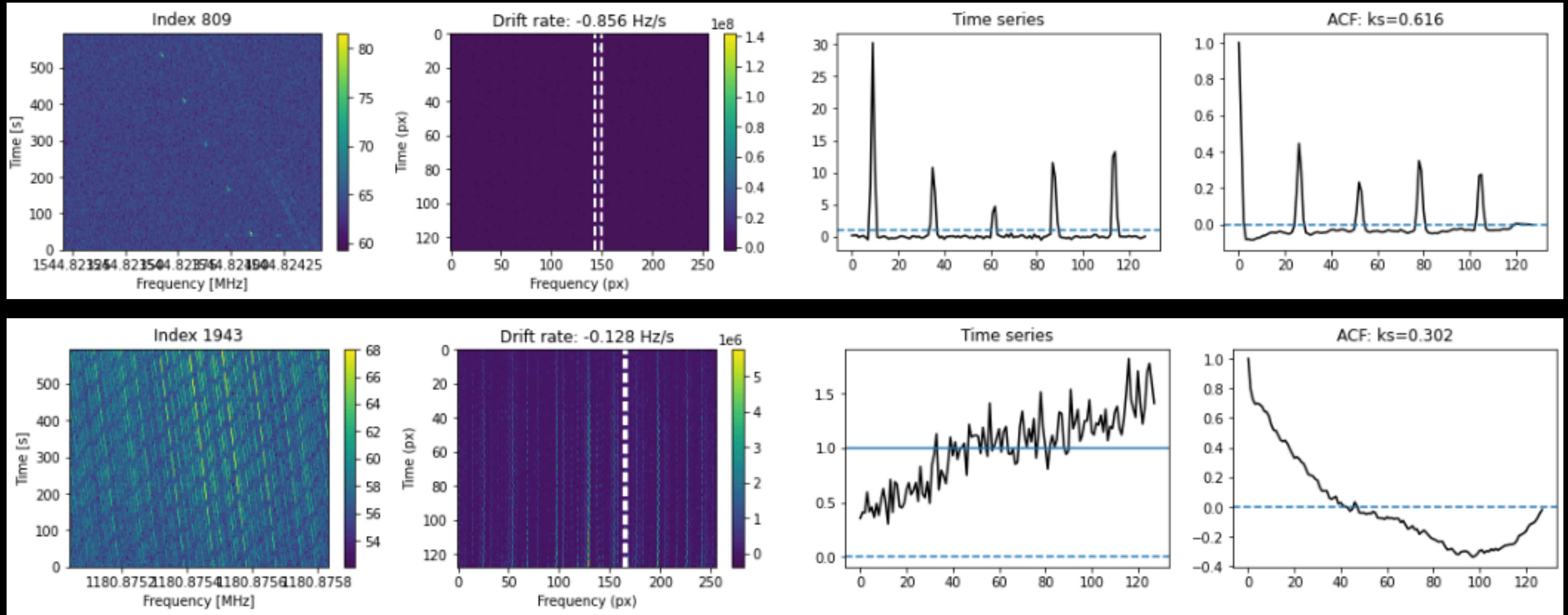
- L and S bands in particular are very noisy
- Non-narrowband signals detected just because they are above the SNR threshold
- Difficult to apply a one-size-fits-all bounding box method
- Perhaps ML can help!



Some examples



Some more examples



Examples of diagnostic statistics

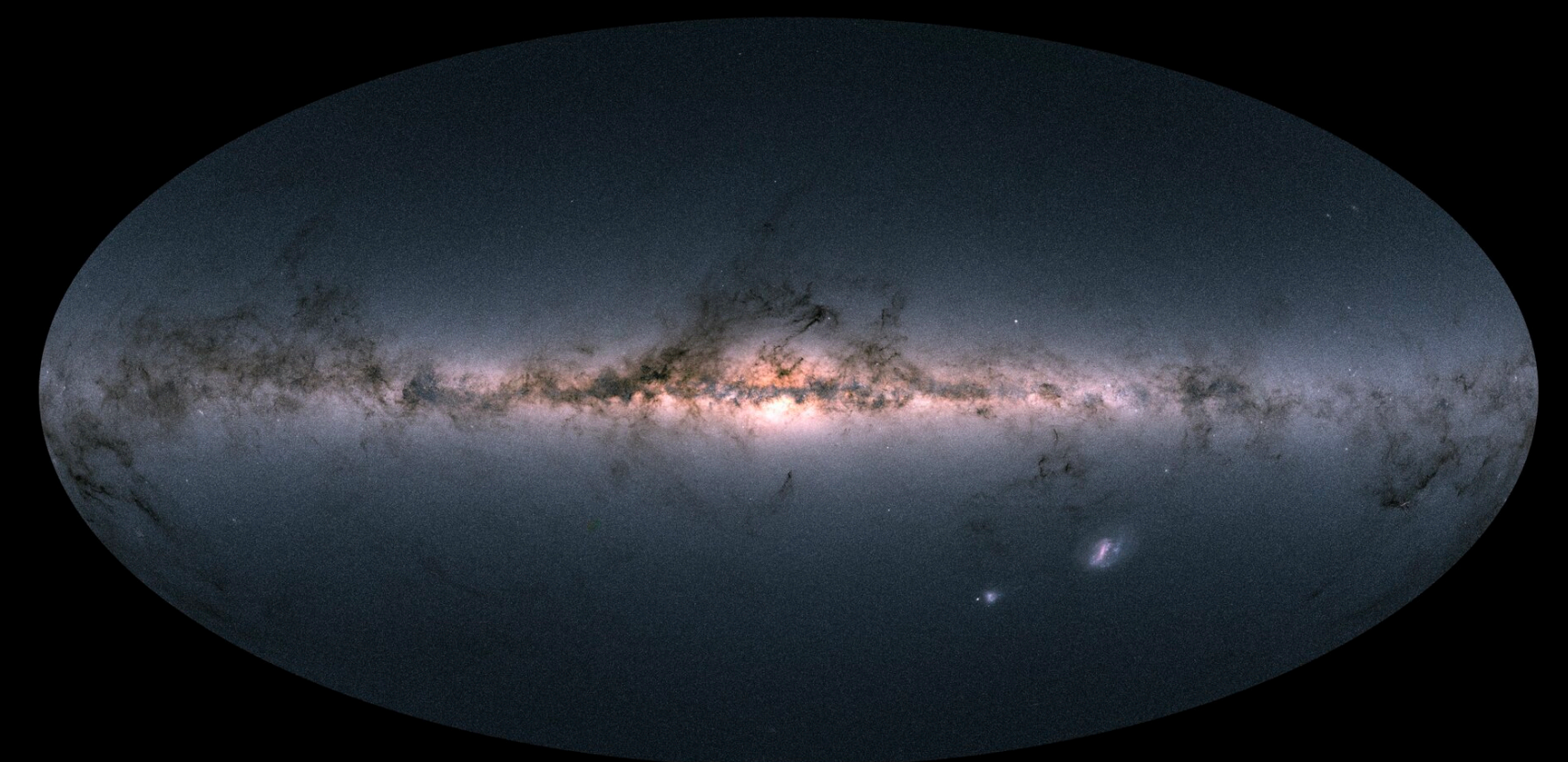
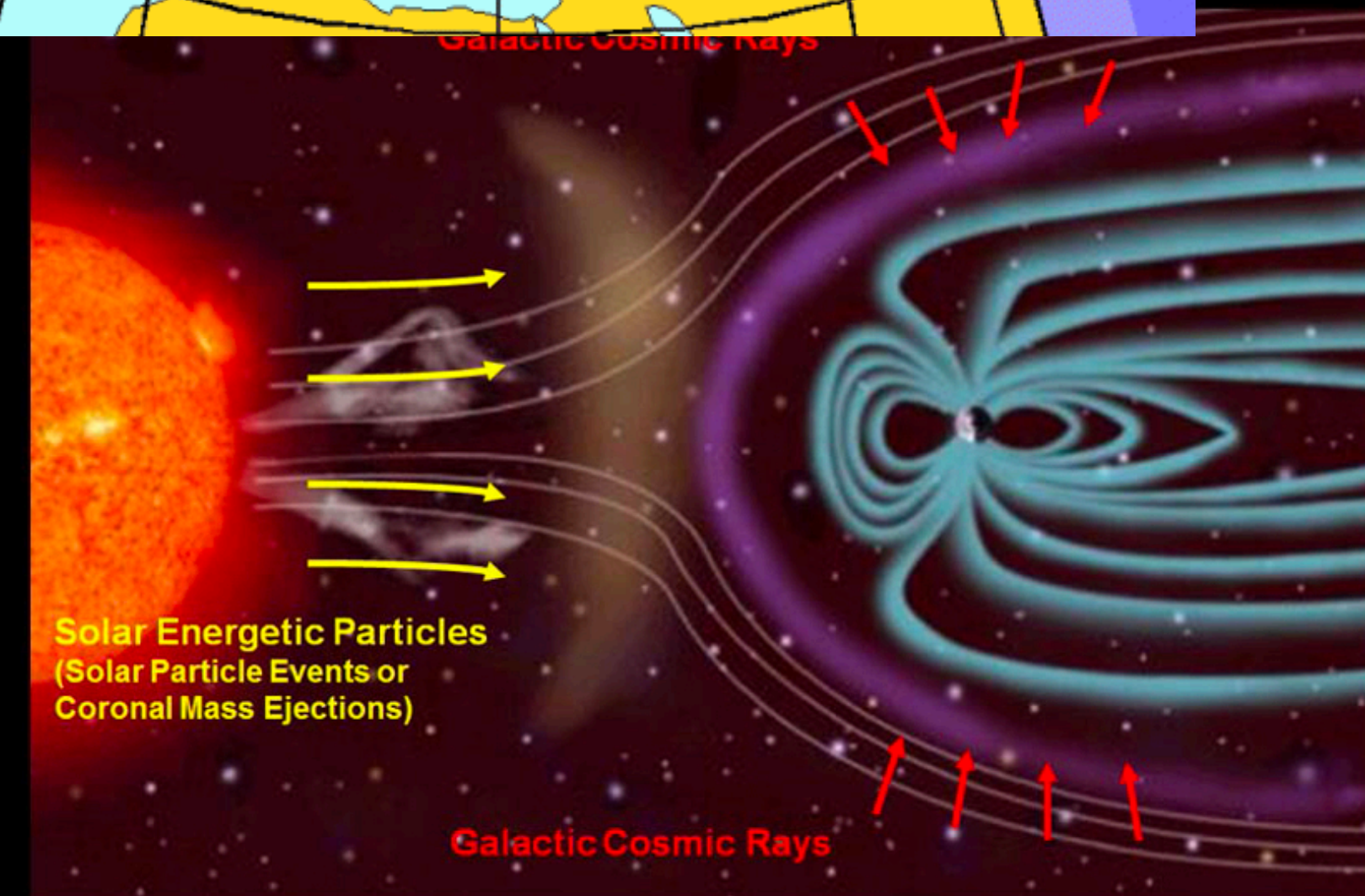
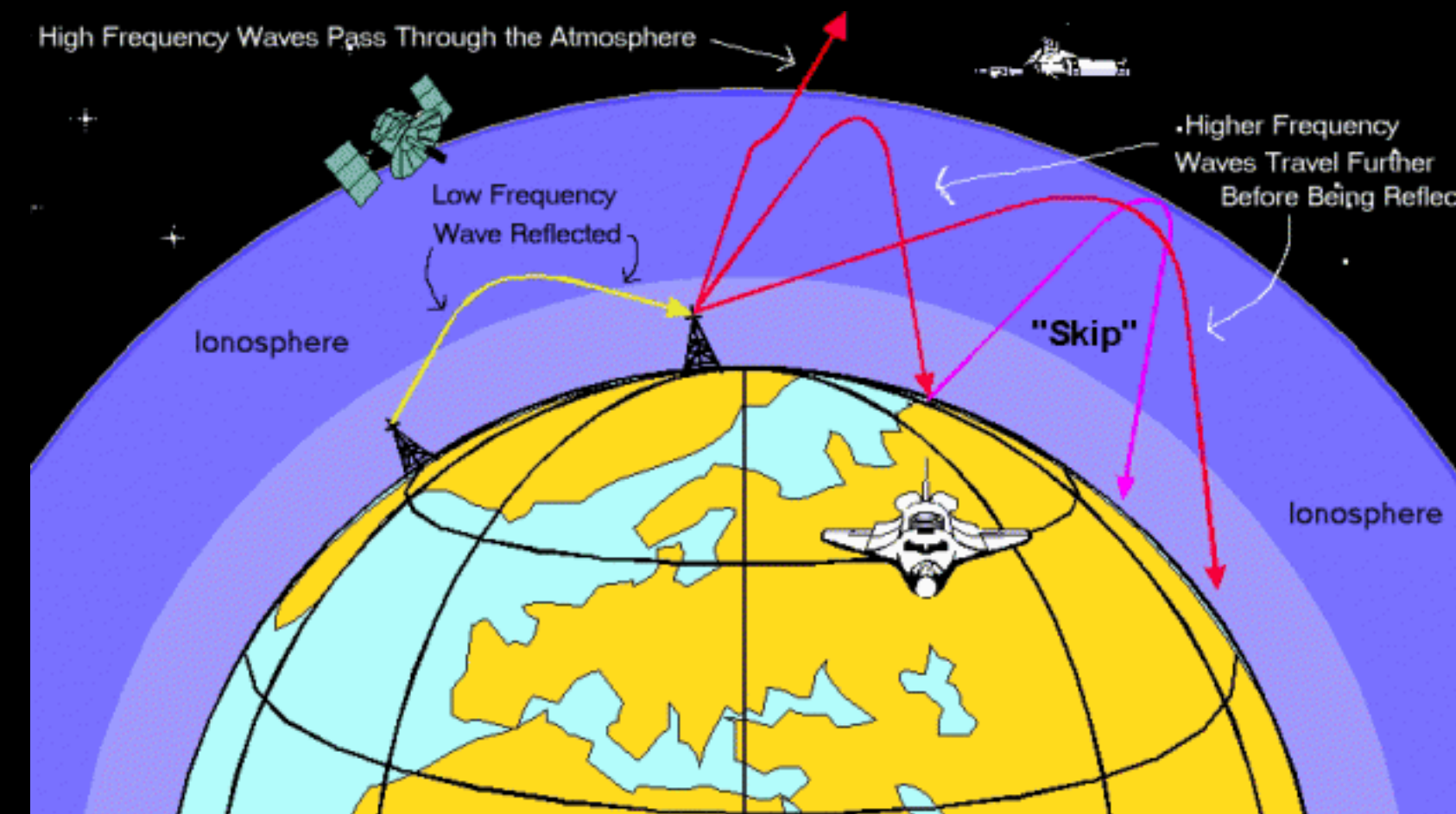
Statistic	Asymptotic Value (with no noise)	Target Distribution
Standard Deviation (RMS)	1	Intensity, exponential
Minimum	0	Intensity, exponential
Kolmogorov-Smirnoff statistic	0	Intensity, exponential
Autocorrelation lag	Variable	Autocorrelation, Gaussian
Least squares fit to autocorrelation	Variable	Autocorrelation, Gaussian

There are a number of constraints...

- Time resolution
 - Sufficiently resolve scintles
- Integration time
 - Collect enough scintles, gain stability
- Signal brightness
 - Compute accurate statistics embedded in noise
- RFI environment
 - Bad normalization, false narrowband detections, confounding modulation

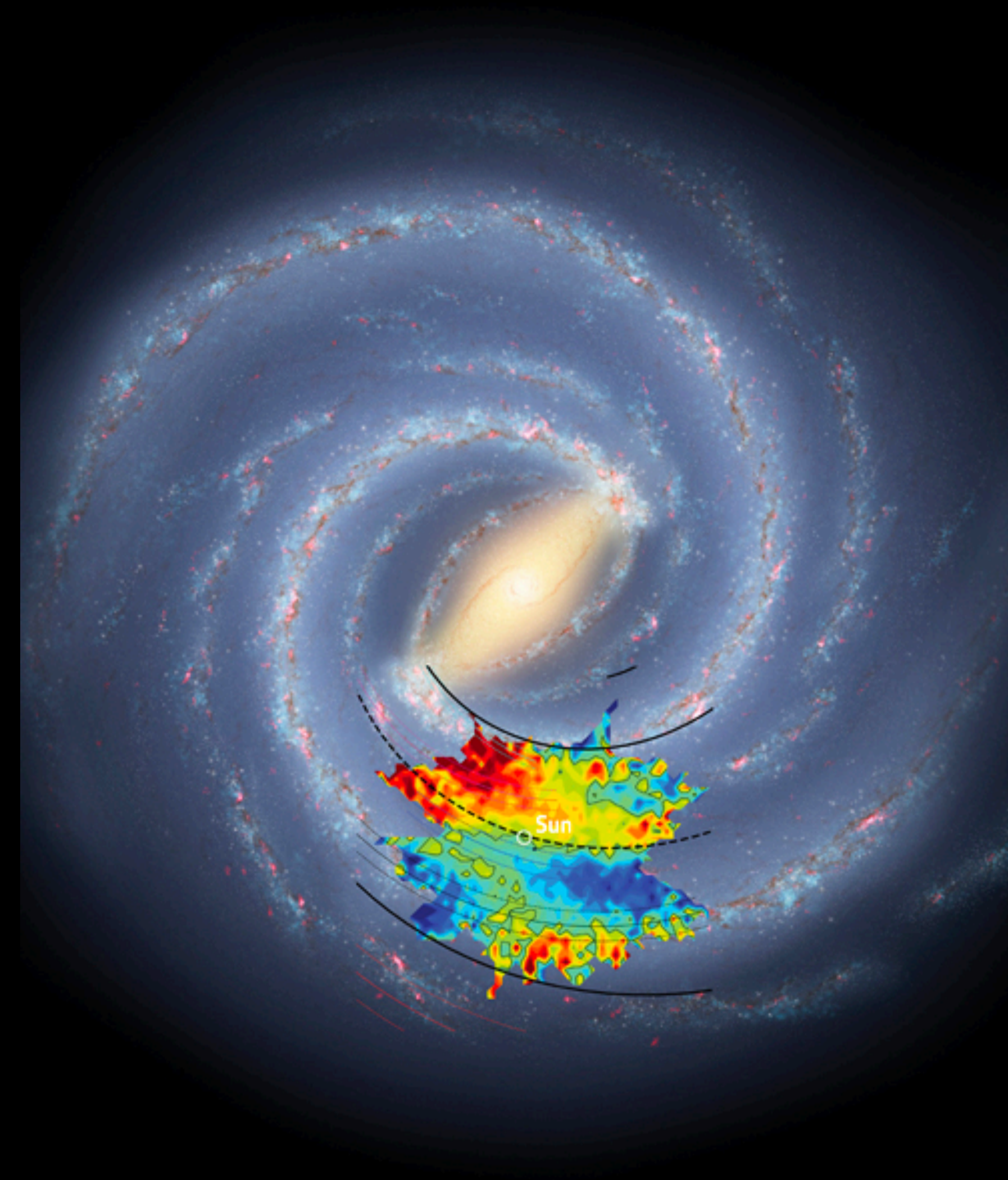
Regions of ionized plasma

- Ionosphere
- Interplanetary Medium (IPM)
- Interstellar Medium (ISM)



Next steps: a Galactic Center / Galactic Plane survey

- Target most promising sections of parameter space
- Survey of Galactic plane with interesting targets
- Gaia DR3?



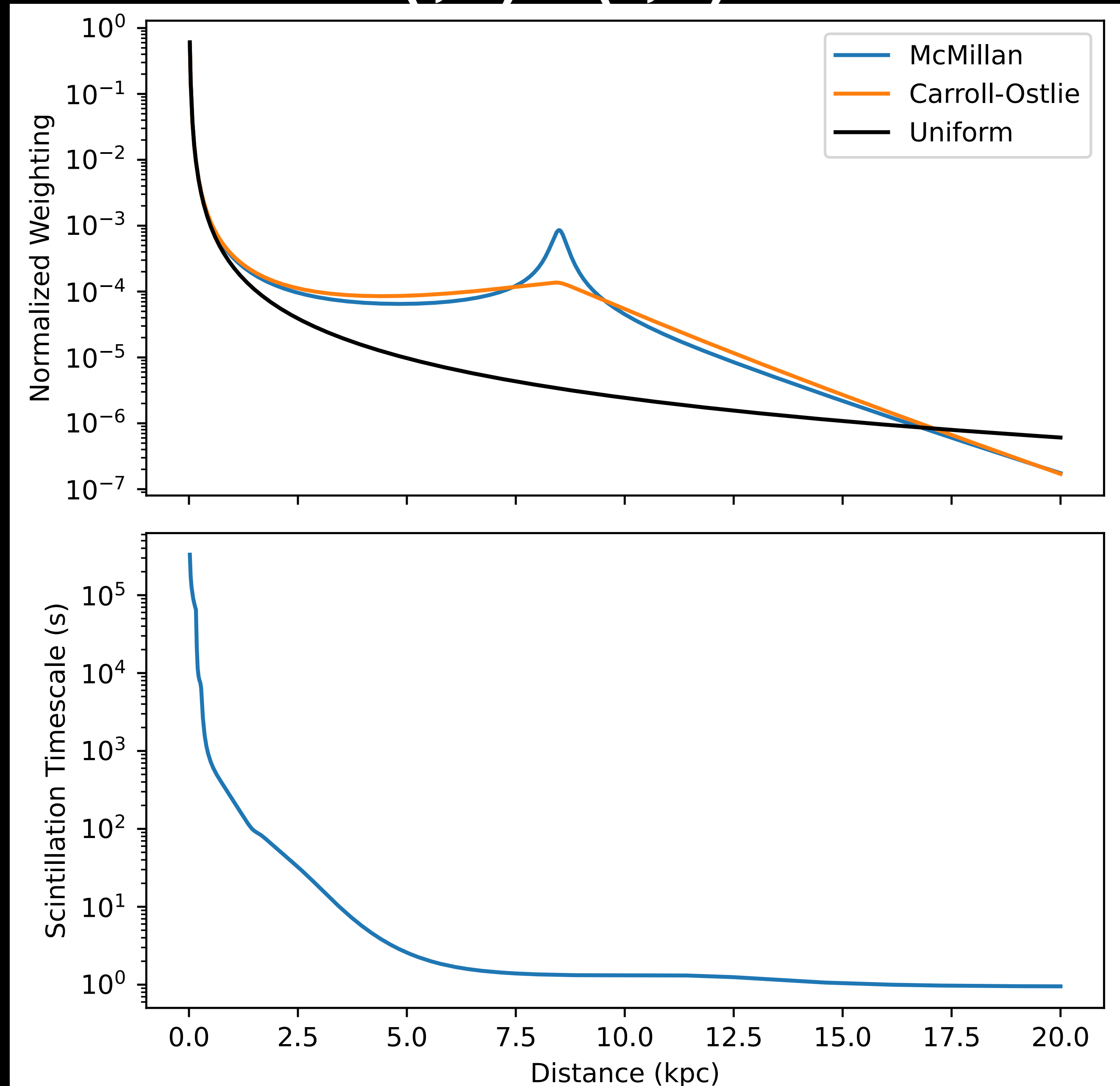
Gaia

Density-based sampling

Modulating by the inverse square-law for detectability:

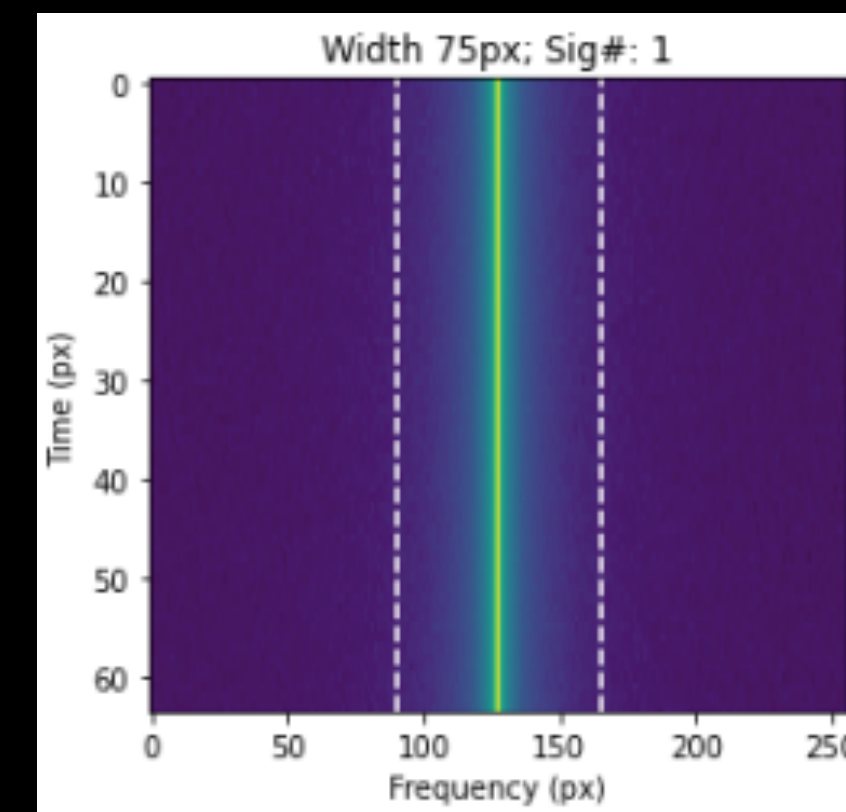
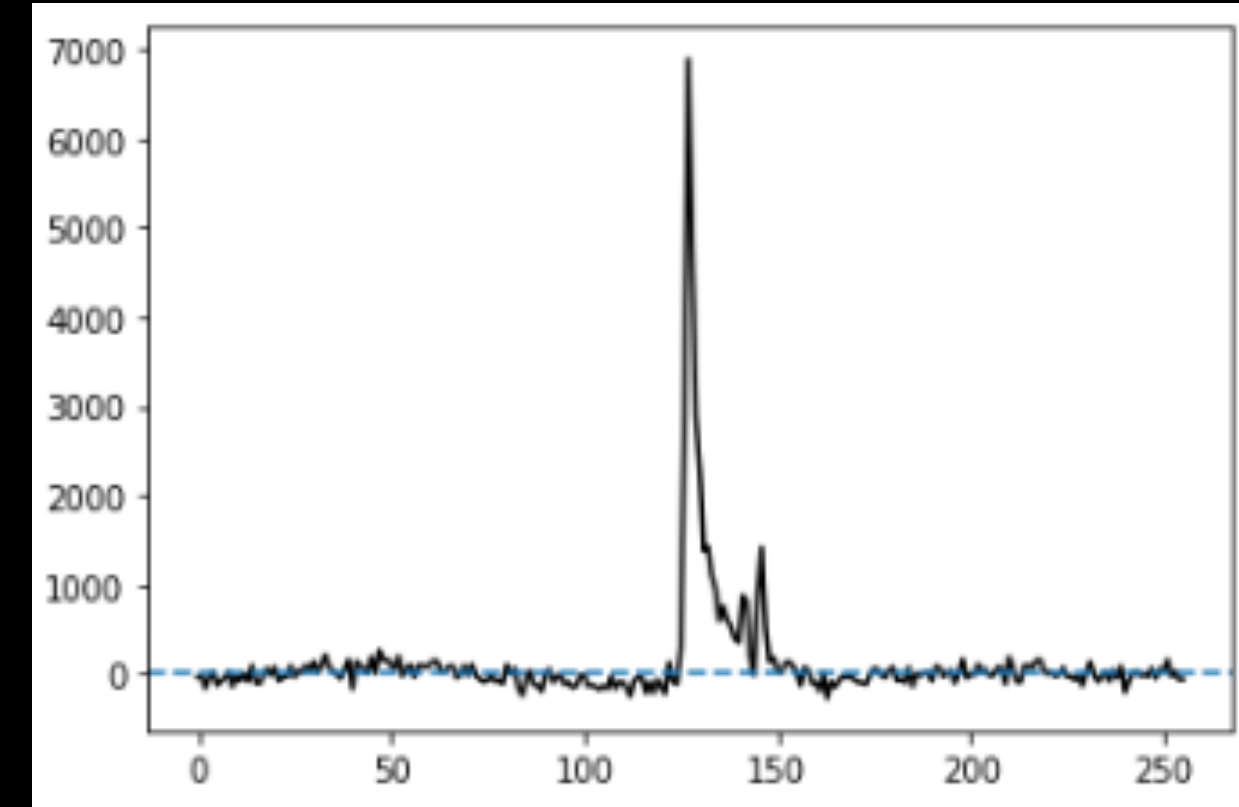
Depends on the assumptions made about transmission power and resources.

$$(l, b) = (1, 0)$$

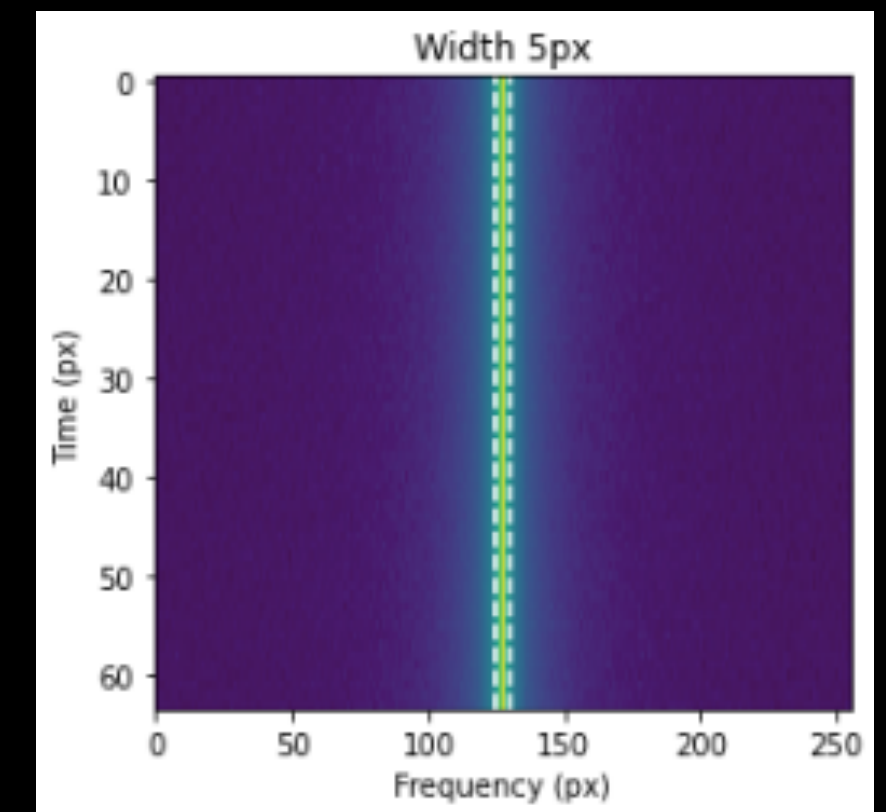


Selecting bounding boxes

- After experimentation with various methods, the final pipeline uses a combination of baseline fitting and peak detection to calculate the right size of frame to use
- The final bounds are created using a thresholding method, similar to PSRCHIVE
- Take the final bounded signal and integrate in the frequency direction to derive our raw time series — then we normalize to mean of 1 before calculating our scattering statistics



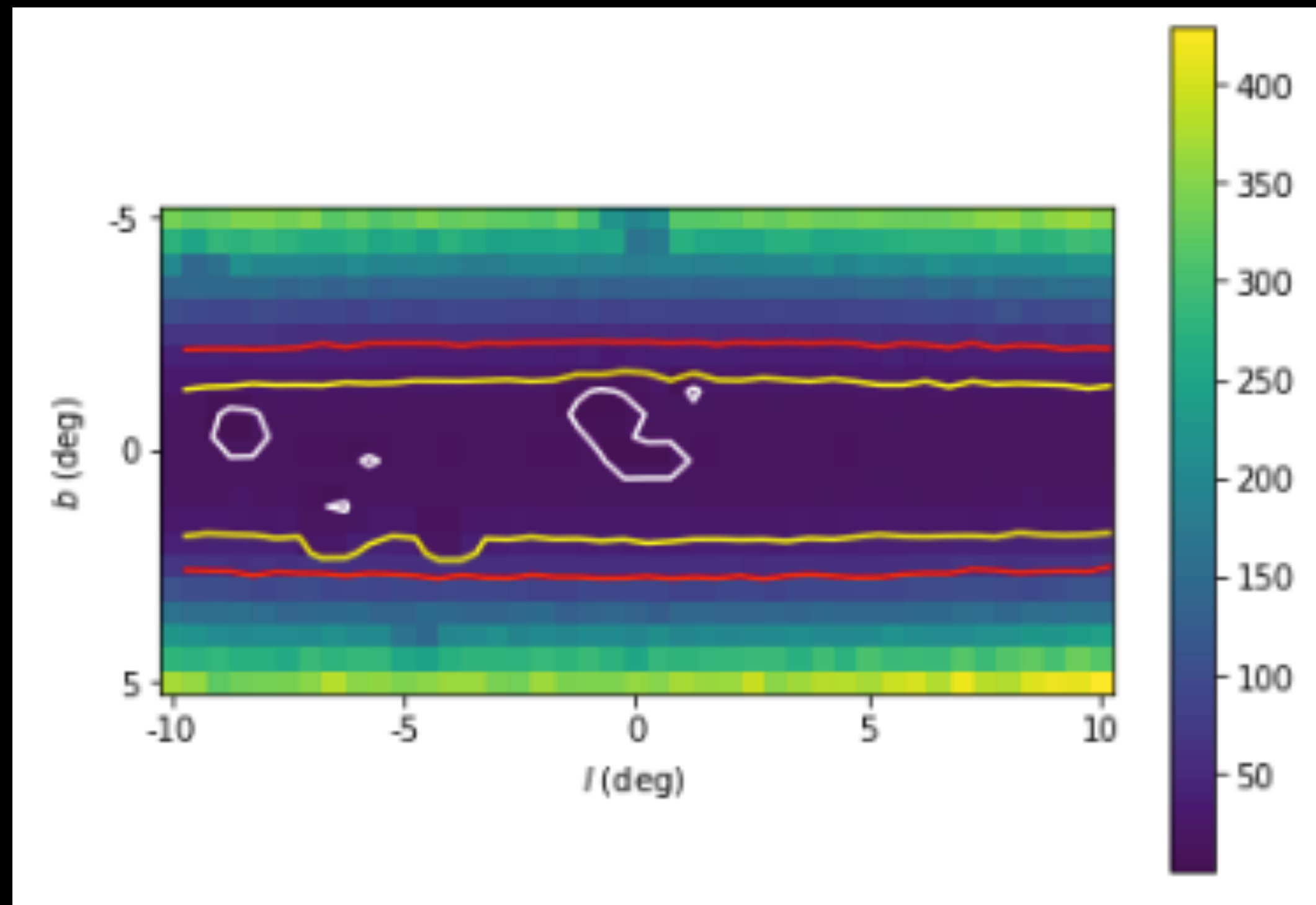
Polynomial fit



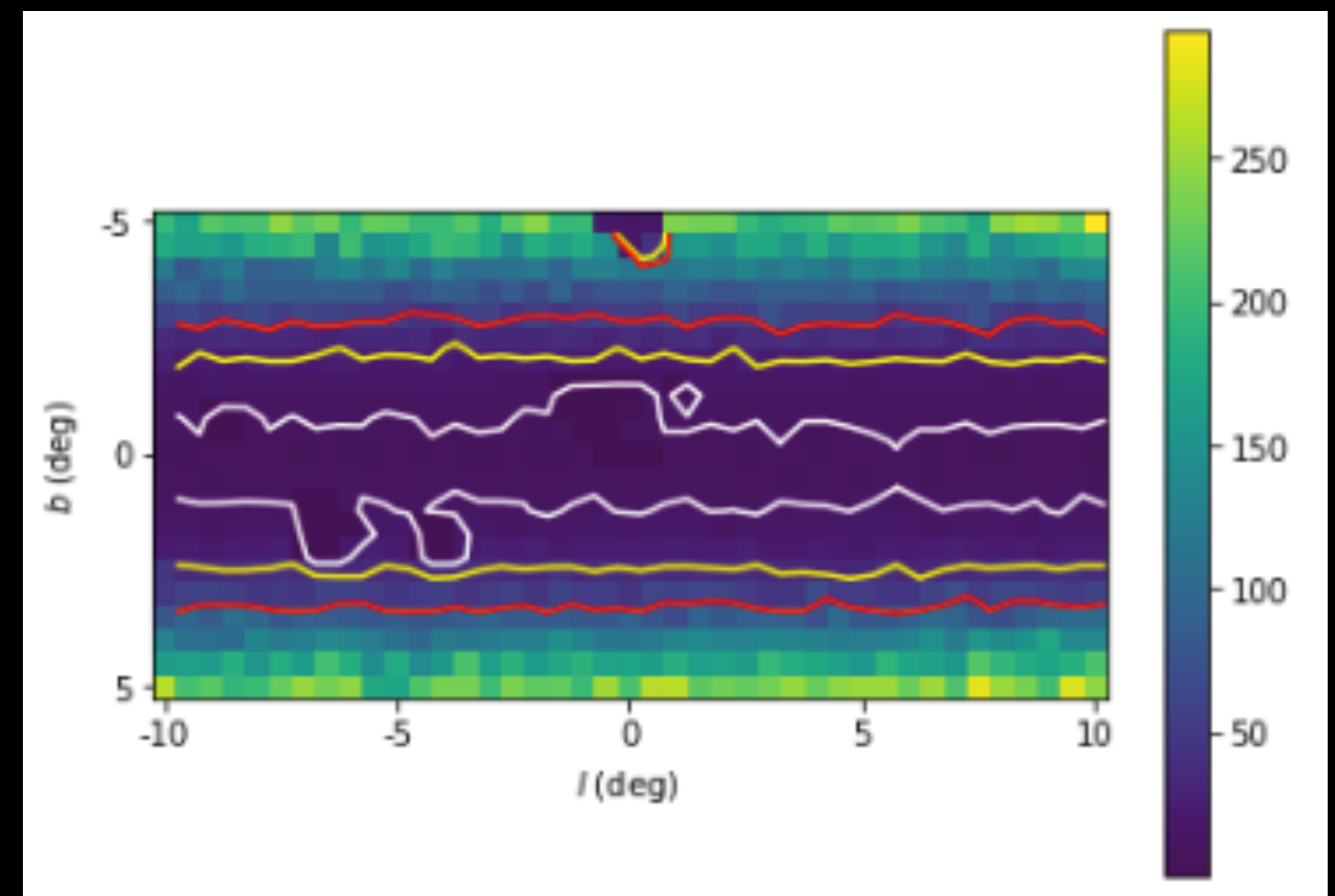
Threshold fit

Scintillation maps around the GC at C-band

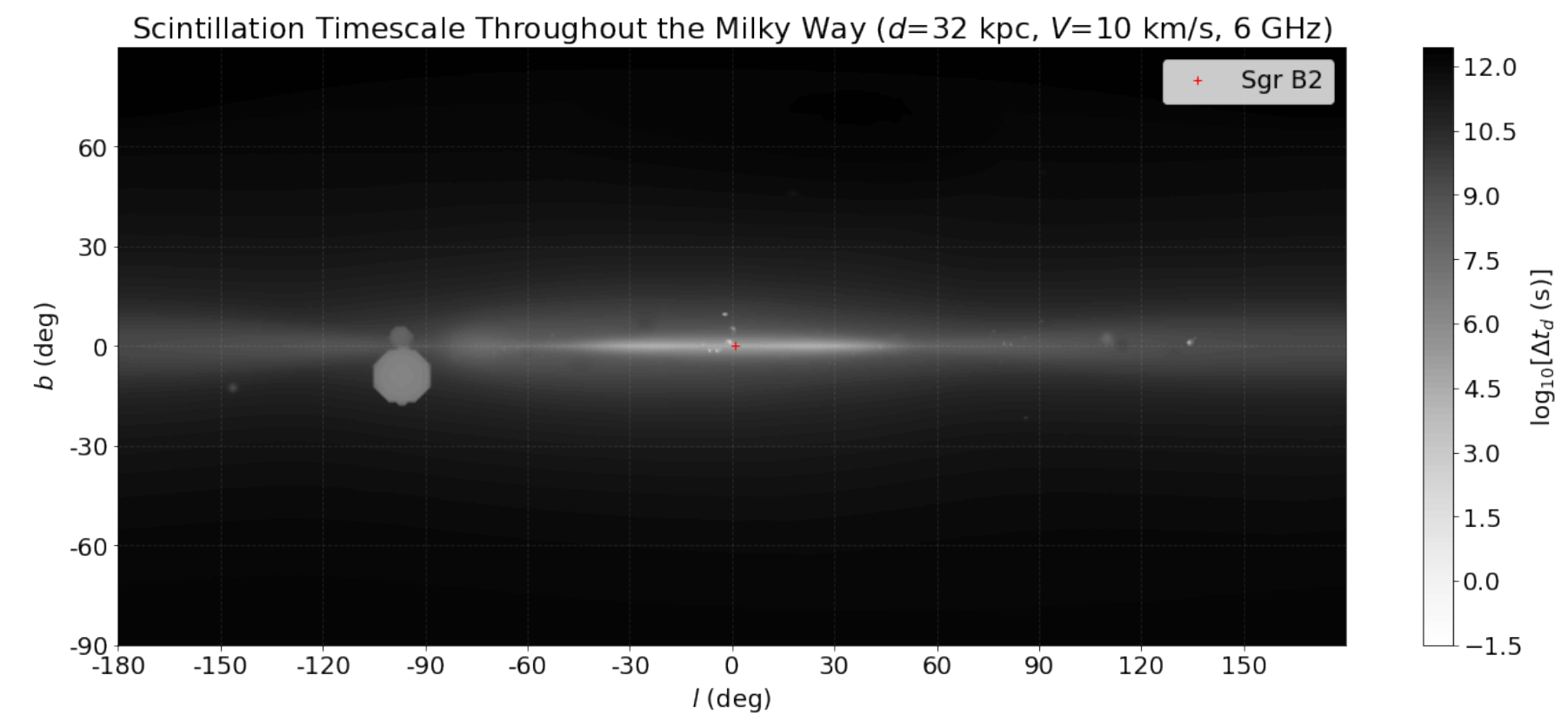
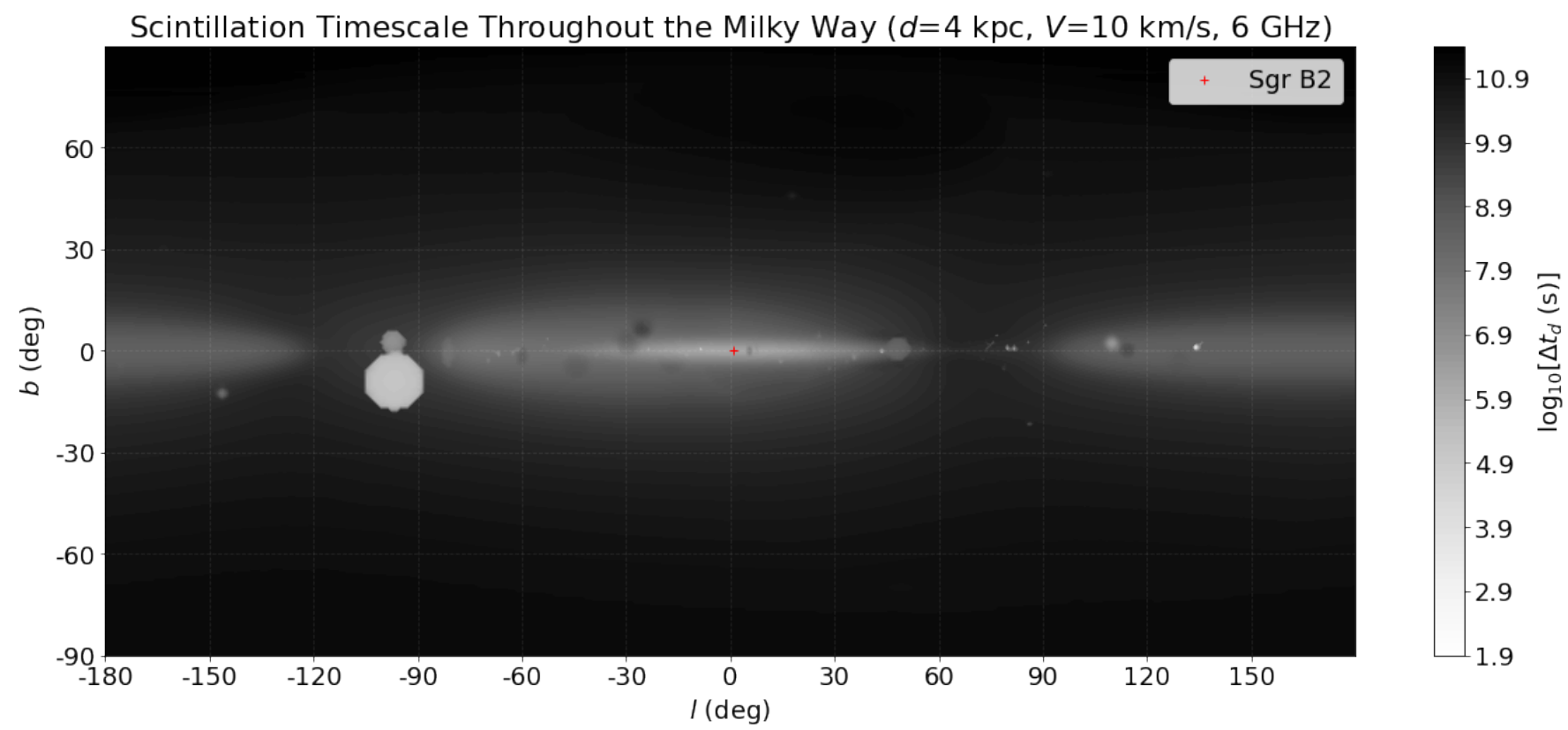
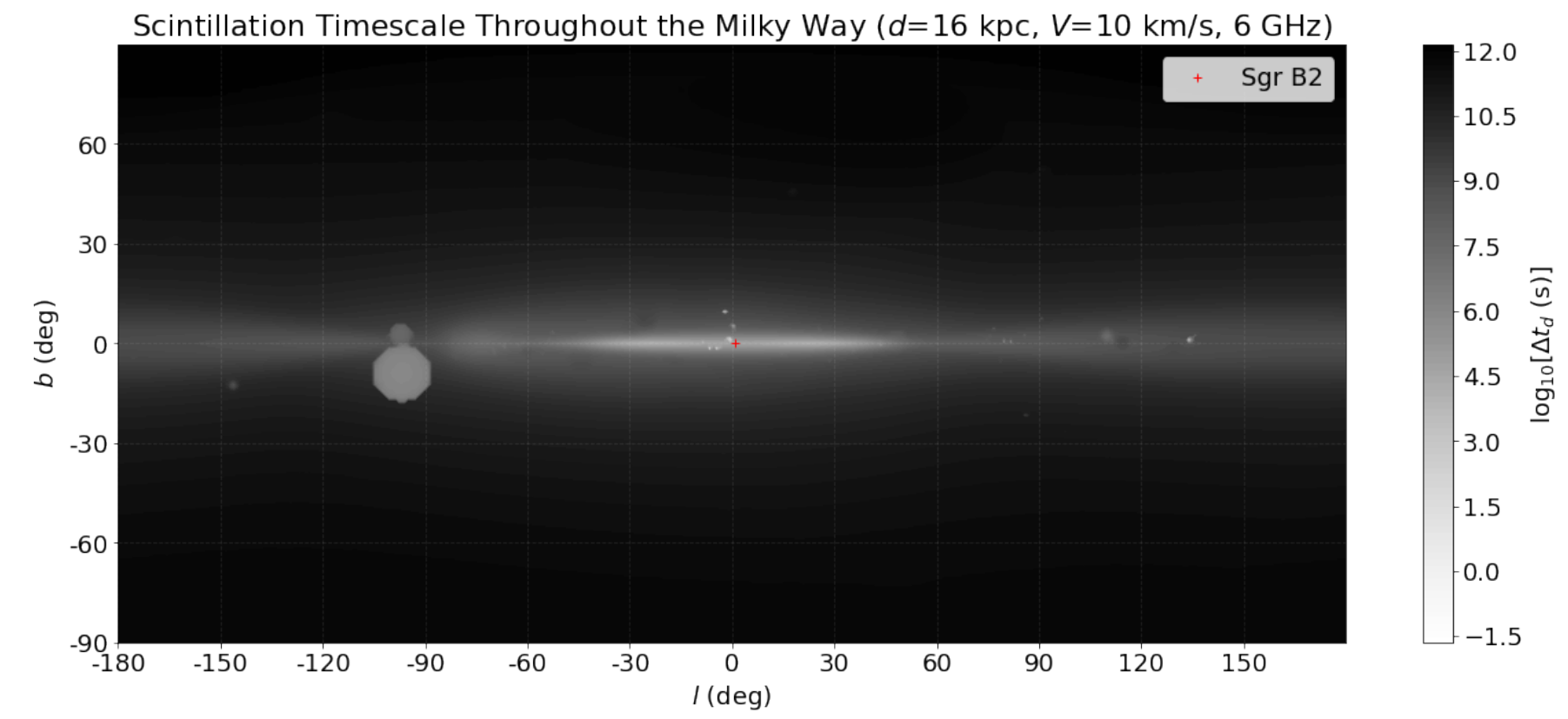
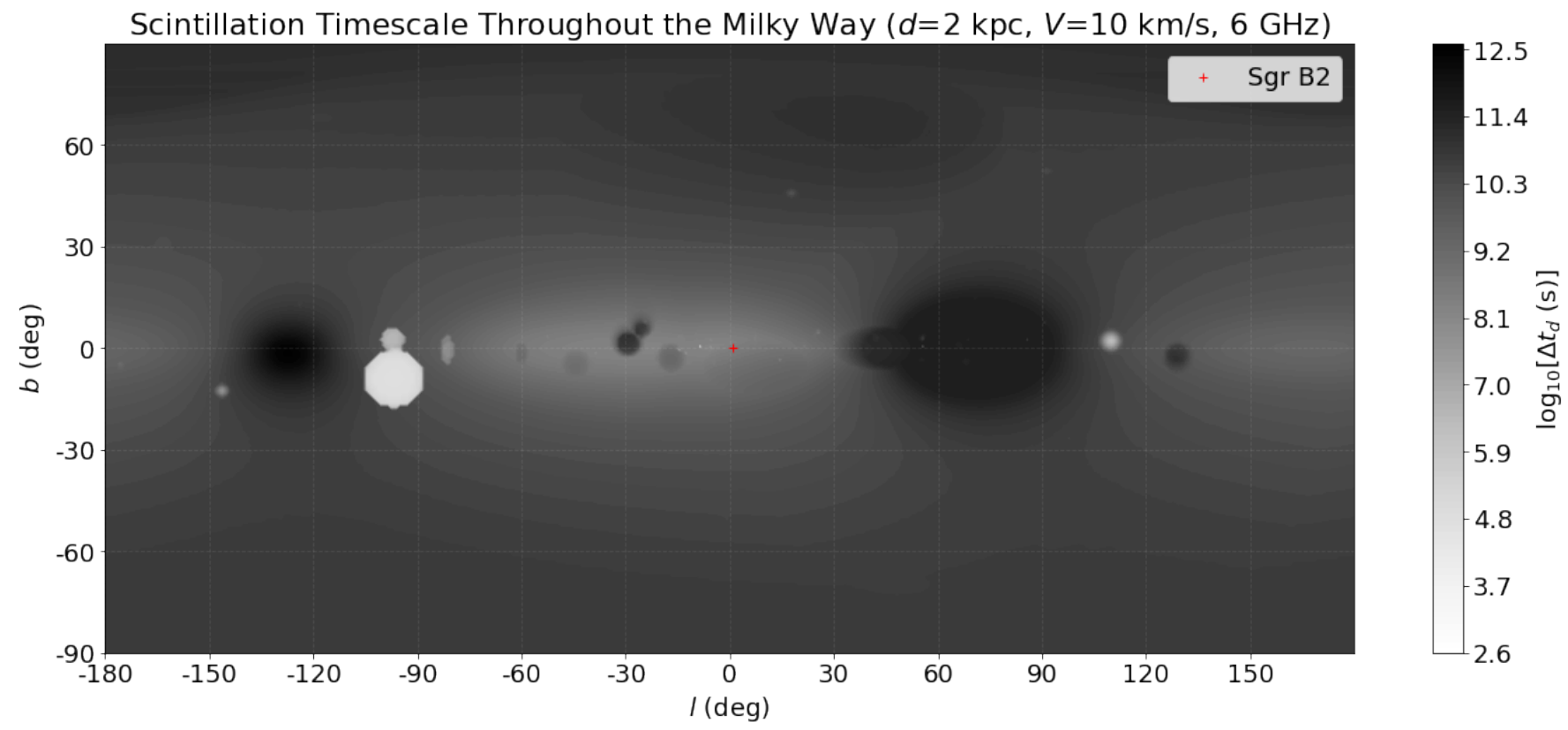
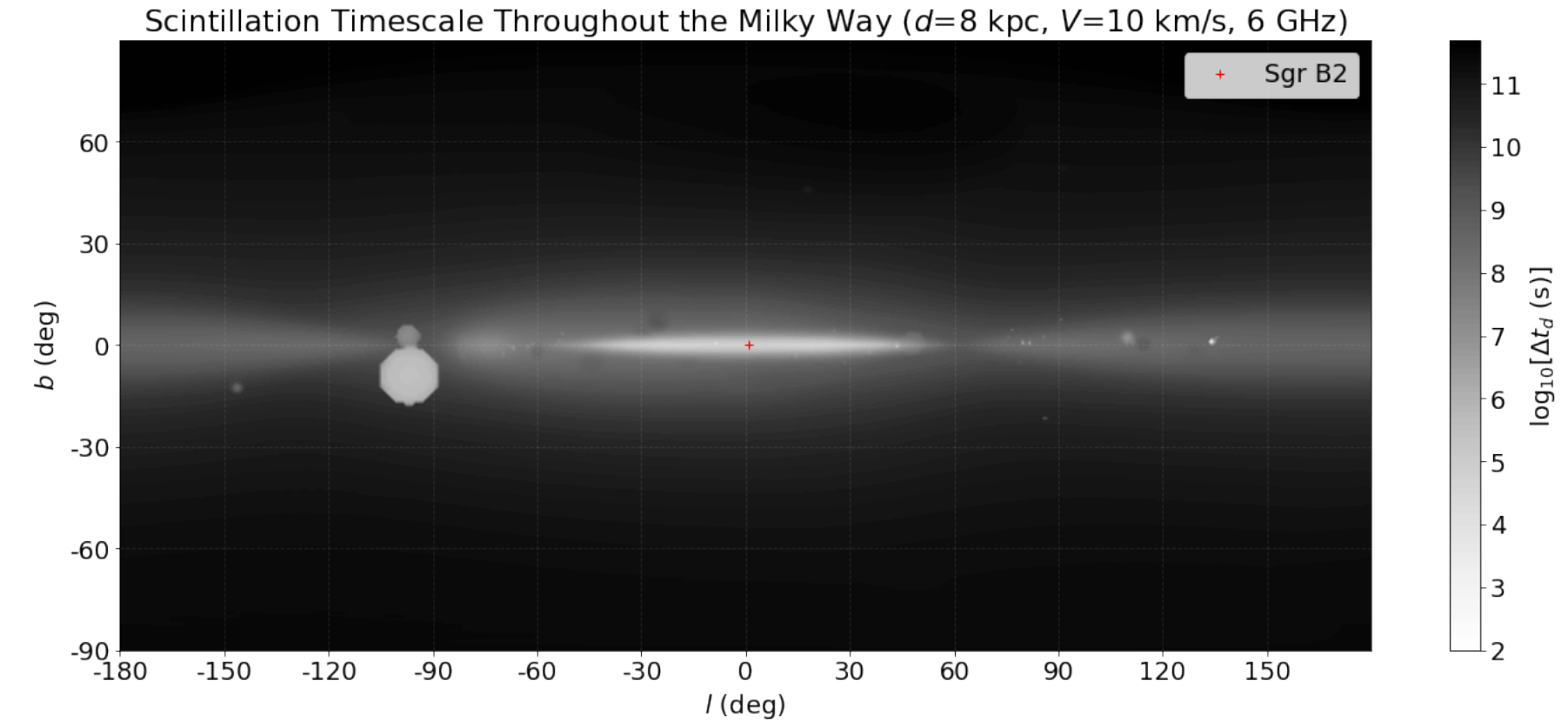
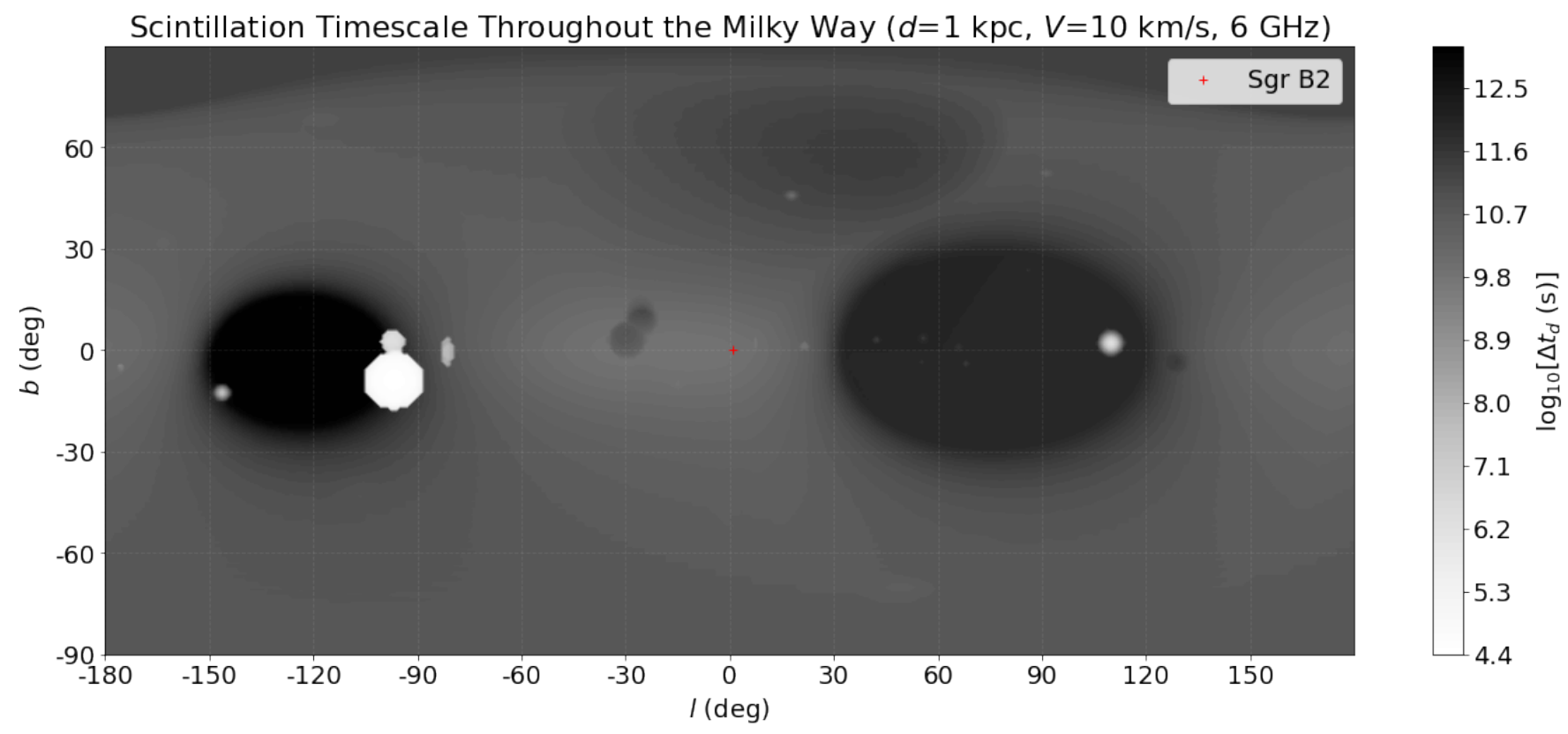
Median



Mode



10 s, 30 s, 60 s contours

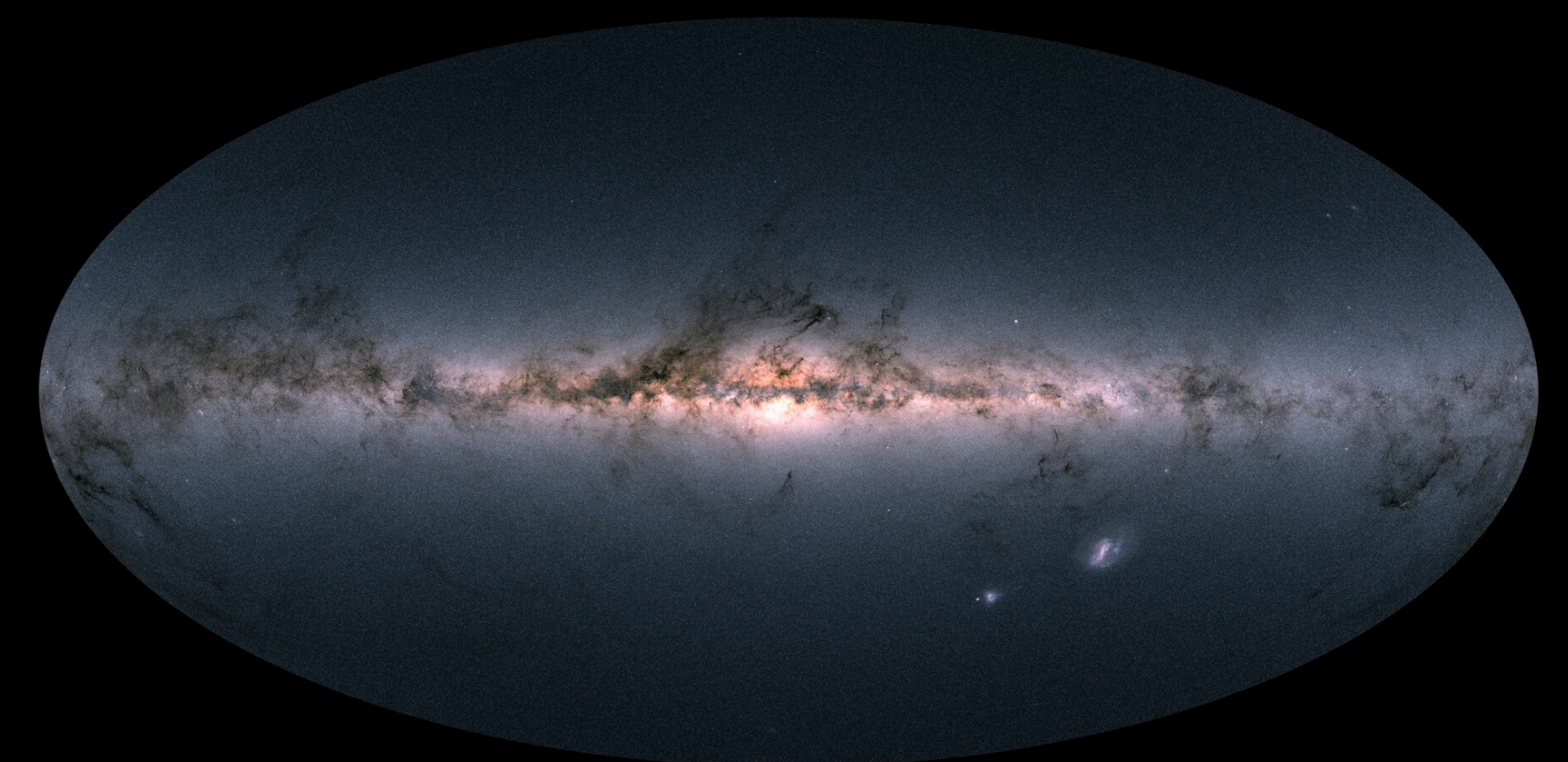
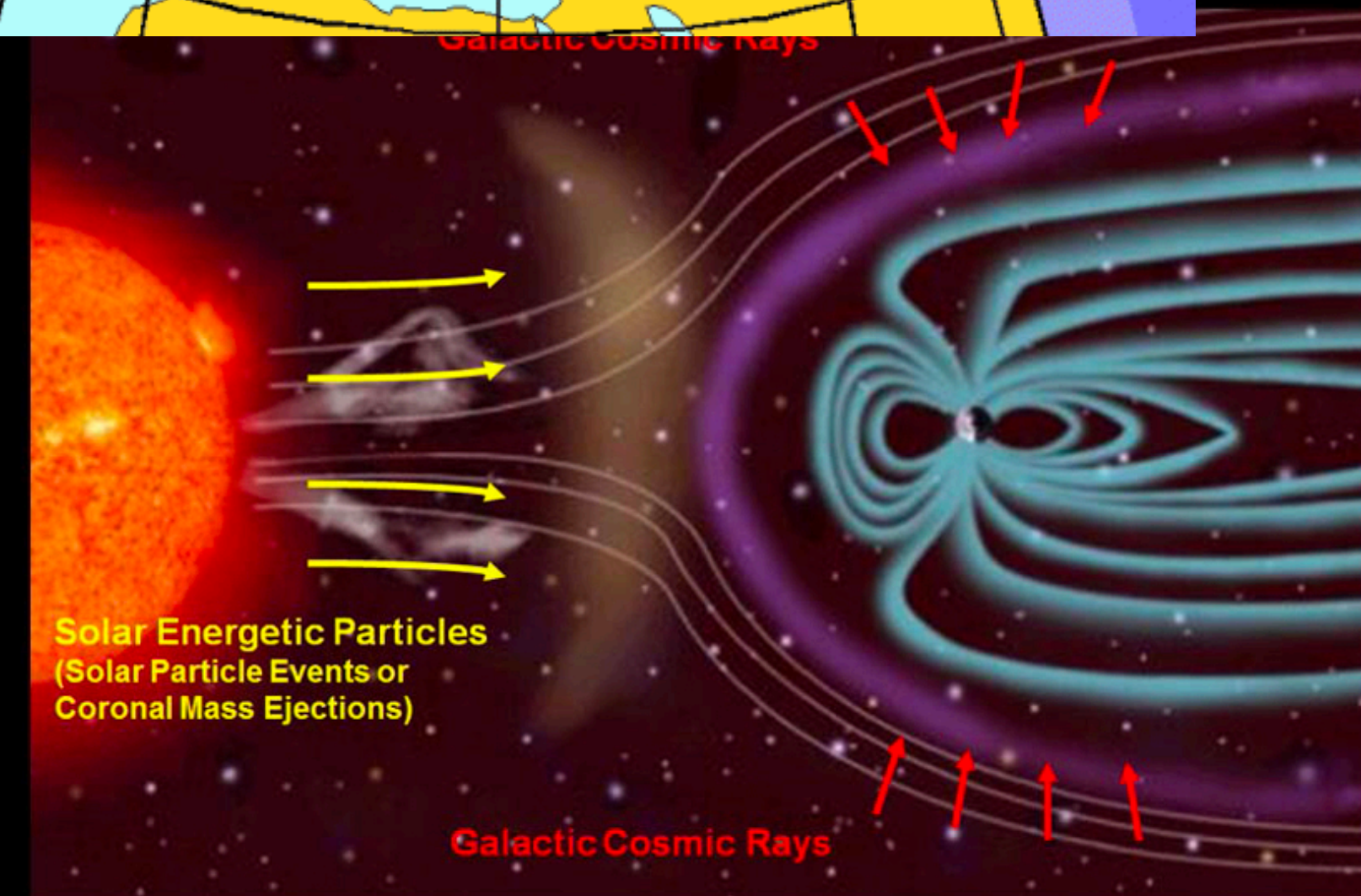
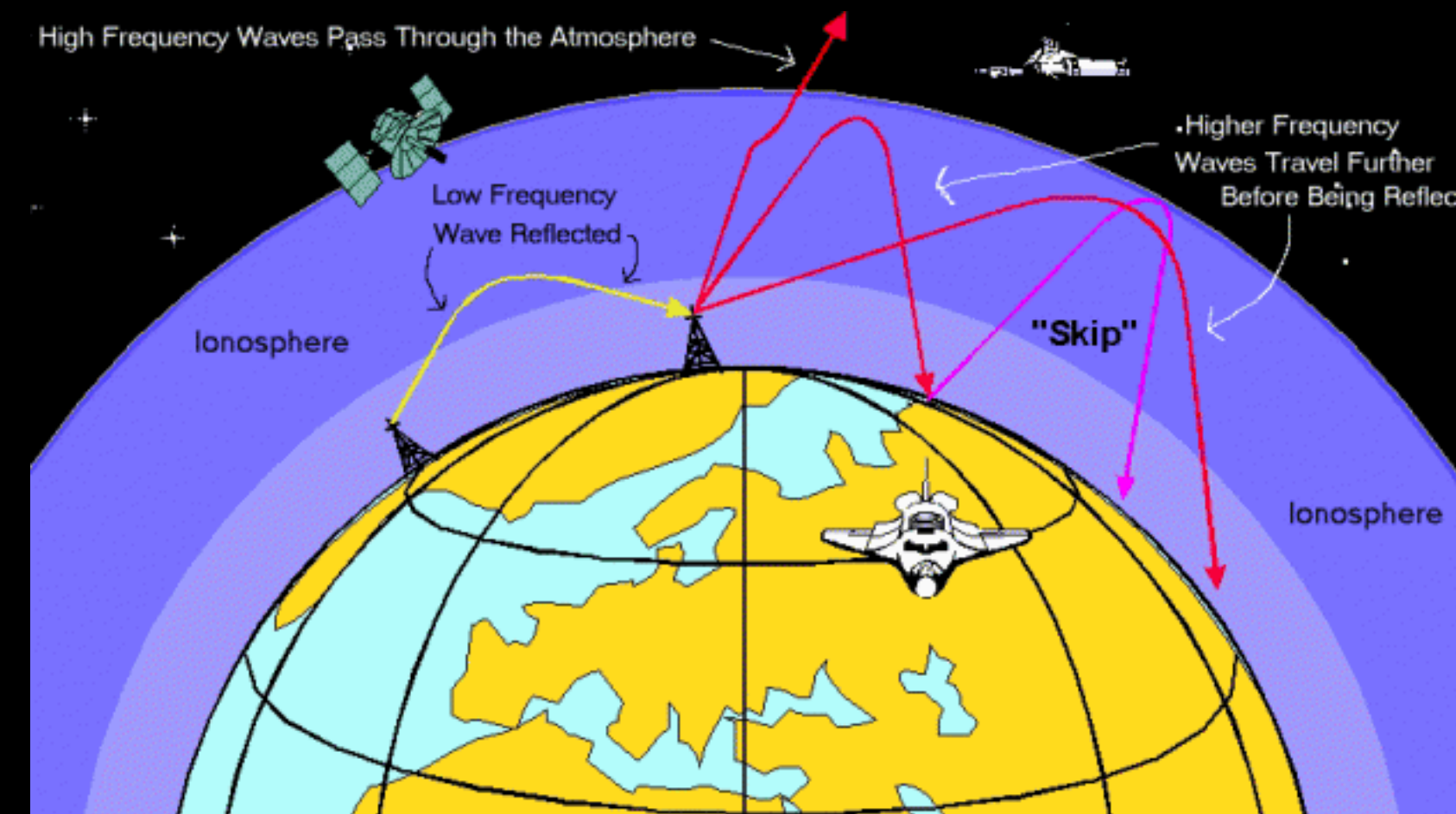


Scattering intensity

- Ionosphere — weak
- IPM — mostly weak
- ISM — can be strong!

$$m_d \ll 1$$

$$m_d \approx 1$$



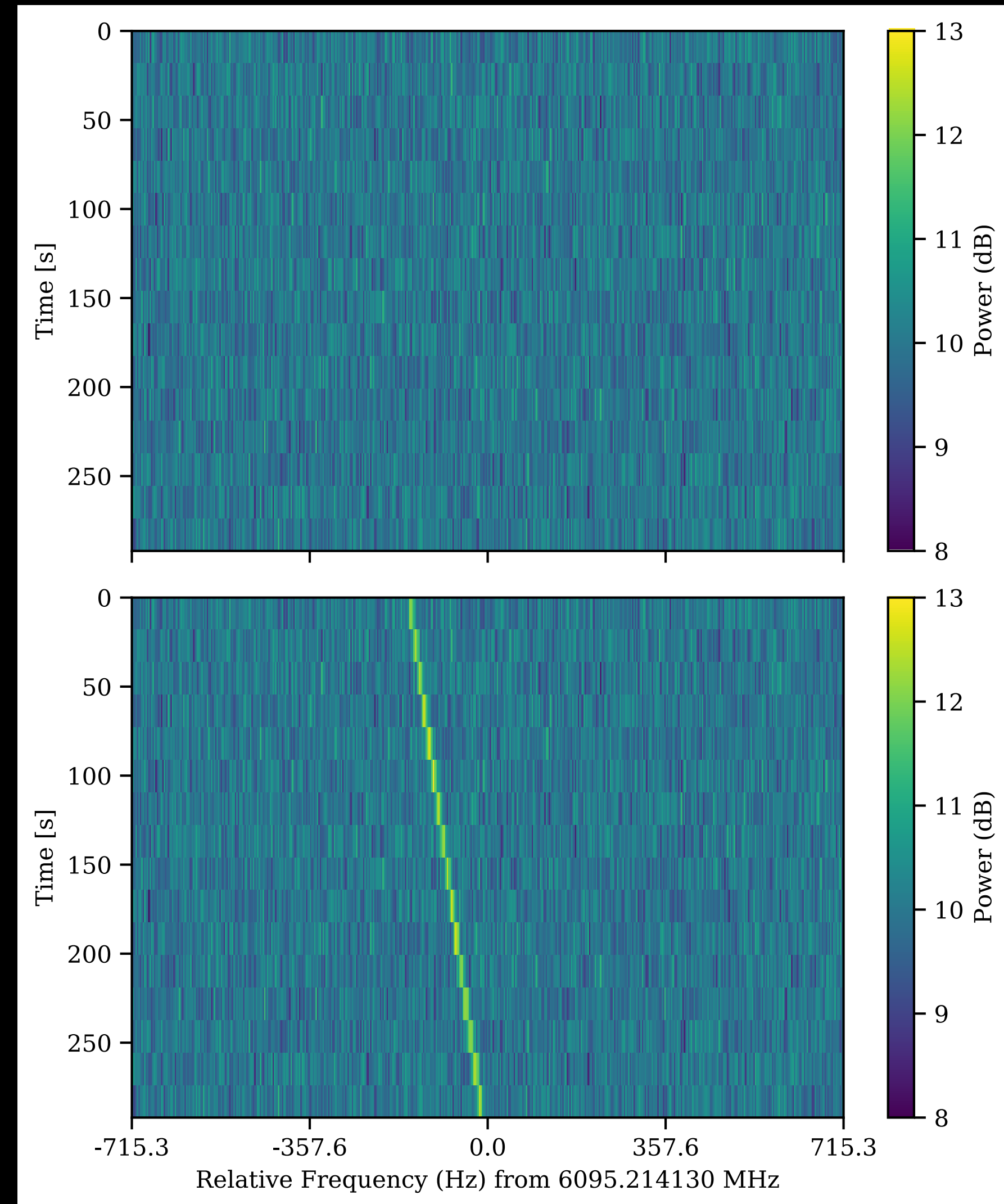
Setigen

- Python library for synthetic spectrogram and voltage data
- Specific focus on narrowband signal generation and injection

 **setigen** Public

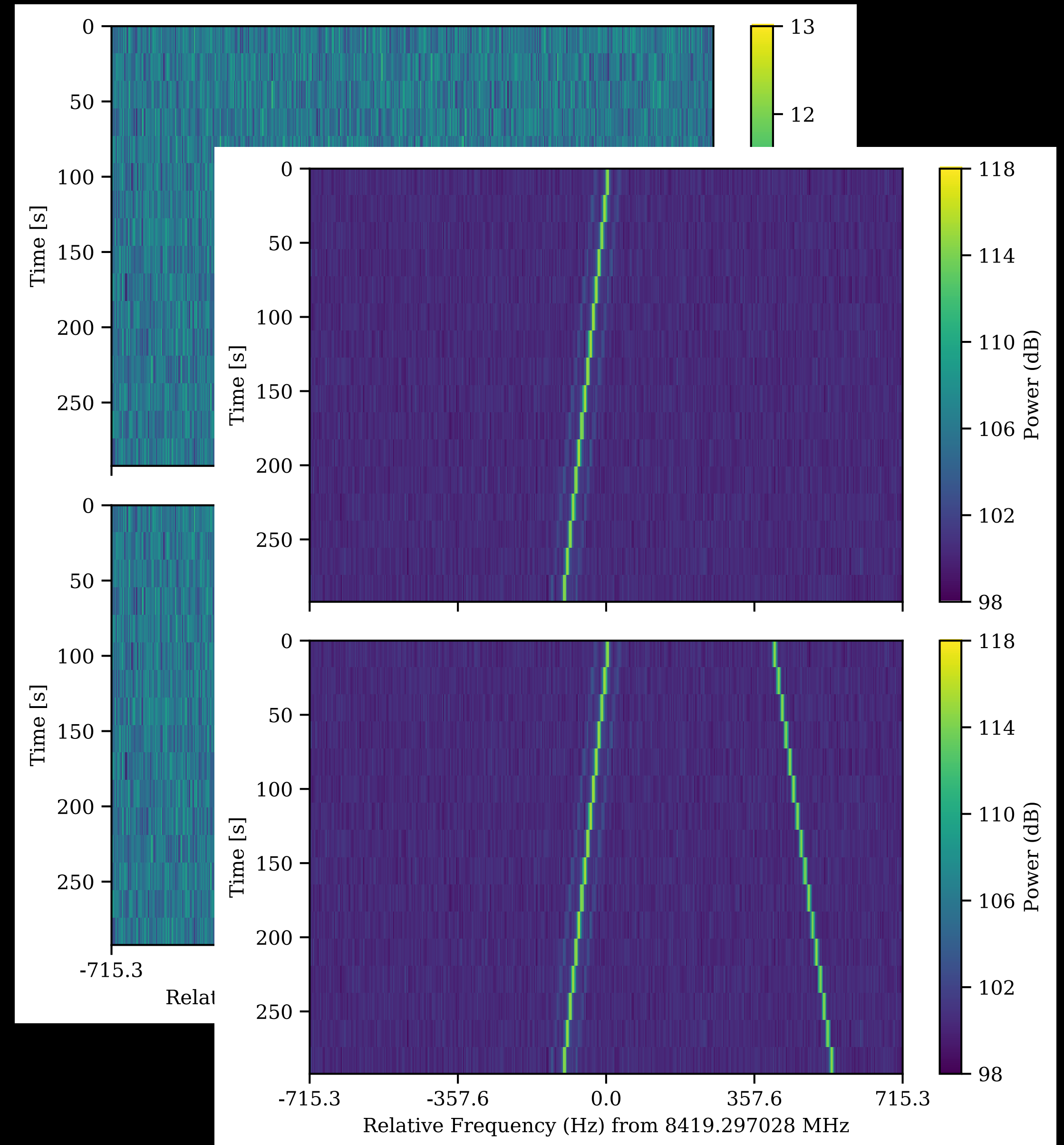
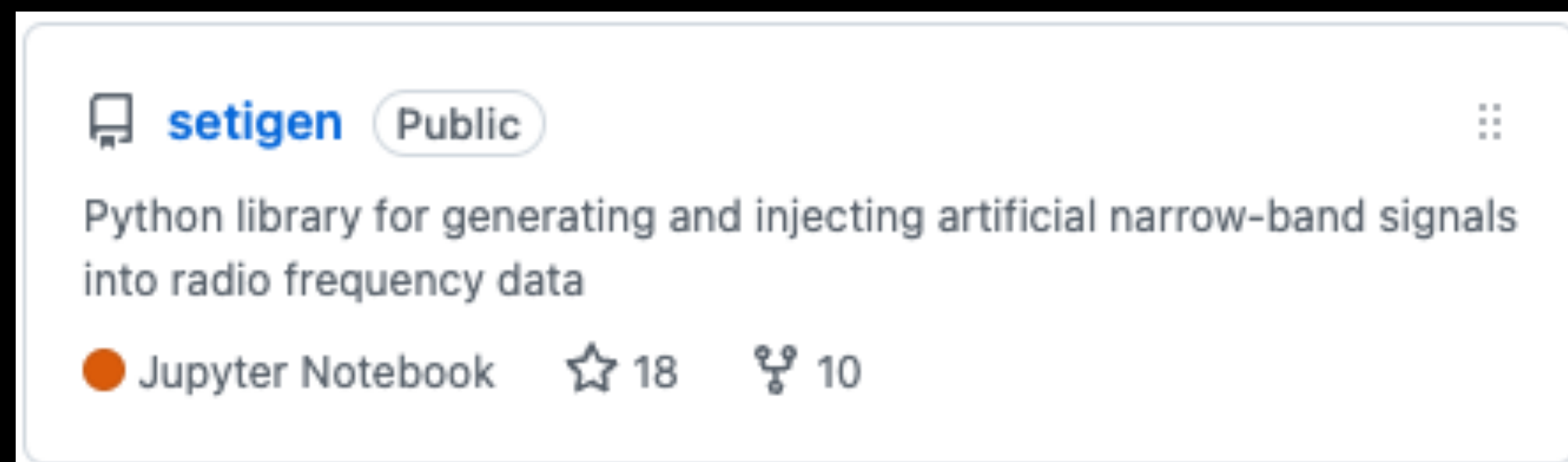
Python library for generating and injecting artificial narrow-band signals into radio frequency data

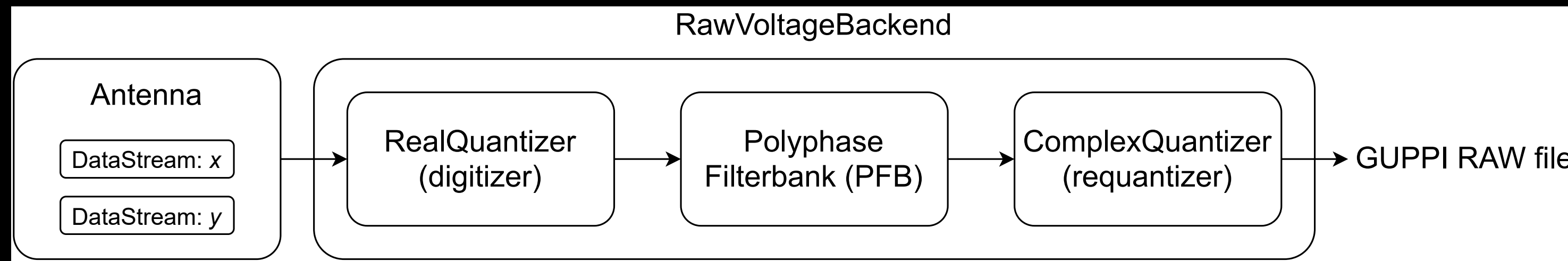
 Jupyter Notebook  18  10



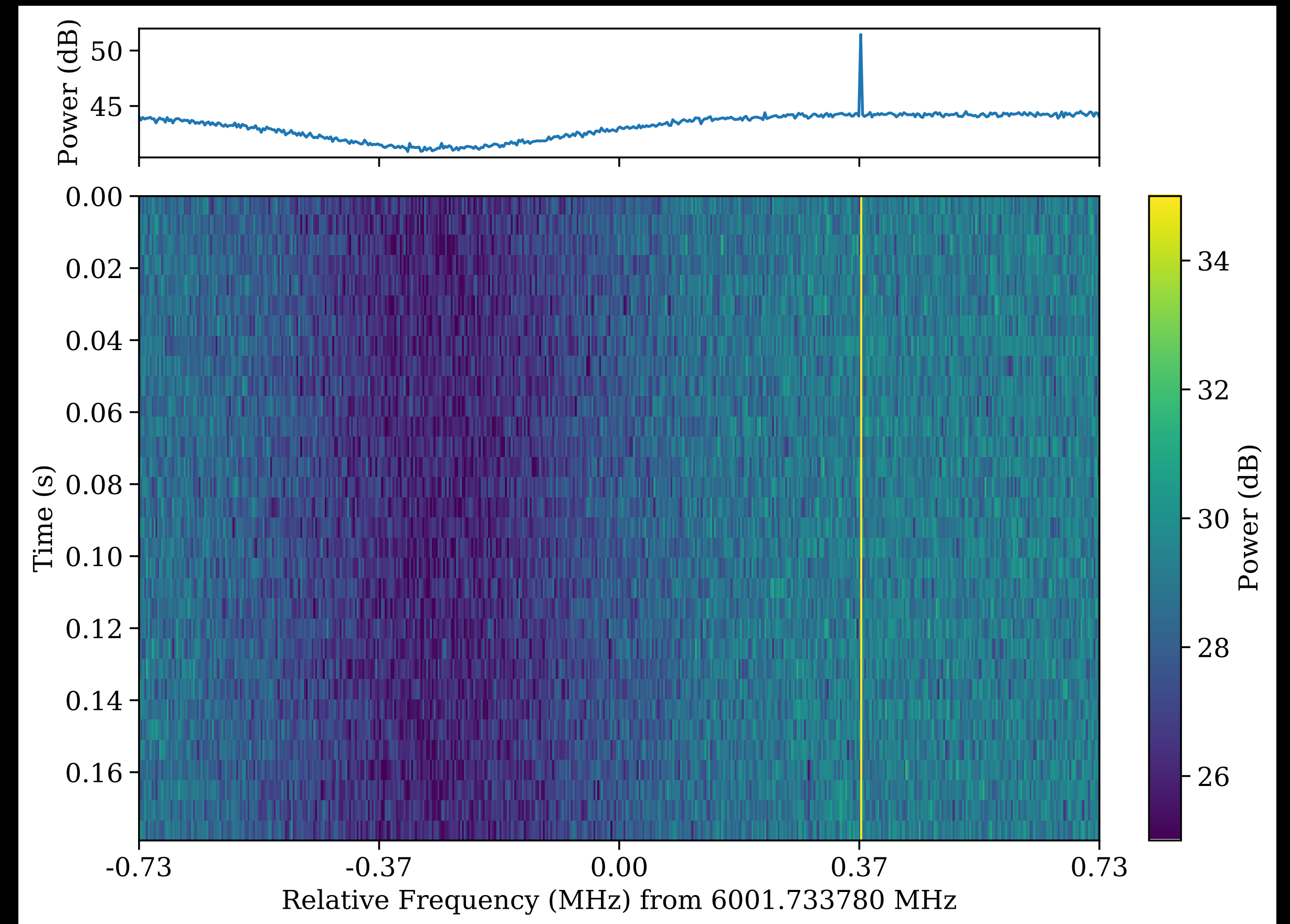
Setigen

- Python library for synthetic spectrogram and voltage data
- Specific focus on narrowband signal generation and injection



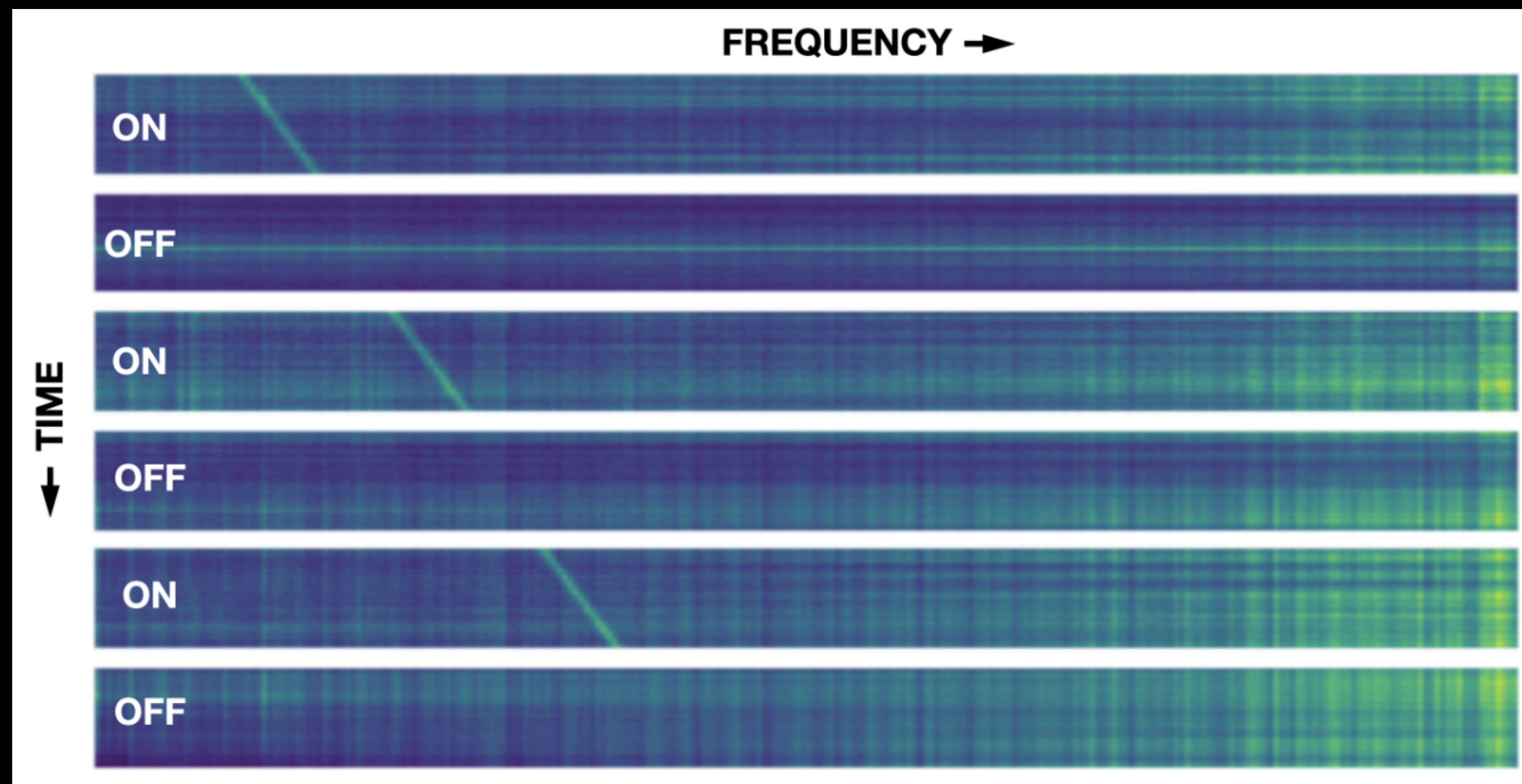


- Synthetic complex voltage data
- Simple models of backend components, such as a polyphase filterbank



Applications of Setigen beyond my research

- Injection — recovery testing
- ML dataset production (e.g. Kaggle)
- Multibeam search surveys
- Development of software for the Allen Telescope Array



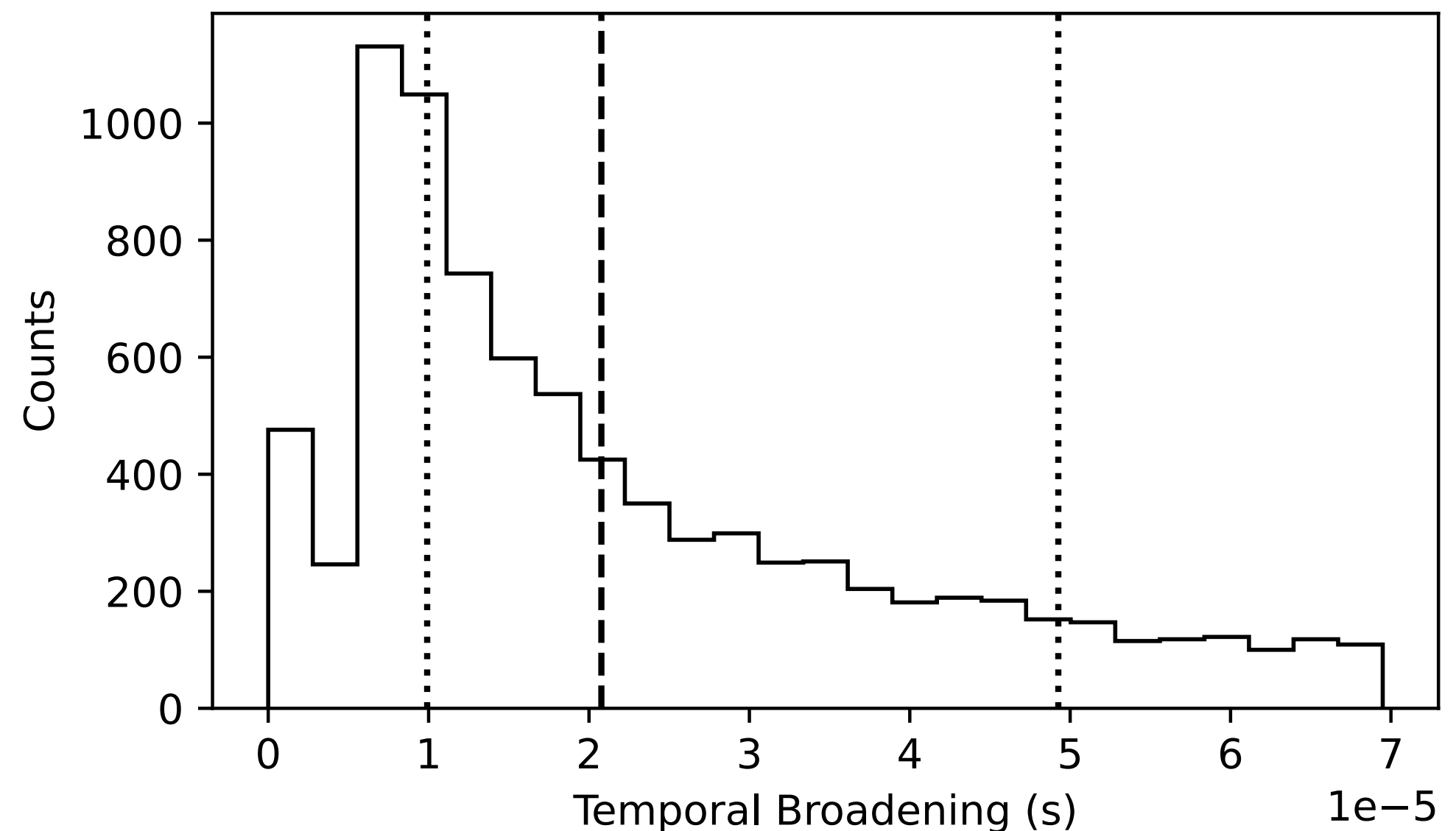
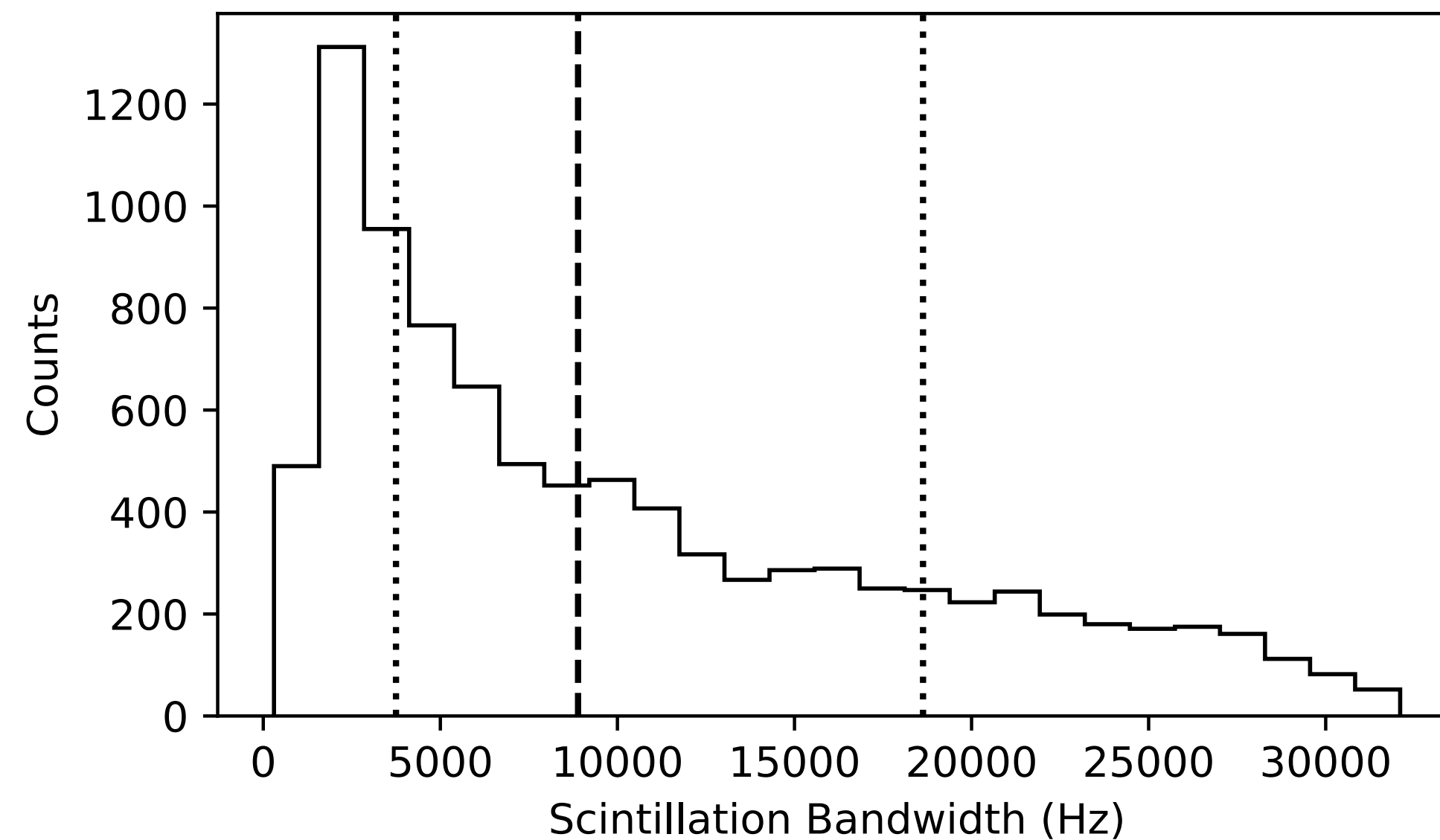
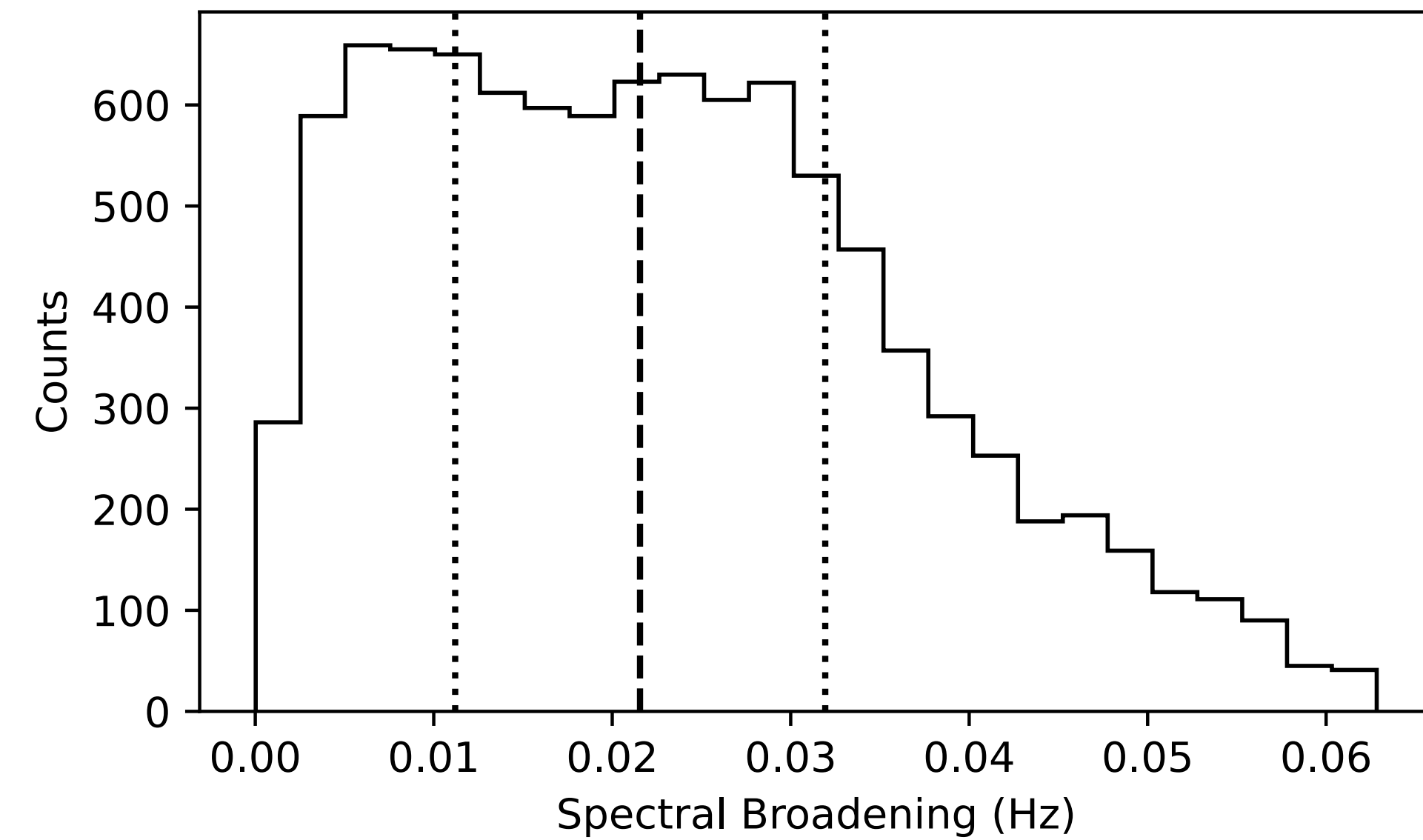
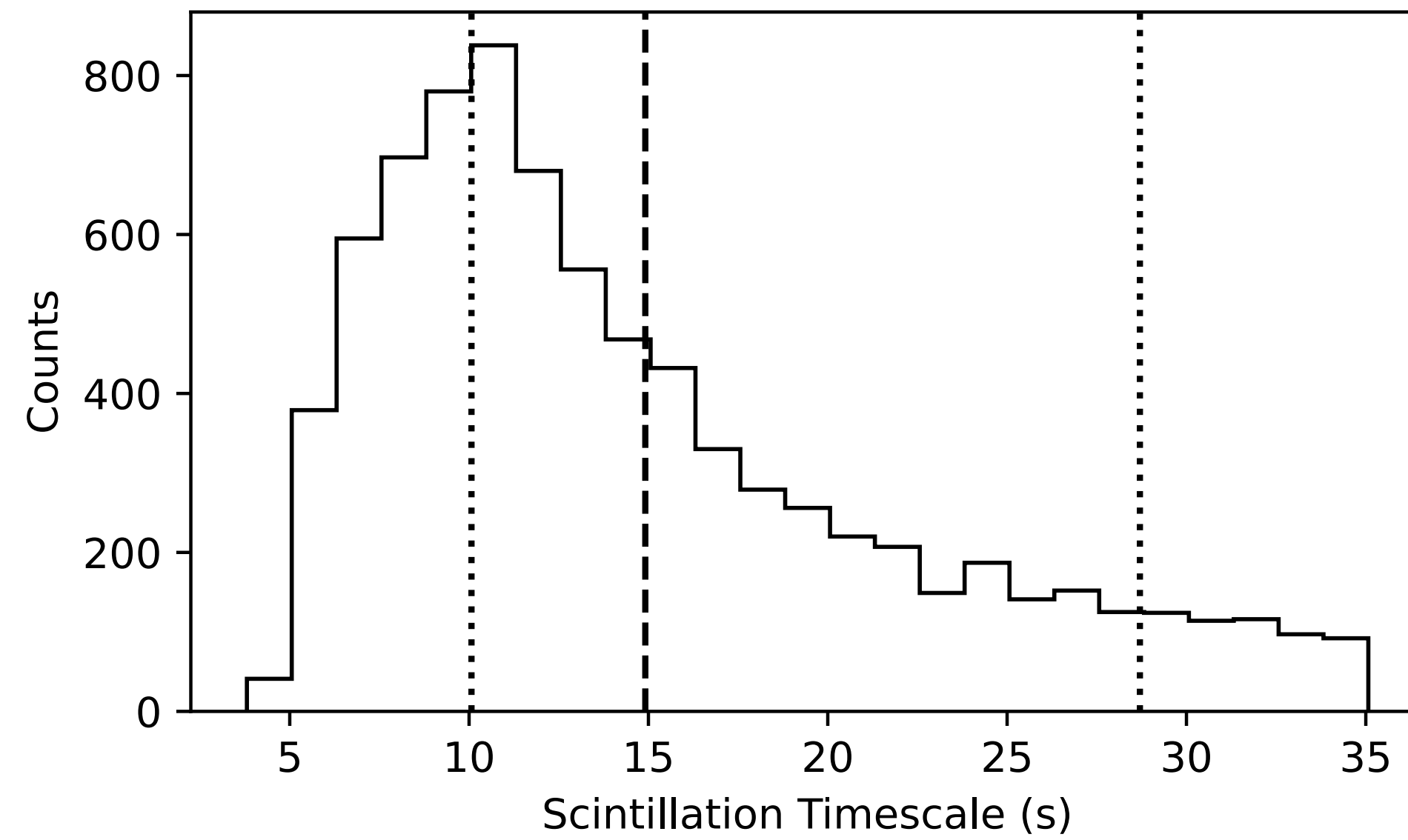
Breakthrough Listen x Kaggle 2021

Inter-quartile range

Median

C-band

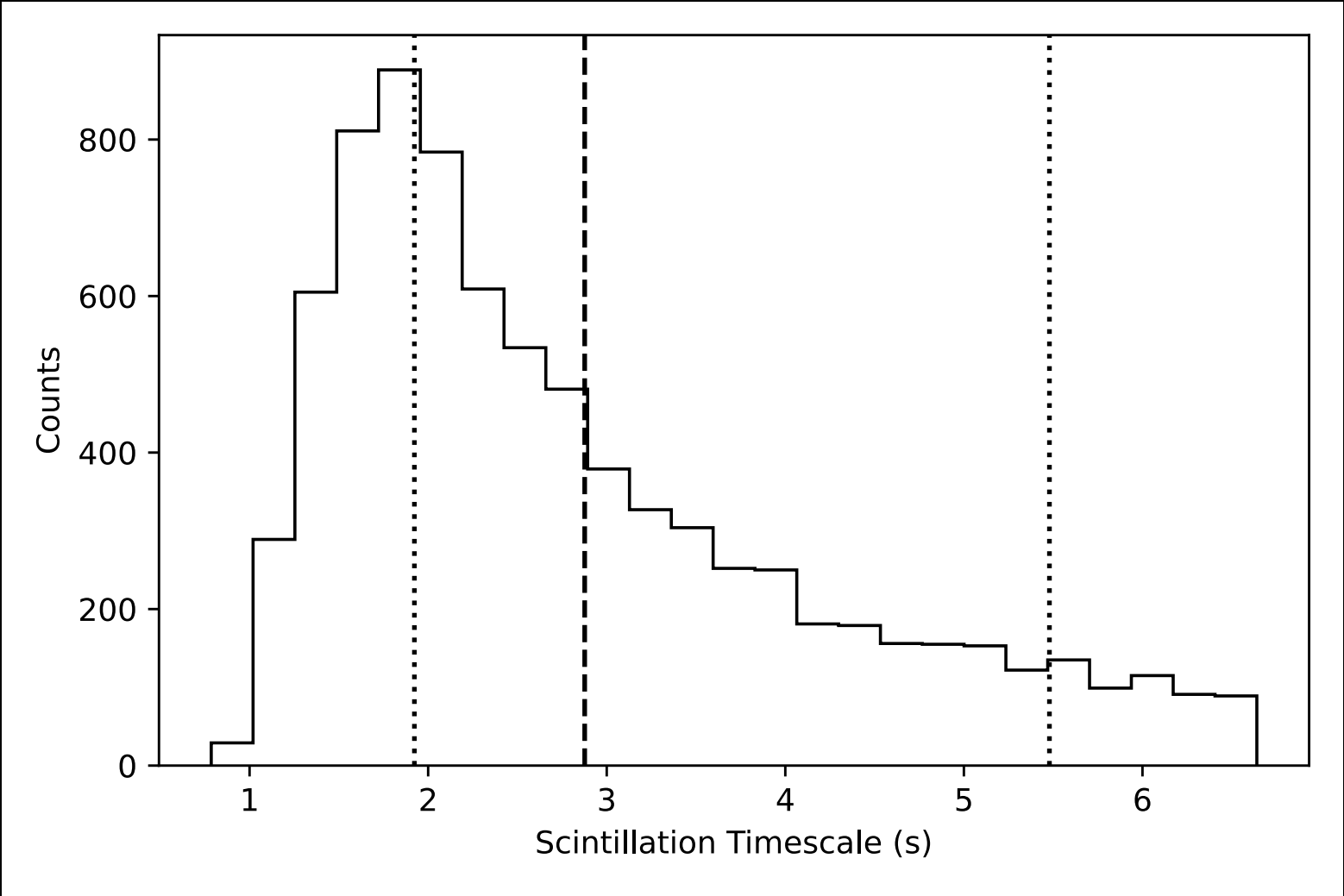
$(l, b) = (1, 0)$



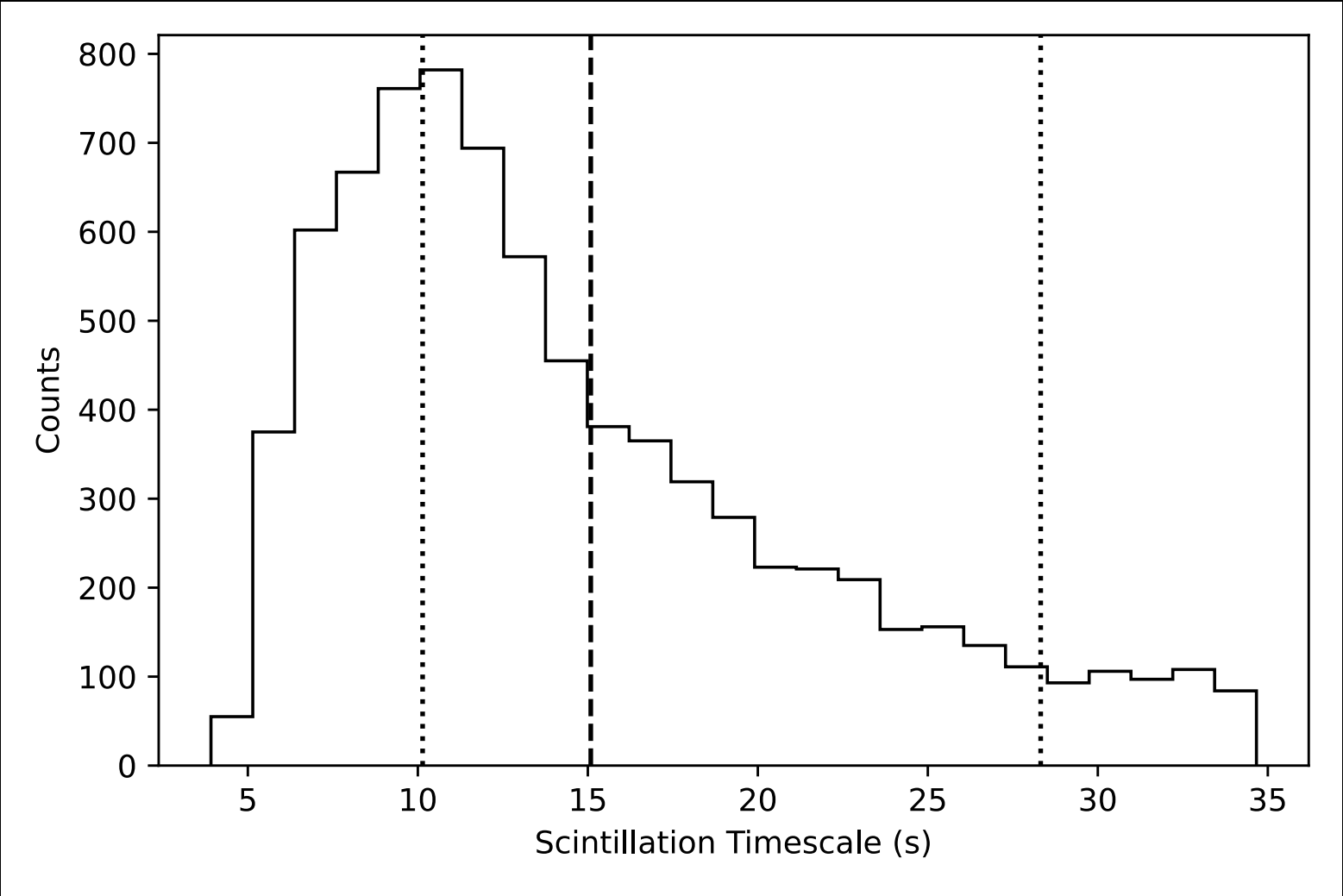
Monte Carlo-sampled timescales

Density

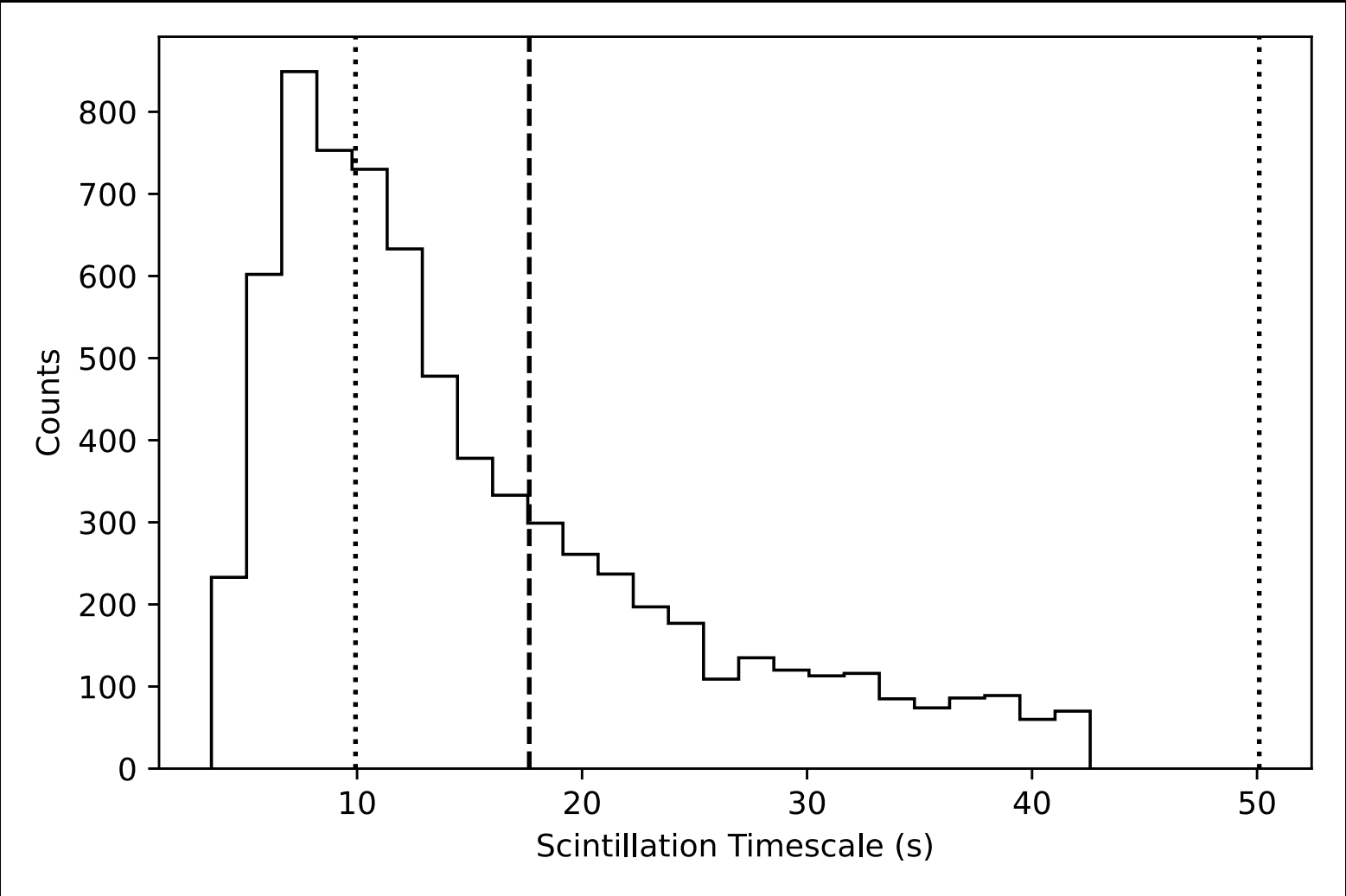
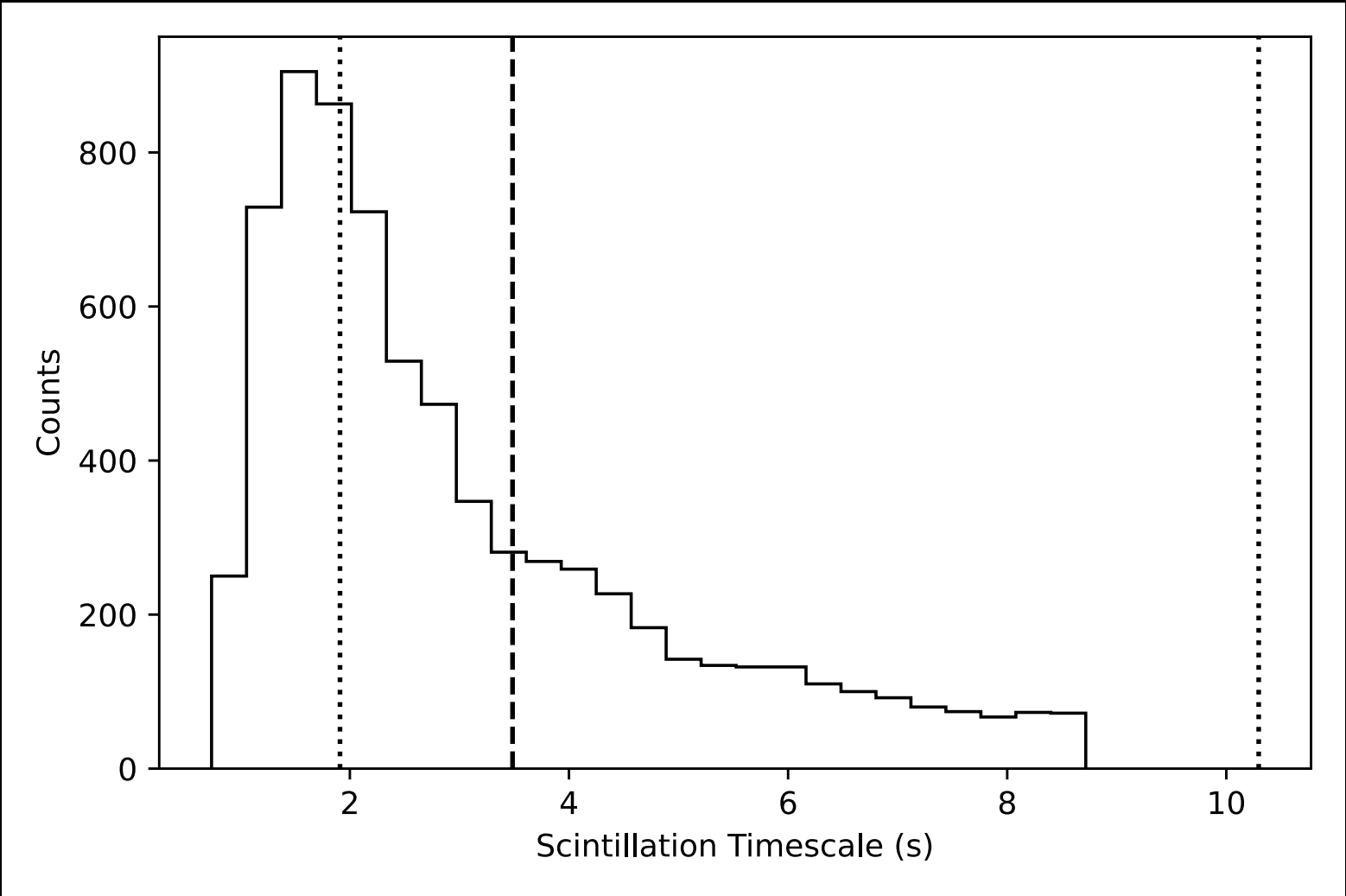
L band



C band



Uniform



Distance sampling



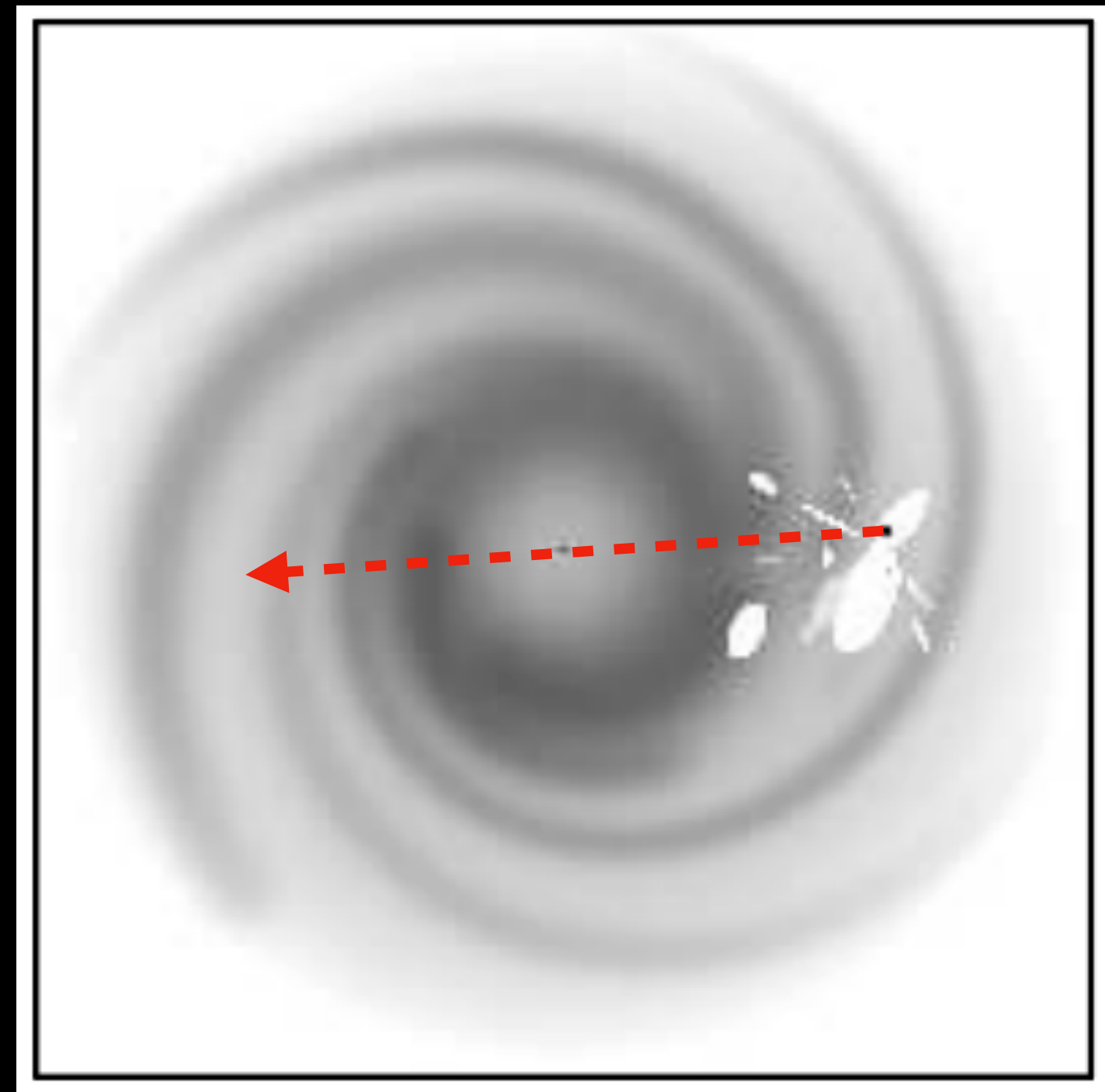
Statistics at different bands

$(l, b) = (1, 0)$

Band	Frequency (GHz)	Median (s)	IQR (s)	Mode (s)
LOFAR	0.110 — 0.240	0.22	0.14 — 0.41	0.14
L	1.1 — 1.9	2.9	1.9 — 5.6	1.9
S	1.8 — 2.8	4.8	3.3 — 9.0	3.1
C	3.95 — 8	15	10 — 28	11
X	8 — 11.6	28	19 — 52	16

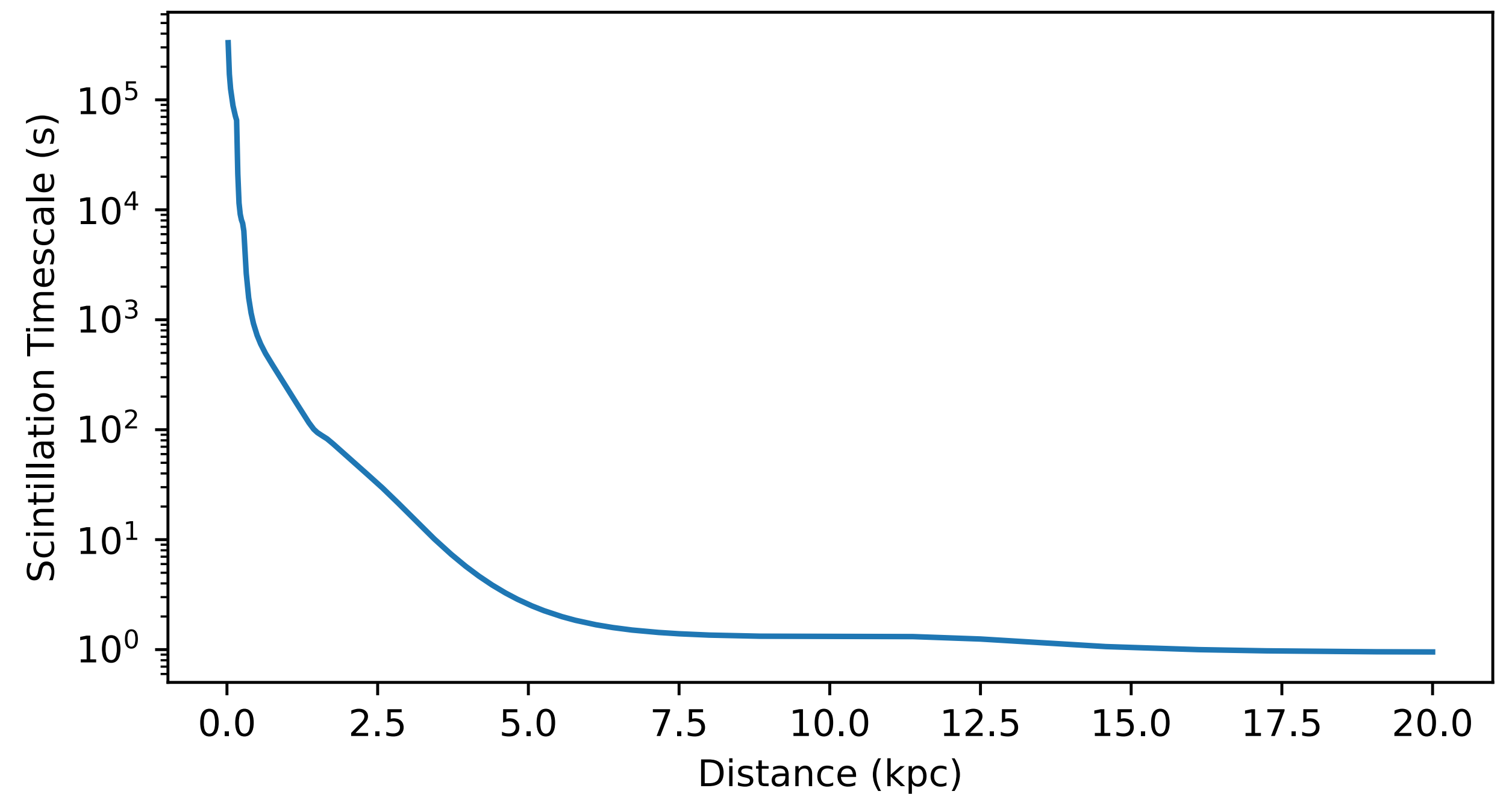
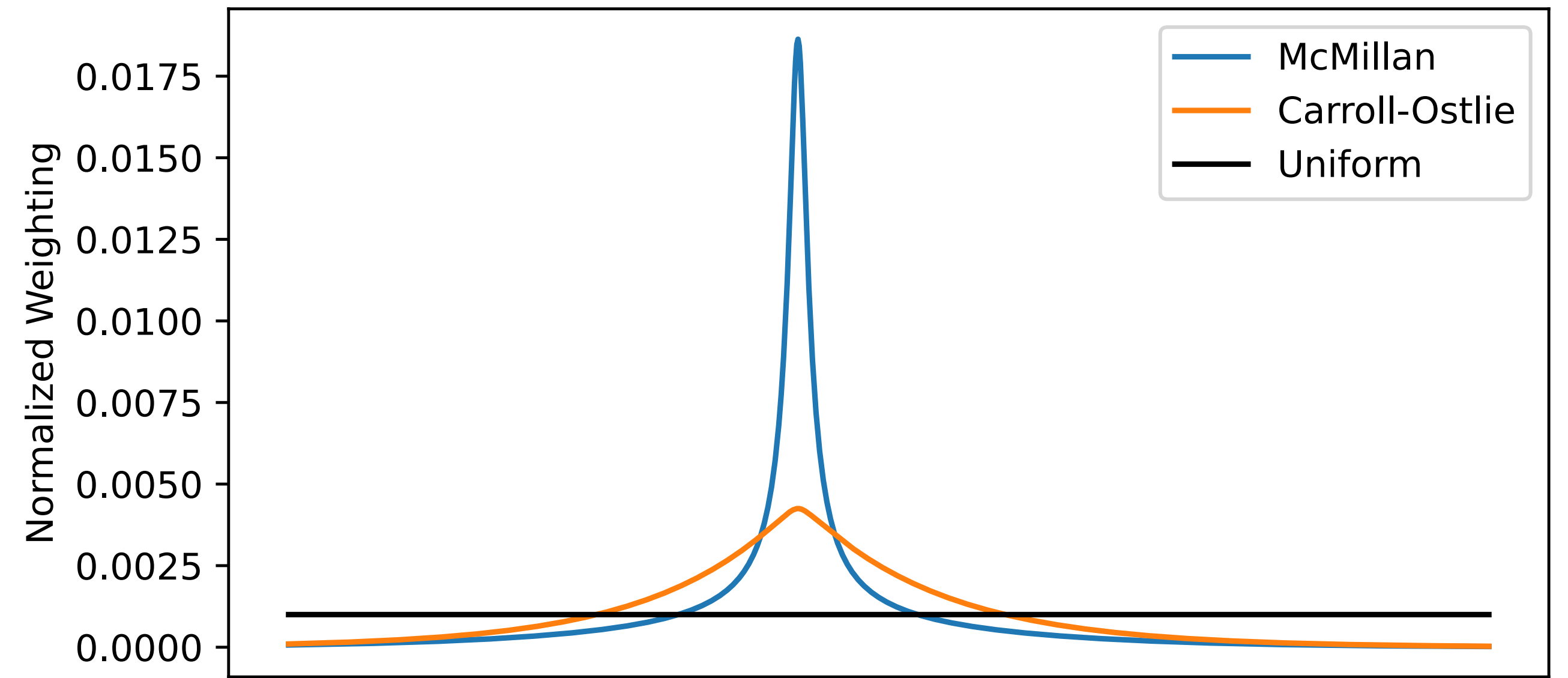
$\Delta t_d \propto \nu^{6/5} \nu_T^{-1}$

Density-based sampling



Cordes & Lazio 2002

$$(l, b) = (1, 0)$$

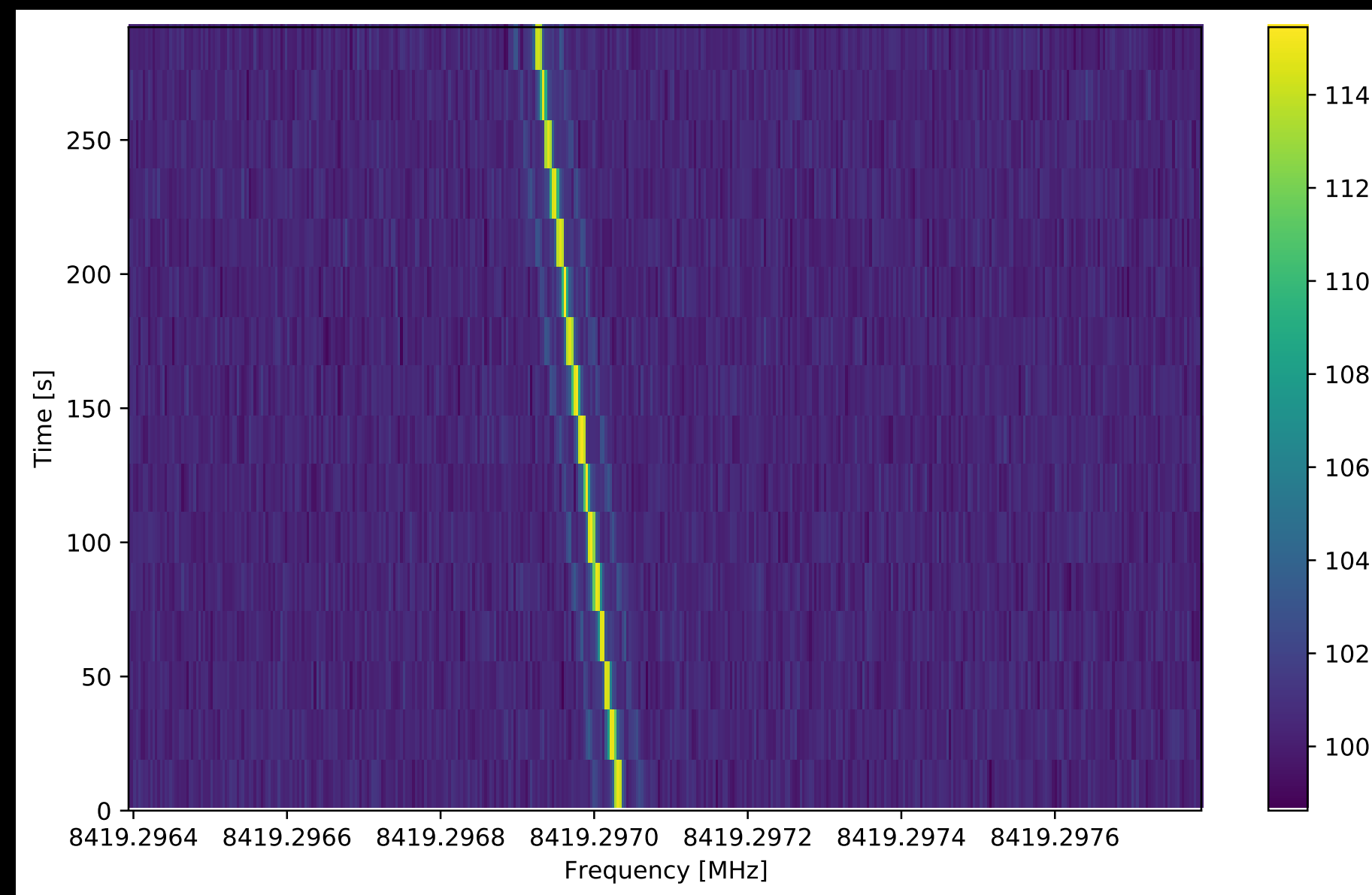


What would strongly scattered signals look like?

- Temporal scintillation
- Spectral broadening
- Pulse broadening
- Spectral de-correlation

What would strongly scattered signals look like?

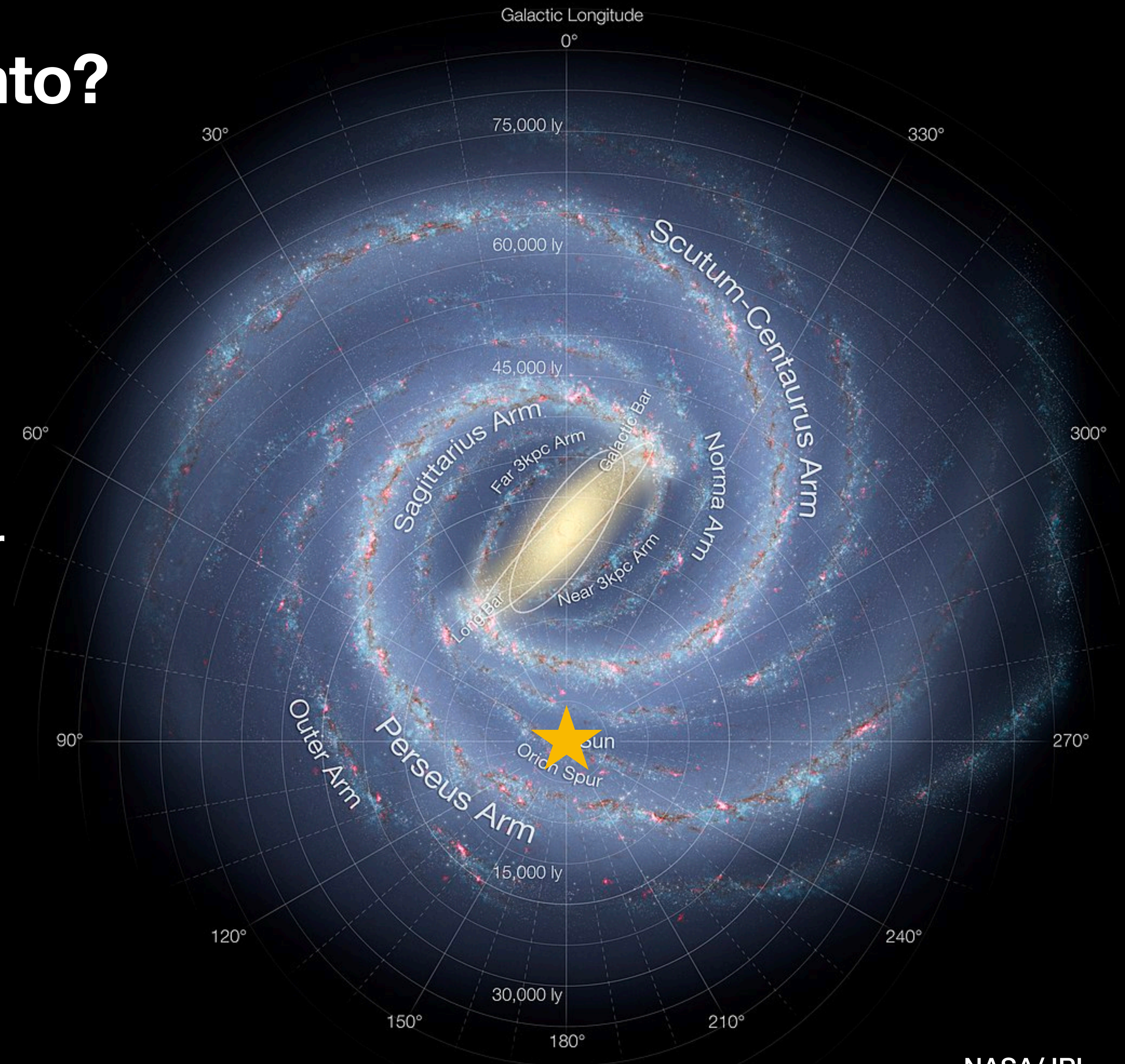
- Assuming a 100% duty-cycle narrowband transmitter



- Temporal scintillation
- Spectral broadening
- Pulse broadening
- Spectral de-correlation

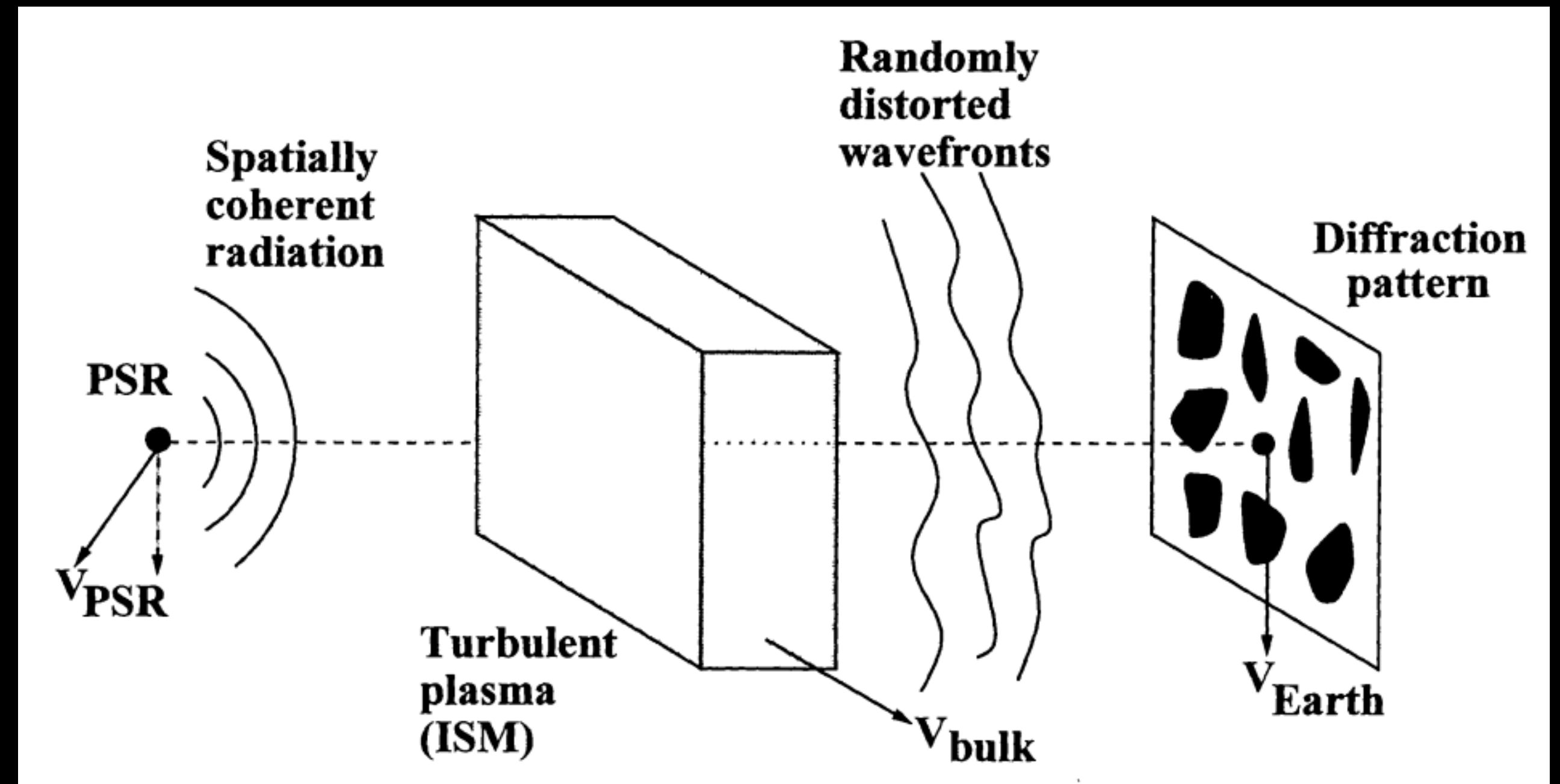
Why is this worth looking into?

- Astrophysical modulation as a filter for technosignature candidates
- Looking towards the Galactic Center is well motivated by SETI
- Could provide a framework for using more of the actual signals during narrowband analysis



We focus on so-called diffractive scintillations

- Electron density fluctuations give rise to phase fluctuations
- Multi-path propagation
- Interference pattern with characteristic spatial and spectral scales
- Can lead to 100% intensity modulation on characteristic temporal scales Δt_d



Cordes 2002

Parameter space exploration of scattering parameters

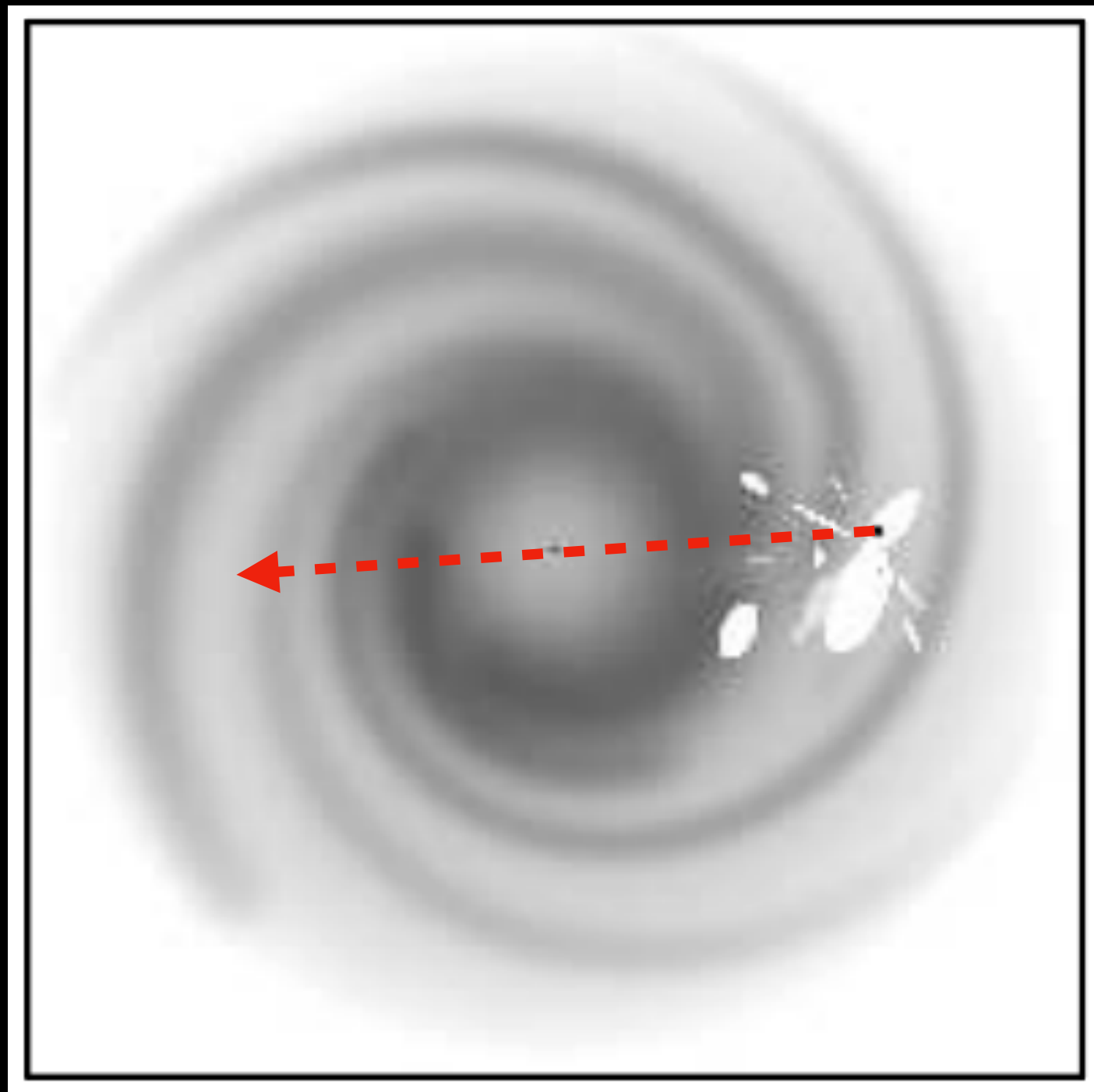
- A priori, we do not know:
 - Sky direction
 - Frequency
 - Distance
 - Transverse velocity

Monte Carlo sampling!

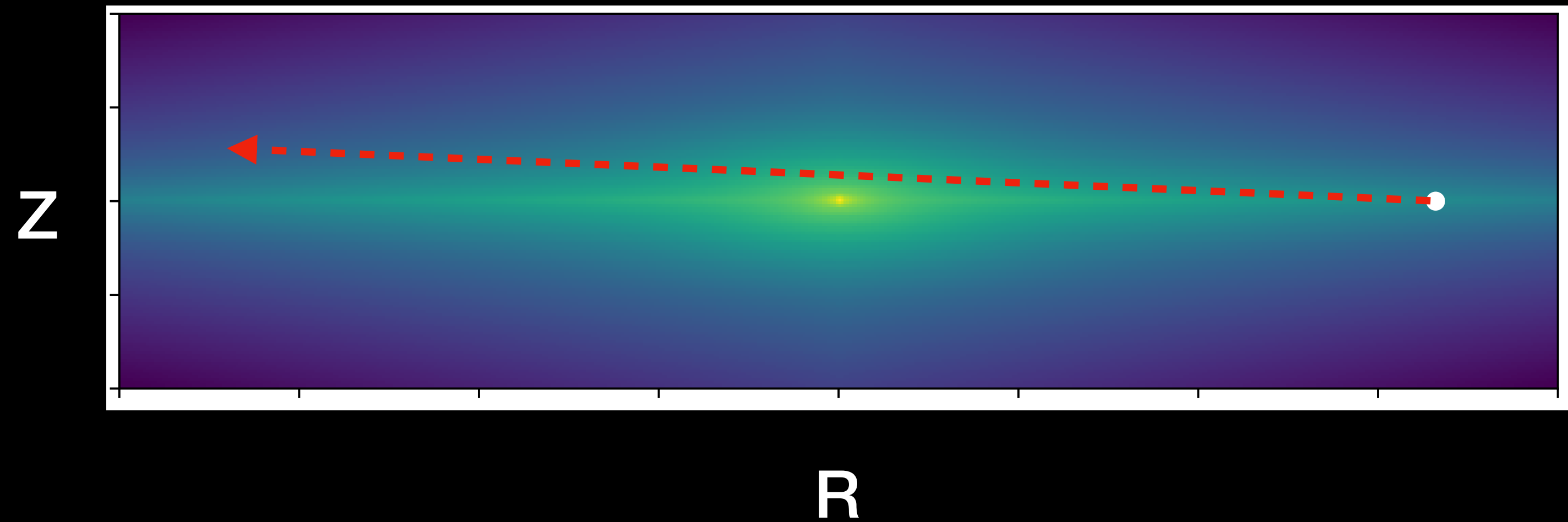
$$\Delta t_d \propto \nu^{6/5} \nu_T^{-1}$$

- Sky direction → Chosen parameter
- Frequency → Uniform sampling within chosen band
- Distance → Uniform or density based sampling
- Transverse velocity → Uniform sampling

Density-based sampling

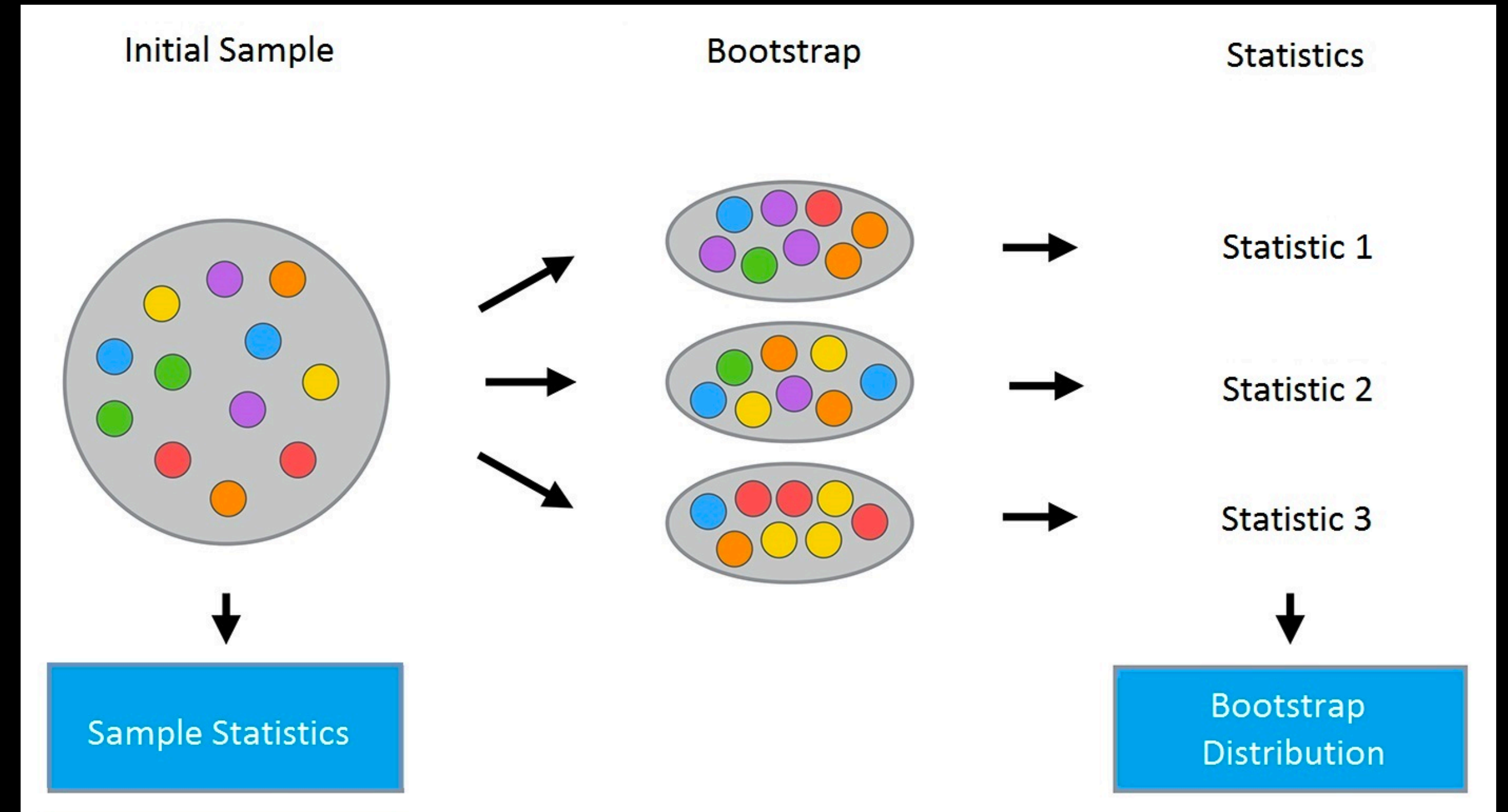


Cordes & Lazio 2002



Low sample regime

- Spread of values around the asymptotic “truth”
- Both correlated and uncorrelated samples within the same observation

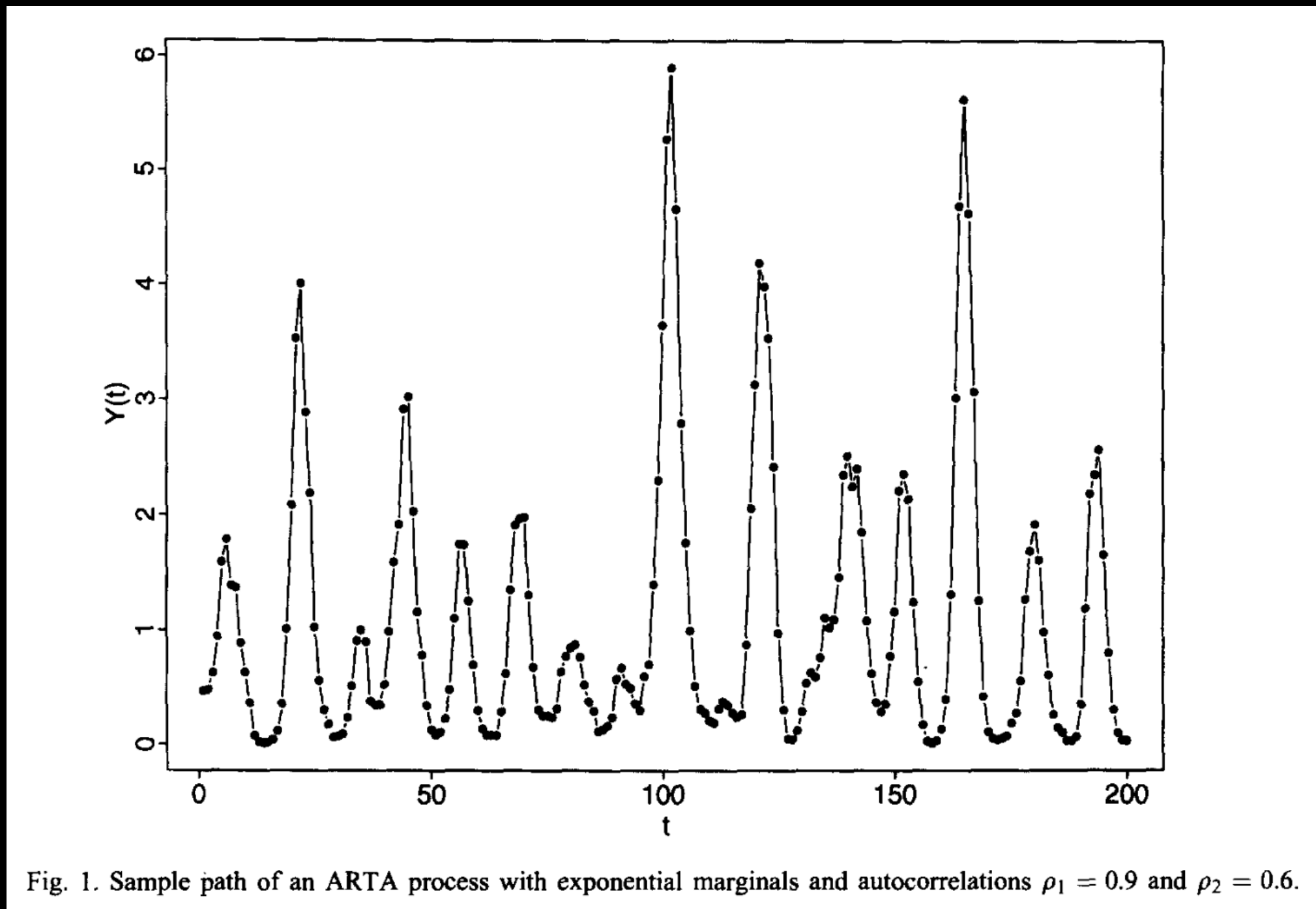


Kaggle

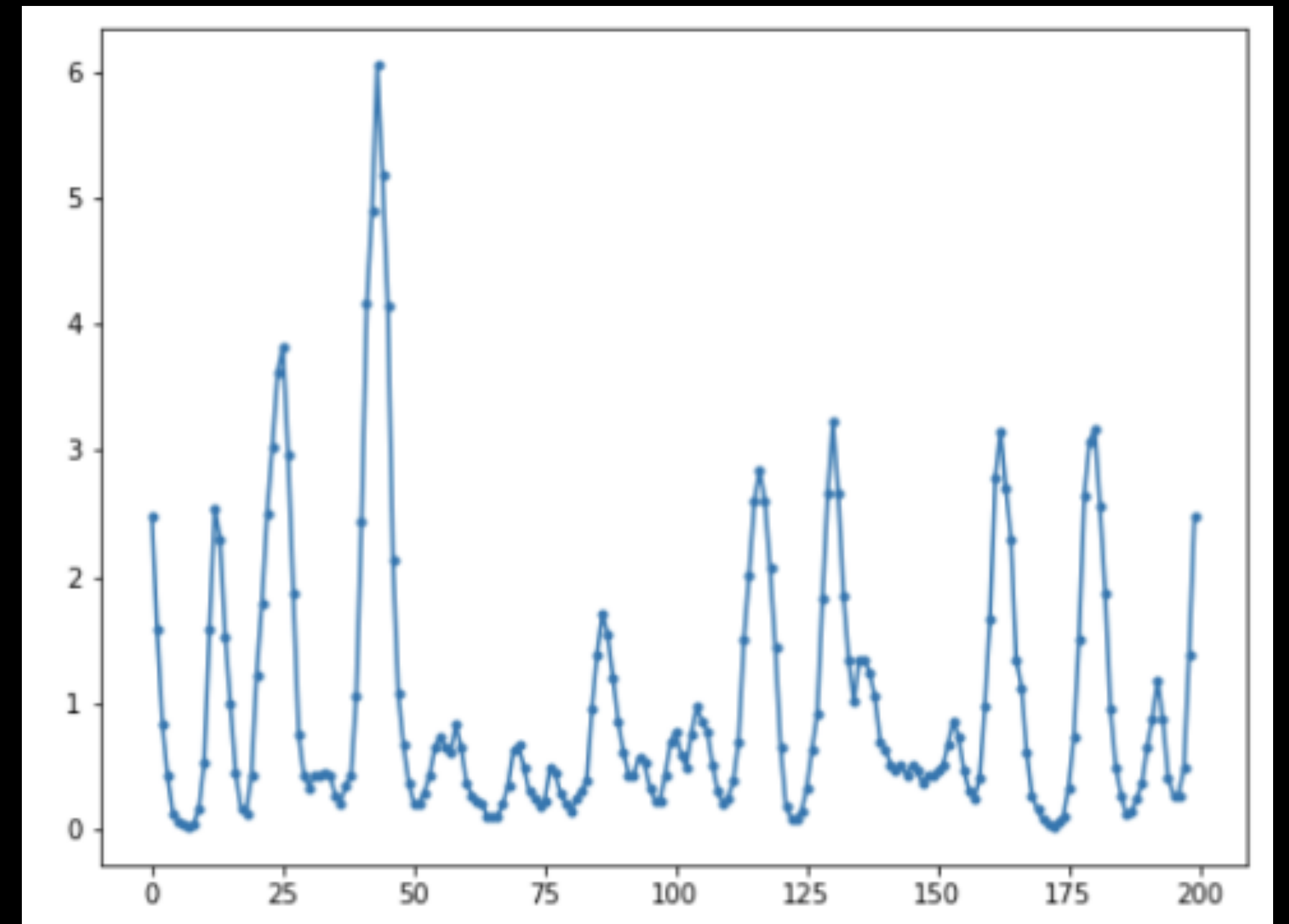
- How can we evaluate this?

Quick way to produce synthetic data with asymptotic statistics

- (Cario & Nelson 1996) The ARTA random process:
 - Matches a target intensity distribution
 - Matches a target autocorrelation structure (with custom asymptotic precision)



Cario & Nelson 1996



Our implementation